DIGITAL TRANSFORMATION OF KKBN BUSINESS PROCESS USING ODOO: A CAPSTONE PROJECT

A Capstone Project
presented to the Faculty of
College of Computer Studies
Camarines Sur Polytechnic Colleges

In Partial Fulfillment of the Requirements
for the degree Bachelor of Science in Information System

By
Gerald C. Buergo
John Edward M. Hontalba
Sharrycel B. Magno

December, 2025

DEDICATION

This capstone project is gratefully dedicated by the researchers to their beloved families, friends, instructors, classmates, Almighty God, and to all who extended help to the researchers in order to see the successful completion of this study. Their unwavering support, understanding, and encouragement served as the driving force that inspired the researchers to persevere despite the challenges encountered along the way. To their families, for the unconditional love, guidance, and motivation that have empowered them at every step of this academic journey; to their mentors and advisors, for sharing knowledge, wisdom, and patience with them and guiding them toward successful completion of this project; above all, to God, for granting them wisdom, strength, and perseverance to accomplish this study with faith and gratitude in their hearts.

Team Aport

ACKNOWLEDGMENTS

The proponents extend their earnest appreciation to the people who assisted and offered their undying support and guidance to make this capstone project possible.

To **Rosel O. Onesa** Dean of the college of computer studies for her permission and cooperation in allowing the proponent to conduct the Study

To **Ichelle F. Baluis** capstone Subject Adviser for her unwavering support, endless patience, and constant encouragement. Her wise guidance, valuable study, and considerate understanding have been crucial in accomplishing this study.

To **Josefina H. Llagas**, our capstone project adviser, for her selfless sharing of a lot of ideas, tireless checking of our study, advice, and unwavering guidance for the better of this study.

To **Jeremy Jireh S. Neo** our capstone project consultant, for his invaluable support and ideas, thoughtful suggestions, and expert opinion that really help in the completion of this study.

To **Angelie Ella L. Laynesa** our Grammarian, for her expertise and meticulous efforts in editing the manuscript. Her professional skills and dedication greatly improved the quality of this study.

To our panelists headed by **Niño Jeffrey L. Luzon**, together with **Jayvee Neil Sias** and **Dr. Jocelyn Lipata** for their valuable recommendations, insightful comments, and suggestions, which further helped us to more improve this study.

To **KAYAMANA KALUSUGAN BASTA NATURAL (KKBN)** specifically to the owner for accepting our request as our client and for outstanding cooperation to make this study successfully.

To **Justine D. Labios** the designated Secretary for his continued assistance in documenting the panel's recommendations, ideas, and comments during the capstone project defense.

To our **Friends, Classmates**, and **Loves ones** for their inspiring words and for staying overnight just to study and for coming to school to support each other, thoughtful ideas, and heartfelt prayers. Their encouragement, and unwavering presence gave us the strength and motivation needed throughout the development of this study.

To Odoo **ERP & IT Experts** for participating as respondents in the survey and for sharing their valued insights and technical expertise. Their contributions enriched the study immensely, especially in the analysis of ERP System features. Your commitment and cooperation allowed us to gain a better understanding and full analysis of the system.

To **All**, whose names may not be mentioned here but were instrumental in the completion of this study.

To all **Respondent**, thank you for your candid answers and invaluable collaboration, which made this study possible. Finally, the researchers are grateful to the Almighty God for providing them with His grace, strength, hope, and peace of mind, which allowed them to overcome the difficulties and challenges encountered throughout this study.

Thank you very much!

ABSTRACT

Title:	Digital Transformation of KKBN Business Process Using Odoo: A Capstone Project
Authors:	John Edward M. Hontalba Sharrycel B. Magno Gerlad C. Buergo
Number of Pages:	193
School:	Camarines Sur Polytechnic Colleges
Degree Conferred:	Bachelor of Science in Information System
Keywords:	Digital Transformation, Enterprise Resource Planning (ERP), Odoo, ISO 25010, Small and Medium Enterprises (SMEs), Business Operations, Odoo Modules, System Implementation

In today's competitive environment, small and medium enterprises (SMEs) must embrace digital transformation to enhance operational efficiency and streamline processes. This study focused on implementing the Odoo Enterprise Resource Planning (ERP) system in Kayamanan, Kalusugan, Basta Natural (KKBN). The objectives include analyzing existing business processes, configuring key Odoo modules—Inventory, Sales, Accounting, Point of Sale, Purchase, and Employee Management with Attendances, and evaluate the system's acceptability based on ISO 25010 quality characteristics, including Functional Suitability, Performance Efficiency, Usability, Maintainability, Reliability, and Security. A mixed-method approach was utilized, combining surveys and interviews with stakeholders such as the business owner, secretary, sales personnel, ERP professionals, and IT specialists. The system's effectiveness was measured using a weighted mean to determine how well it met operational needs. Results revealed high acceptability across all quality criteria, with the strongest performance in security and functional suitability. The overall weighted mean of 4.75 indicated that the system surpassed expectations and complied fully with required standards. The study concludes that the customized Odoo ERP solution is highly effective in automating and optimizing SME operations. It further recommends exploring additional Odoo modules, enhancing customization to meet evolving business needs, and conducting comparative analyses with other ERP systems to identify best practices for SMEs.

TABLE OF CONTENT

LIST OF TABLES

LIST OF FIGURES

CHAPTER 1
INTRODUCTION

This chapter present the project background of the research: The implementation of Odoo ERP Software for KKBN (Kalusugan, Kayamanan, Basta Natural) Business Operations. It includes the Project Context, Purpose and Description of the Project, Objectives of the project, Significance of the Study, Scope and Limitations of the Study, Project Dictionary, and Notes for citations.

1.1 Project Context

In this technological age, where almost everything is transformed into digital format, organizations have been the first to embrace this change by deploying software and other devices to facilitate their work, make it faster, and improve efficiency. Nowadays, manual processes are one of the challenges that most small-scale businesses have to deal with. These old ways of doing things are often associated with paperwork and manual processes that lead to a decrease in accuracy, efficiency, and scalability. Based on this, many small firms fail to be resilient in a digital business landscape. The hindrances caused by the conventional ways of working serve as a wake-up call for small businesses to incorporate digital transformation as a way of staying alive and successfully operating in an environment where sustainability and innovation are prerequisites. Software called Enterprise Resource Planning (ERP) is what companies turn to in order to handle day-to-day operations such as accounting, purchasing, financial planning and analysis, project management, risk management, and supply chain activities. ERP companies bring together a collection of financial and operational applications that create a shared data structure, therefore, the only correct source of data is the one that the user has inputted.

ERP systems are program bundles that cover a number of areas that are concerned with financial and operational aspects and are combined through a common data model thus, confirming the data that the user enters is the only one available [1]. Hence, ERP systems make sure departments have precise and up to date information, and consequently they enable better decision making, transparency, and overall effectiveness of the organization.

Through ERP deployment, organizations are able to access several main advantages. It integrates various processes into one to eliminate the need to have too many systems, trims down the waste and improves the quality of management by giving precise and timely information and offers solid ground for strategic planning. While conversely, companies that do not use ERP, have problems like data duplication, lack of consistent records, and they make bad decisions as they are lacking real-time information.

In the market of ERP systems, Odoo ERP is one of the most trusted and versatile systems especially for SMEs. The use of Odoo has been proven to increase productivity as it permits the execution of more tasks in a shorter time, it is an effective training tool, it helps in decision-making, it cuts human errors, and it gives the user a customizable module to manage inventory, sales, and human resources. In the end, it lowers the costs of operation and increases efficiency [5].

KKBN is a local counterpart that demonstrates the mentioned issues, is a Nabua and Iriga City-based company that is into selling herbal products like Shatsu, CMD, and Kape ni Manoy. The company uses paper-based means which are the main trigger for problems such as stock-outs, lost sales, unreliable financial records, and cash issues. Moreover, these features the problem of inventory management, the problem of purchase orders, and the problem of sales monitoring, etc. all can be found in the Odoo ERP implementation by KKBN. This single solution can automate the redundant tasks, decrease the amount of paperwork, and raise the overall productivity. What is more, Odoo's modular structure facilitates flexible adjustments as per the company's operational requirements thus, improving data management and efficiency in the process. With the continuation of the use of manual and paper-based methods, KKBN has become stagnant in terms of growth, failure to adjust to market movements and the inability to make fact-based decisions. In such a situation, a business could be saved by adopting Odoo ERP through automation, centralized data storage, and real-time data access. This research emphasizes that ERP adoption is quite a significant thing for small businesses like KKBN in gaining operational effectiveness, improving decision-making, and going back to a better horizon in terms of sustainable growth in the digital time.

1.2 Purpose and Description of the Project

To analyze current state of KKBN business processes, pinpoint what can be improved, and apply Odoo ERP modules according to the set requirements is the main goal of this study. The desired outcome was to digitalize KKBN main operations, thus, improve data accuracy, streamline workflows, and promote evidence-based decisions.

The project was about documenting and assessing business processes, identifying suitable Odoo modules such as Inventory, Sales, Accounting, POS, Purchase, and Employee Management and Attendance, and configuring them according to KKBN's requirements. The deployed system was examined in accordance with the ISO 25010 software quality standards to ascertain the positive effects on Performance Efficiency, Functional Suitability, Usability, Maintainability, Reliability, and Security. The assessment was made by asking users for their opinions on whether the system would assist them in their daily tasks.

A distinct characteristic of this study is that it has to do with actual business operational issues that KKBN, which is a small local business, may go through. Unlike other studies that concentrate on large organizations with the most advanced technology potential, it must be highlighted that a small company such as KKBN has its own operational attributes like few workers, a small budget, and many manual processes. Therefore, in this study, the byproduct of the implementation of the real-world Odoo ERP deployment at a small local business, is the proof that it is not only achievable but also beneficial for the company to use ERP system effectively.

1.3 Objectives of the Project

Generally, this study aims to implement ODOO ERP at KKBN Iriga and Nabua Branch and business processes for efficient work.

1. Examine the business processes and challenges in KKBN Iriga and Nabua.
2. To select and configure the necessary ODOO module for KKBN.
   - 2.1.1. Inventory
   - 2.1.2. Sales
   - 2.1.3. Accounting
   - 2.1.4. Point of sale
   - 2.1.5. Purchase
   - 2.1.6. Employee Management and Attendance
3. To evaluate the configured ODOO module using ISO 25010 specifically using the following:

3.1.1. Functional Suitability;

3.1.2. Performance Efficiency;

3.1.3. Usability;

3.1.4. Maintainability;

3.1.5. Reliability; and

3.1.6. Security

1.4 Significance of the Study

The researchers believe that this study will help KKBN to optimize its business operations for present and future needs. It also provides fast transaction processing and easier tracking for the owner.

Business Establishment/Management. This study contributes to KKBN's productivity and efficiency. Time is saved, mistakes are reduced, and decision- making skills of the management are improved.

Customers. The system increases customer satisfaction by ensuring correct billing, prompt order processing, and better communication. Consumers Customers benefit from improved service and a smoother shopping experience.

Researchers. The study allowed researchers to improve their academic and professional skills in research and writing. It also enhanced their analytical thinking, problem-solving abilities, and understanding of real-world information system implementation.

Information System Field. This study advances ERP best practices, particularly for SMEs, by showing the practical benefits of ERP adoption.

Future Researchers. The study provides references and insights that may serve as a foundation for further improvements.

Other Businesses. This study may serve as a guide for SMEs considering digital transformation, showcasing Odoo's potential in reducing manual processes and modernizing workflows.

1.5 Scope and Limitations of the Project

The research inquired into KKBN's present state to determine its business processes and requirements. After the study, the researcher selected and rearranged the Odoo Modules for reconfiguration such as Inventory, Sales, Accounting, POS, Purchase, and Employee Management and Attendance. The research further included an assessment of an ISO 25010 model to check whether the configured Odoo system is appropriate based on Functional Suitability, Performance Efficiency, Usability, Maintainability, Reliability, and Security. Odoo was a platform that presented numerous open-source business applications that would support companies of any size to handle different elements of their business practices. It provided numerous modules for which users need to manage their organizations like project management, production, stock control, sales, accounting, and human resources, to name a few. Businesses have the opportunity to select the first few modules they need and then add new ones whenever they desire. He was aware of it since it had a friendly user interface, was packed with many different customization options, and provide easy integration with other third-party applications. Odoo ERP helped enterprises to improve their operations, grow productivity, and become data-driven by the use of technological innovations.

The research scope is confined to Odoo ERP, and it does not contain other ERP systems. The survey is mainly within the KKBN Iriga and Nabua branches and only covers 10 respondents the owner, some staff, 3 IT experts, and 2 ERP experts. The research discusses instantaneous effects on the Odoo ERP incorporation and does not look through the wider organization impact. In addition, as the Odoo community edition is adopted, customization alternatives are limited, and internet surf is needed to run the system.

1.6 Project Dictionary

To make understanding easier, this research explains technical words with clear definitions, simple expressions, and practical examples.

Enterprise Resource Planning (ERP). The ERP refers to a type of digital tool used by business to manage day-to-day operations. The researcher applied one of the ERP systems in this study, which a single platform to boost productivity, facilitate decision-making, and aid in the general expansion and prosperity of the company [3].

ISO 25010. This standard gives guidelines and recommendations to assess software quality of products. The study evaluates key quality characteristics of the implemented ODOO system including functional suitability, Performance Efficiency, Usability, Maintainability, Reliability, and Security in alignment with the 2023 updated version of ISO 25010 parameters [2].

Functional Suitability. This evaluates how well the system functions meet the specified requirements and perform the desired tasks correctly and completely. It ensures that Odoo modules such as Sales and Inventory provide correct result that align with business goal.

Performance Efficiency. This evaluates the system's operation, stability, and responsiveness under conditions. It ensures Odoo's ability to handle data and process transactions successfully without experiencing any system lags or delays.

Usability. This feature evaluates how easy and satisfying it is for users to operate and become familiar with the system. It focuses on Odoo's accessibility, navigation, and user interface to ensure that even non-technical users can use it properly, efficiently, effectively, and with minimal training required.

Maintainability. This evaluates to the system's ability to be customized, corrected, or enhanced with minimal time or effort. It ensures that Odoo modules are flexible and easy to update as the business grows or new requirements emerge.

Reliability. This evaluates its ability to carry out its operations reliably, precisely, and consistently over time. It ensures that Odoo will continue to be dependable when managing daily operations, reporting, and data storage.

Security. This evaluates how well the system protects data against unauthorized access, loss, or modification. It includes that system has data integrity and privacy for both the company and employees.

Inventory. This includes both the initial components utilized in manufacture and the products ready for sale. This research serves as one of the modules used to analyze the current condition of inventory management methods, including tracking, replenishment, storage, as well as efficient regulation of inventory stock in this business [4].

Sales. A transaction when a person who buys intangible or material items, goods, or services to a purchaser in exchange for cash is referred to by this term. One of the modules in this study can be used to manage conversations with customers, monitor sales transactions, automate sales process, evaluate sales information, and generate reports.

Accounting. The processes ensure manages all financial transactions, including sales, purchases, and customer invoices, ensuring accurate and real-time financial reporting. It helps efficient record-keeping, generate reports, and make informed financial decisions for businesses.

Point of Sale. The businesses to process sales transactions quickly and easily. It has a direct connection to accounting and inventory, allowing for automatic sales recording and real-time stock level updates for improved accuracy as well as efficiency.

Purchase. Purchasing entails obtaining goods and products or service. This research evaluates how effectively it supports procurement operations, procurement procedures, and guarantees precise and prompt management of supplier transactions inside the company. It also emphasizes the role of purchasing in maintaining smooth business operations and ensuring timely availability of resources.

Employee Management and Attendance. The process focuses on handling employee records, attendance, and performance monitoring to ensure efficient workforce management. It supports automating HR processes, enhancing employee productivity, and supports decision-making for business growth.

Odoo. Open-source enterprise resource planning (ERP) software package that provides a wide range of capabilities through various applications which may be enabled as per businesses requirements. This is among the greatest solutions for enterprises of all sizes of businesses. The researchers conducting this analysis have chosen to implement Odoo as their ERP system at KKBN.

Digital Transformation. Is the process of integrating digital technologies into all areas of an organization, fundamentally changing how it operates and delivers value to customers. It involves adopting and implementing technology to create or modify products, services, and operations, enabling continual, customer-driven innovation.

KKBN. Kayamanan, Kalusugan Basta Natural is a business based in Iriga and Nabua that offers herbal products such as Shatsu, CMD, and *Kape ni Manoy* and other herbal products.

Notes

[1] Anaya, L. and Qutaishat, F. 2022. ERP systems drive businesses towards growth and sustainability. *Procedia Computer Science* (2022), 854–861.

[2] Ivanović, T. and Marić, M. 2021. Application of modern Enterprise Resource Planning (ERP) systems in the era of digital transformation. *Strategic Management*. 26, 4 (2021), 28–36. https://doi.org/10.5937/straman2104028i.

[3] Panduwiyasa, H. et al. 2021. Accounting and Smart System: Functional Evaluation of ISO/IEC 25010:2011 Quality Model (a Case Study). *IOP Conference Series: Materials Science and Engineering*. 1092, 1 (Mar. 2021), 012065. https://doi.org/10.1088/1757-899x/1092/1/012065.

[4] Sauvant, K.P. et al. 2022. An Inventory of Concrete Measures to Facilitate the Flow of Sustainable FDI: What? Why? How? Third edition.

[5] Utami, R.S.D.D. 2024. Implementation of Odoo Based on Enterprise Resource Planning System with Sales and Purchasing Module Using Rapid Application Development.

CHAPTER 2
REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter provides a review of related literature and systems. It presents relevant studies, synthesizes the state of the art, and identifies the gap that this study aims to address. The review supports the implementation of ERP software in KKBN to enhance business operations.

2.1 Related Literature and Systems

The system and literature review usually serve to establish a historical framework for the solid argument of the research problem and also the central ideas. This part is supposed to review various articles, journals, and studies and literature both from local and foreign sources published during the last five years to provide a strong basis in support of the current study.

The literature reviews that were carried out are as follows: (1) Odoo for ERP Solution Implementation; (2) ERP for Business Process Implementation; (3) ERP Software Implementation Methodology/Framework; (4) ERP Software Impact on Business Operations. (5) Customized Odoo Module Evaluation/Assessment.

2.1.1 Odoo for ERP Solution Implementation

The challenges that the X Beauty business faces which are a good starting point for the discussion of Erp features and fi are the manual inventory process that caused frequent stockouts and inaccurate records, the sales process which disrupted sales and reduced customer trust, and the purchase issue which was when a product was out of stock. By using traditional tools like spreadsheets and manual tracking which are inefficient as errors easily occur and inconsistent flow of information in the supply chain, automation not only is the reason for the crypto inventory restocking delays but also need to contact suppliers when the stock runs low. The resulting experience is unsatisfied customers and a loss of money showing that manual processes are a major threat to competitiveness that was proposed in the article hence being [16].

As the researchers Panduwiyasa et al. [2021], went on to say, the ISO 25010 quality model is a comprehensive guide that could be used to gauge software quality as it outlines the twelve functionalities that are required by the program including functional performance, compatibility, usability, reliability, security, maintainability, and portability. These factors together determine if a system is meeting both the technical requirements as well as the users. Their findings have shown that the application of this model for assessing an intelligent accounting system has not only lead to improvements in the system overall functionality, reliability, and customer satisfaction but also provided a regulated system for permanent development, making it a significant instrument for identifying the quality and efficiency of ERP and other business management software [15].

Odoo is an ERP platform that is free and open-source, and it comprises both ERP and CMS functions. It is mostly popular among small and midsize enterprises because it is modular the company's business processes. Launched in 2005 as Tiny ERP, it was rebranded as Open ERP in 2008, and finally, in 2014, it was changed to Odoo [8]. Odoo is not only the owner of it but also the community. Starting with Version 9, Odoo has been available in Community and Enterprise Edition: the first is for free, but with fewer features, while the second is paid and has more features and support, like a mobile interface.

It has been revealed that through the traditional ERP approaches might be risky and expensive for small and medium-sized enterprises (SMEs) knowledge of impacting SMEs with ERP systems and their study showed that Odoo was able to make it through a series of business operations that previously were done manually or through basic tools like Excel, oxygen. Cross-functional flow charts were used to show that Odoo improved integration, allowed for real-time monitoring, and increased process efficiency. "Open-source ERP systems may be regarded as an alternative for SMEs with limited budget and human resources." [p.1]. The study also points out that with proper configuration and adequate user training, Odoo can, in fact, not just efficiently run commercial activities but also enhance the educational aspect [23].

Odoo ERP system together with the different modules manufactured by PPIC is a good example to be used in the case study to showcase its effectiveness while dashboards and real-time reports are additional features made for decision-making purposes [22].

As proof of that, the application of Odoo ERP at AH Mart has integrated business processes through divisions, enabled fast data accessibility, and trimming response time, According to Arvianto et al. [2022] "Across the world, the retail sector is undergoing transformations and advances incredibly fast, as seen by the many traditional retailers adopting new retail models, such as through the use of e-commerce systems and digital corporations)," [p.5]. Outdated retail business processes regarding errors basically are inefficient, as it is in the case of AH Mart, a typical retail shop, where manual processes many errors, stock taking that is time-consuming, and difficulty in generating accurate financial reporting. For example, "AH Mart financial records are documented using Microsoft Excel software, and the financial input procedure is not planned, which causes frequent errors" [p.3]. However, in the study of the ERP system, it was found that the traditional retail business processes can be effectively modernized through the ERP system and point out the need for restructuring AH Mart's manual processes to computerized and integrated system [4].

Valve it may be, but commercial returns are not the issue most of the time. As a result of of Woodymoodijakarta ERP Odoo implementation and module integration and development efficiency has increased significantly.

The methodology was a systematic business needs analysis and system prototyping, data migration, Black Box testing, deployment and user feedback. The implementation was effective in streamlining processes; however, it had an impact on accuracy and efficiency. Different criteria largely accepted by users of the system saw rates from 90% to 97%, which is high. Findings are such that the Odoo ERP system has been remarkable in improving birthing process operational efficiency in other studies and further research and development towards more complex systems are recommended. Examples of these companies show that most of the businesses are embracing the ERP software to enhance their output [26].

2.1.2 ERP for Business Process Implementation

At present, the vast majority of enterprises are using Enterprise Resource Planning (ERP) Systems to address their business problems due to the numerous benefits and supports that they provide, such as the cost reduction, of which a survey done by Putra et al. [2021]. The ERP System positively affects the business processes which result in better business performance. This research shows the relationship between the installation of the ERP system and the key financial metrics namely the Return on Assets (ROA) and the Return on Equity (ROE). "ERP system implementation brings an enormous benefit to the overall organization and its capabilities, and it also has effects on the performance of the company" [p.2]. This reveals the necessity of strong inner organizational capabilities since the financial and operational flexibility both can increase the chance of success in ERP implementation by a large extent. Hence, ERP systems could be used to create a competitive advantage and augment the long-term success [17].

In accordance with this, the installation of an ERP system in manufacturing companies results in a leap of performance, maximally reducing production planning, inventory control, and logistical steps. According to the study, ERP brings down the production cycle time, consumption of resources, while increasing the inventory turnover. Employee surveys and regression analysis are the main factor advantages, but the qualitative feedback describes process automation and decreasing manual intervention as the main success implications [20]. After all, ERP systems succeed in the long run through operations reengineering based on effective change management and proper training.

By the same token, upgrading the enterprise's ERP system could be the trigger for it to be more efficient, with enhanced quality products, and increased productivity with real-time data, which in turn reduces costs and boosts customer satisfaction.. Odoo is one such open-source ERP and is the beneficiary of both low cost and low vendor dependence. The studies proved that commercial ERP software deficiencies exist, thus stretching a user's desire for an adaptation of Odoo ERP version 15.0. to their preferences. Reported by Qowindra et al. [2023], the findings of the investigation elucidate a large enhancement in system quality, information quality, service quality, and user satisfaction with Odoo customized ERP based on SAP Business One evaluation recommendations. The said paper stresses that the Enterprise Resource Planning (ERP) systems are instrumental in streamlining business processes and enhancing communication [18].

In their search for technological advancement and higher value, companies should embrace ERP systems. These systems not only simplify the tasks but also boost decision-making by allocating effective resources toward proper implicit knowledge leading to the overall improvement in performance and cost cutting. In the growth of the global market, the effective management of the business is success-dependent on the use of advanced information technology [13].

The implementation of an ERP system can revolutionize a company to be more efficient, produce quality products, and achieve higher productivity by utilizing real-time data, which in turn will reduce costs and increase customer satisfaction. The two types of ERP software are proprietary and free. Odoo is open-source software that is low-priced and is associated with low vendor issues. The findings of this research demonstrate that Odoo's personalized ERP based on the recommendation from SAP Business One has made noticeable progress in terms of system quality, information quality, service quality, and user satisfaction. The study ERPs are the backbone of business operations and communication [21].

2.1.3 ERP Software Implementation Methodology/Framework

In the fast-paced and competitive business landscape, where the staff and resources are often lacking, small and medium-sized enterprises (SMEs) are now adopting the Enterprise Resource Planning (ERP) systems for automating business processes and attaining an upper hand over rivals. The article offers a new model for programming ERP software in small and medium businesses. As was pointed out by Mansour et al. [2022], it recommends a four-step procedure, (1) top management commitment (2) identifying problems (3) system scan and data collection (4) implementation with an agile approach.

The report clarifies how this framework can help small and medium enterprises in enhancing efficient operation of the company, and inventory management [11].

The deployment of the ERP system in small and medium sizes facilities begins with the right analysis of the ICT and the business process maturity of that organization. The realization allows the finding of the chance of improvement and is the basis for the success of the ERP implementation. The process then transforms into a human-centered design that is structured into three phases: problem identification, solution development, implementation and evaluation, and lastly, reflection and learning. This method ensures that the ERP solution is tailored to the specific requirements of the SME, thereby making the implementation process iterative and flexible [19].The work examines the challenges and advantages small and medium enterprises (SMEs) may experience when utilizing ERP systems. Setiawan et al. [2024] present that ERP can stimulate the growth of SMEs, but it may stop functioning properly if not properly managed, particularly because SMEs usually have a shortage of resources and perhaps are reluctant to embrace new technologies. They suggest two simple steps: (1) examine the readiness of the business and its technology, and find the areas to improve (2) begin using the ERP system. It's necessary to comprehend and enhance the business processes before starting the ERP system.

Consequentially, the focus of this research is the troubles SMEs have and how ERP systems can be used to explore those. Heightened by a study by Alaskari et al. [2021], the study proposes a nine-implemented ERP system for SMEs, which is outlined by its own steps, inputs, processes, and outputs. The prototype should be easy, logical, and understandable, and may be used as a roadmap for SMEs to access ERP systems. This was mixed by a study of a company that successfully used the framework to adopt ERP [1].

The majority of the firms consider the implementation of an ERP system as very expensive and risky, but such views have been rebutted by this research since it proves that proper planning raises both productivity and competitiveness to a high degree. In the opinion of Momani et al. [2022], the data reported is based on surveys and theoretical studies that show that 88.3% ERP of companies positively affected performance, and 94.7% of managers realized its importance as a strategy. The conclusion is that the key to success would not depend so much on the software but on business process knowledge, alignment of goals, and risk management, like in the case of inadequate training and funds. Consequently, the research recommends that following a structured methodology that is best tailored to each company's unique needs is one of the most effective ways to both minimize the chances of failure and achieve successful ERP implementation [12]. Introducing ERP is a massive venture for any organization, and it happens to be successful through the execution of the right framework and methodology. This is not a matter of technical work but a strategic effort. Reports indicate that different enterprises make use of different ERP systems like SAP, Oracle, Odoo, each of them having its own methodical process that normally includes planning, installation, testing, and training. Open-source options, such as Odoo, are less expensive for small and medium enterprises. Even these technologies, however, require a solid plan to avoid problems. A wrong method or missed steps can be disastrous. The principle of correct ERP deployment is centered around the fact that the firm is genuinely based on the system, which saves money and time [7].

2.1.4 ERP Software Impact on Business Operations

Enterprise Resource Planning is one of the best digital tools in the world; often, it is extremely effective in small and medium-sized enterprises (SMEs) because it combines all the essential functions into one system that not only facilitates but also enhances the performance of the business processes. The successful incorporation of ERP depends largely on effective leadership, satisfied users, and appropriate training. Despite setbacks such as high costs, transformational leadership can help them conquer such obstacles. In summary, ERP is a major contributor to the financial and non-financial performance, and it is a powerful tool for changing business process management [3].

Consequently, it was expressed that while ERP could be a tool for companies to improve their operations, the implementation risks and difficulties should not be forgotten and some companies were cited to prove their point like Wal-Mart, and Samsung as examples of companies that successfully implemented ERP, Nike and Avon as the companies that had failed and finally they advised businesses to first think about all the risks and hurdles before implementing ERP then to take measures to prevent the risks [14].

Business ERP software is an essential resource to the oil and gas companies because it helps to manage multiple departments and facilitate the transfer of information. Presently, M. Teja Sri et al. [2024] hold a view that mentions the fact that though ERP has numerous pros such as time saving and enhancing decision making, the majority of the companies do not have the capability for its effective implementation. The lack of the right culture in the organization and insufficient training are the chief reasons for it. The study explains about these issues and minimal resolution possibilities by proposing solution routes such as better training and the quest for system compatibility with the company's work patterns. The research document is to be perceived as a pragmatic guide for the companies which will help them to reap the most from the practice of ERP system implementation [10].

The solution that ERP brings to businesses is the major productivity boost. Through and by automating different business functions, it becomes operationally efficient and effective. An ERP system is a full package that brings your entire business under one umbrella giving the transparent view of your business operations. This automation results in better decision making, human errors are removed, and labor costs are trimmed. Apart from the monetary aspect, the outcome is a more flat and flexible organizational structure. Companies that are thinking of introducing or migrating to ERP should weigh the potential gains and the strategic value of these systems. That is why ERP is the effective tool for the growth and competition in a long run [2].

So that there will be no more manual counting by the employees, who are getting tired and the business owner will stop chasing them for their work by correcting mistakes that are done in the product & sales and no longer recorded manually by the employees for all the activities. It is thus possible to monitor sales and products by the owner of the KKBN even when he is at home [6].

2.1.5 Customized Odoo Module Assessment/Evaluation

An ERP solution is an excellent example of the drastic change that happens in a business when it is able to streamline its processes due to the software. The acquisition of an ERP system leads companies to better decisions, fewer mistakes, less human labor, and, finally, a strategy that is both leaner and more flexible [4].

The management of a company should thoroughly investigate the benefits before deciding on the implementation or modification of the ERP system. Such is the research done by Fatoni Efendi et al. [2022]. Logistics in terminology is the lifeline for competition in business technology development according to the integral resource requirement. In a company, the management of resources is the most required process that should be adequately done [5].

The procurement of the right inventory and the trade of items is an example of how resource management is done in a company. An ERP system is a broad concept that organizations can adopt to become effective. The Bravo Company Manual System for workspace, monitor inventory level without that is why they take a lot of time and effort and in the end, a lot of mistakes are made., ERP solution can help to organize, minimize and less time consuming [9]. In today's age, almost all enterprises are digitized to assist them. As per Lubis et al. [2021]. Indoporcelain, which is a ceramic manufacturing industry, also employs ERP to enhance the overall efficiency of the works. Indoporcelain will bring in a larger IT staff that will be properly supporting the ERP system. Furthermore, the company would need to create a proper organizational structure, have the support of the leadership, and open some kind of training so that everyone in the organization would know how the ERP system works. In fact, this is a research work wherein the researcher has developed it by means of the Odoo app management module. By employing the Odoo method as a life's best practice model that is designed to help, the project is intended to increase the effectiveness and efficiency of business processes for the purpose of coordination to achieve the targets. This is the way to eliminate the process and alter the potential costs which are spent less time by any specific process in business or organizations [25].

2.2 Synthesis of the State-of-the-Art

In this research, the review literature witnessed some differences as well as commonalities, the related literature that serves as ODOO ERP system guide for the businessmen in the implementation of this business, and the research studies that equally discuss the pros of ODOO implementation for the companies to augment performance, work efficiency, and customer satisfaction. They pointed out that ERP systems had benefits such as efficiency increase, cost reduction, and better data management. Furthermore, they address the problems of ERP systems, such as needing customization, vendor support, and training, and insist that the correct choice of ERP system is crucial to the specific company needs and should be taken into account factors like business size, industry, and financials.

The research studies provide the basis for argumentation concerning the different industry sectors Odoo ERP can be applied to. As evidence, Arvianto et al. [2022], Utami et al. n.d. [2023], and Wu et al. [2020] show that the program is put into operation in retail, manufacturing, and small and medium enterprises. Kc et al. n.d. [2020] makes a comparison between Odoo and Microsoft Dynamics NAV, whereas Zahra et al. [2023] evaluates Odoo's features. It is shown in these articles that Odoo is a cost-effective, and user-friendly solution, though it often demands custom development and training for users. The effect of such a system on the performance of a company is different; while Natalia Br Ginting et al. n.d. [2021] state that there are significant improvements seen, Tongsuksai et al. [2021] discuss only moderate improvements. The successful ERP implementation is majorly aided by efficient project management Qowindra et al. [2023], user training Syamsuddin et al. [2023], organizational capability, and strong IT infrastructure Putra et al. [2021]. Thus, the above-mentioned discursive statements imply the need for meticulous arrangement, proper equipment, and readiness of the body for the of the ERP systems.

These studies are meant to introduce the impact of Enterprise Resource Planning (ERP) systems on the efficiency as well as productivity of a company, talking about the benefits as well as the issues faced by the companies in the case of ERP deployment, particularly small and medium-sized enterprises (SMEs). The research studies generally differ in terms of methodology as well as focus since, for example, Alhayek and Abu Odeh [2024] research is mainly focused on the benefits of ERP systems, whereas Syamsuddin et al. [2023] are concerned more with the disadvantages of implementing ERP systems, in addition, Pabustan et al. [2021] attain this qualitatively. The research work adumbrating the considerable benefits of ERP systems whereas some colleagues shout the issues of implementation. These reports are the components in the structure of the comprehensive reading, functioning mainly by ERP implementation problems, which in turn leads to the concept of making a robust adoption framework, and demonstrating the operational consequences of open-source ERP solutions such as Odoo. In addition, the techniques and assessment methods featured in the literature allow for examining ERP modules and understanding their influence on corporate operations.

2.3 Gap Bridged by the Study

Digital transformations have become the norm in today's world with many companies making the switch from traditional manual processes to digital systems that are more efficient and environmentally friendly. Data we gathered from our RRL have indicated the manner ERP systems such as Odoo can be of assistance in terms of improving a variety of business areas like inventory, sales, accounting, Point of sales, purchase, and employee management. For instance, some studies have asserted that the operating of manual processes most of the time will lead to the difficulty of human errors, inefficiency, and time-consuming monitoring tasks, which are some typical issues of small businesses such as KKBN. According to SOP1, that is to say, it is obvious that manual processes in

KKBN such as inventory tracking, sales monitoring, and manual recording give a challenge that led to errors, inefficiency. Furthermore, new research concentrated on the ways ERP software could automate those functions whereas, in the meantime, they presented the idea of centripetal and chain integration all over program processes. For the most part, these studies are usually geared to Odoo implementation, naming it specifically, but they do not include detailed explanations concerning its comparison with other ERPs in terms of customization and flexibility. Even though previous studies emphasize the advantages of ERP implementation, they tend to fall short on the issue of whether customization of Odoo can meet the requirements of a small business in a unique way that other ERP solutions can't.

To deal with existing voids, in this research the KKBN's actual business processes are examined through the analysis of the problems caused by manual methods and the presentation of the Odoo ERP configuration and customization according to their personal preferences. Unlike earlier studies, this one not only illustrates the use of ERP systems but also assesses them according to ISO 25010 standards of quality. The research not only makes a fair comparison of functional suitability, performance efficiency, usability, maintainability, reliability, and security of the system based on actual user experience, but it also helps to provide a better understanding of the subject matter. We opine that this strategy will be useful not only to KKBN but also to numerous small businesses due to exposing them to the customizations that Odoo ERP has undergone, its comparisons and evaluations are taken in real-life situations.

Notes

[1] Alaskari, O. et al. 2021. Framework for implementation of enterprise resource planning (ERP) systems in small and medium enterprises (SMEs): A case study. Procedia Manufacturing (2021), 424–430.

[2] Alhayek, W.Y. and Odeh, R.A.A. 2024. Exploring the Impact of ERP Systems on Business Efficiency.

[3] AlMuhayfith, S. and Shaiti, H. 2020. The impact of enterprise resource planning on business performance: With the discussion on its relationship with open innovation. Journal of Open Innovation: Technology, Market, and Complexity. 6, 3 (Sep. 2020).

[4] Arvianto, A. et al. 2022. ERP ODOO IMPLEMENTATION IN SMALL RETAILERS. International Journal of Applied Science and Engineering Review. 03, 06 (2022), 66–85. https://doi.org/10.52267/ijaser.2022.3605.

[5] Fatoni Efendi, H. and Aditya, A. 2022. Business Process Analysis and Implementation of Odoo Open Source ERP System in Inventory, Purchasing and Sales Activities (Case Study: Captain Gadget Store) Analisis Proses Bisnis dan Penerapan Sistem ERP Odoo Open Source Pada Aktivitas Persediaan, Pembelian dan Penjualan (Studi Kasus: Captain Gadget).

[6] Ian Pabustan, S.D. et al. 2021. Analyzing the Impact of Enterprise Resource Planning (ERP) Application to the Business Operations in SIA Engineering (Philippines) Corporation.

[7] Ibrahim, M.F. et al. 2024. A literature review on ERP implementation: Methodologies, module, software, and policy. AIP Conference Proceedings. https://doi.org/10.1063/5.0192339

[8] Kc, D. and Hagen, T. 2021. ERP software system comparison between Odoo and Microsoft Dynamics NAV Is Odoo able to replace the existing Microsoft Dynamics NAV core functionalities?

[9] Lubis, M. et al. 2021. Enterprise Resource Planning (ERP) System Customization with the Inventory Management Module: Case Study of Rumah Yatim. ACM International Conference Proceeding Series (Apr. 2021), 7–12.

[10] M. Teja Sri et al. 2024. Integration of Erp in Optimizing Business Process. International Research Journal on Advanced Engineering and Management (IRJAEM). 2, 05 (May 2024), 1584–1587.

[11] Mansour, K. et al. 2022. Novel Integrated Framework for ERP Selection and Implementation. Journal Europeen des Systemes Automatises. 55, 6 (Dec. 2022), 787–792. https://doi.org/10.18280/jesa.550611.

[12] Momani, F. et al. 2022. Erp Implementation Process Improvement: A Methodological Proposals.

[13] Natalia Br Ginting, D. et al. 2021. The role of enterprise resource planning systems (ERP) in improving company performance. 2, (2021), 46–48.

[14] Pandey, A.K. et al. 2022. Examining the Role of Enterprise Resource Planning (ERP) in Improving Business Operations in Companies. ECS Transactions. 107, 1 (Apr. 2022), 2681–2696. https://doi.org/10.1149/10701.2681ecst.

[15] Panduwiyasa, H. et al. 2021. Accounting and Smart System: Functional Evaluation of ISO/IEC 25010:2011 Quality Model (a Case Study). IOP Conference Series: Materials Science and Engineering. 1092, 1 (Mar. 2021), 012065. https://doi.org/10.1088/1757-899x/1092/1/012065.

[16] Penelitian, J. et al. 2023. Challenges and Strategies for Inventory Management in Small and Medium-Sized Cosmetic Enterprises: A Review.

[17] Putra, D.G. et al. 2021. The Influence of Enterprise Resource Planning (ERP) Implementation System on Company Performance Mediated by Organizational Capabilities. Journal of Accounting and Investment. 22, 2 (Jan. 2021), 221–241.

[18] Qowindra, M.R.G. and Wiratama, J. 2023. Development of Enterprise Resource Planning (ERP) using the Rapid Application Development (RAD) Method for the Garment Industry in Indonesia. G-Tech: Jurnal Teknologi Terapan. 7, 2 (Mar. 2023), 504–513. https://doi.org/10.33379/gtech.v7i2.2296.

[19] Setiawan, D. et al. 2024. A Proposed Framework for ERP System Implementation in SMEs. International Journal of Artificial Intelligence Research. 7, 2 (Jan. 2024), 181. https://doi.org/10.29099/ijair.v7i2.1102.

[20] Syamsuddin, S. et al. 2023. Implementation of an Enterprise Resource Planning (ERP) System and its Impact on Manufacturing Company Operational Efficiency. Global International Journal of Innovative Research. 1, 2 (Nov. 2023), 194–199. https://doi.org/10.59613/global.v1i2.29.

[21] Tongsuksai, S. et al. 2021. Critical success factors and challenges for cloud ERP system implementations in SMEs: A vendors' perspective. 2021 IEEE Asia-Pacific Conference on Computer Science and Data Engineering, CSDE 2021 (2021).

[22] Utami, M.P. and Kharisma, I.L. 2023. Implementation of Enterprise Resource Planning (ERP) Based Information System Using Odoo Software. The Eastasouth Journal of Information System and Computer Science. 1, 02, 55–64. https://doi.org/10.58812/esiscs.v1i02.

[23] Wu, J.Y. and Chen, L.T. 2020. Odoo ERP with Business Intelligence Tool for a Small-Medium Enterprise: A Scenario Case Study. ACM International Conference Proceeding Series (Oct. 2020), 323–327.

[24] Yoo, B.K. and Kim, S.H. 2021. A Suggestion for ERP Software Customization Model Using Module Modification Factors. Lecture Notes in Electrical Engineering (2021), 563–568.

[25] Yosevine, P. et al. 2021. Enterprise Resource Planning (ERP) Evaluation and Implementation: A Case Study. Journal of Multidisciplinary Issues. 1, 1 (May 2021), 49–66. https://doi.org/10.53748/jmis.v1i1.10.

[26] Zahra, S.L.Z. et al. 2023. Implementation of Odoo-Based ERP in The Case Study of Micro, Small, and Medium Enterprises (MSME) "Woody Moody Jakarta." Intelmatics. 3, 2 (Aug. 2023), 68–77. https://doi.org/10.25105/itm.v3i2.17590.

CHAPTER 3
METHODOLOGY

This research methodology, statistical tools, and the research design were given in-depth discussions in this chapter. Additionally, the chapter showed a clear map of the Business Process Management (BPM) life cycle framework describing its parts and functions that were used in the project implementation.

3.1 Business Process Management Framework

The execution of the project at KKBN according to the BPM life cycle was done to implement the Odoo ERP System. The introduction of BPM was a significant achievement for the business because it resulted in improved efficiency, productivity, and customer satisfaction. It was also the approach that the organization used to find critical processes, and it was the paving stone that helped the company to perform and be more profitable. Furthermore, the adoption of BPM led to operational simplification, reduced waste, and increased teamwork and communication, thus making the business more competitive.

Business Process Management (BPM) framework uses a clear, step-by-step plan, which reduces the chances of errors, delays, or going over budget during implementation. The six phases of Business Process Management are (1) process identification where key processes are selected for improvement (2) process discovery which involves documenting the current state (3) process analysis where inefficiencies and issues are examined (4) process design or redesign focused on creating an improved version (5) process implementation where the new process is executed (6) process monitoring and controlling which ensures the process performs as expected and allows for continuous improvement.

This framework was appropriate for the study entitle: "Digital transformation of

KKBN business process using ODOO: A capstone project." BPM framework can help the researcher to see the step by step process in more easy way and it also help the researcher to find the one or what is need for the business to change, by utilizing BPM frameworks it helps the researcher to track how well processes are working and find areas to improve also BPM frameworks help the business to improve their processes and run smoothly.

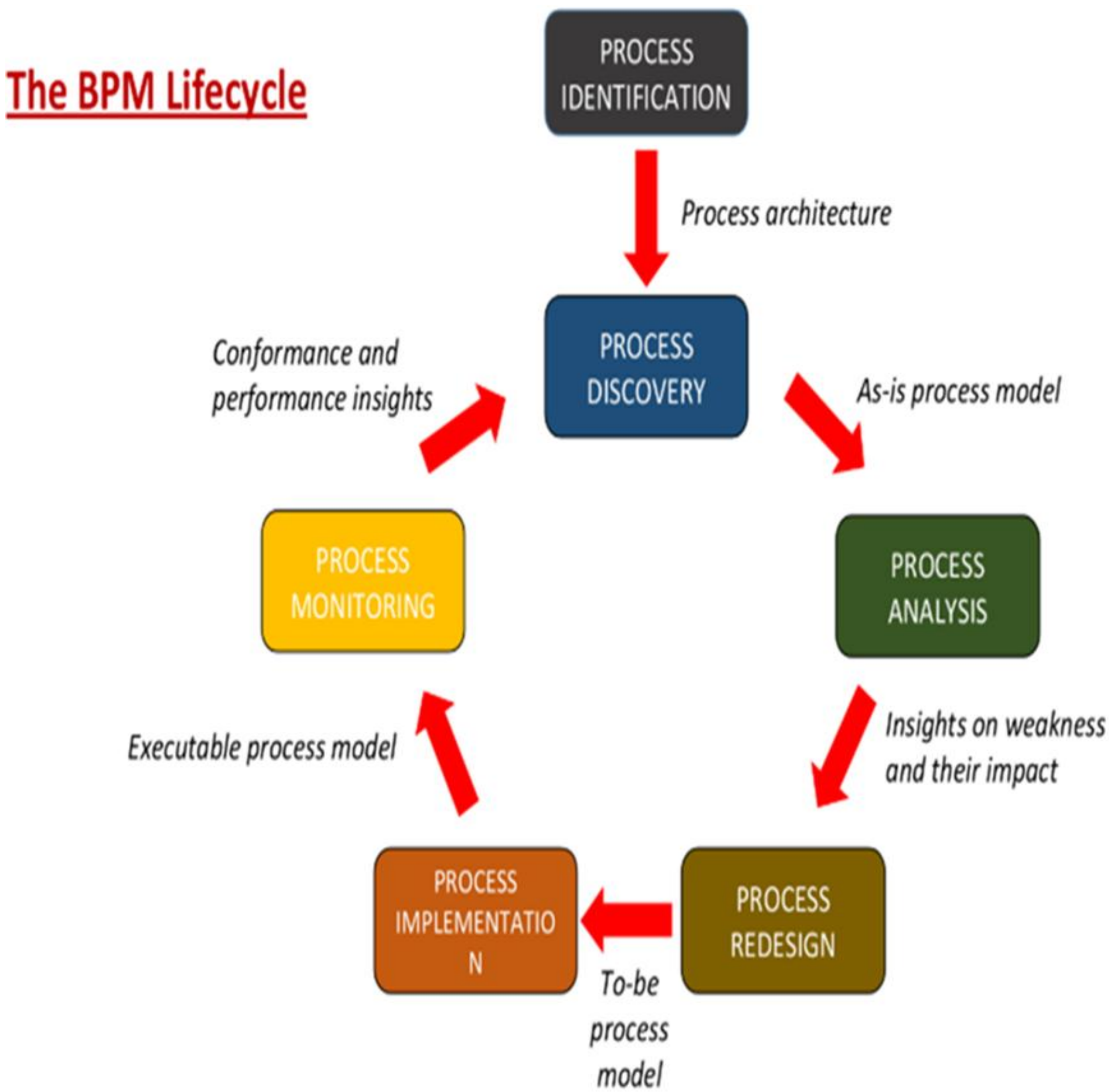


Figure 1. <u>Business Process Management</u>[1]

Figure 1, show the Business Process Management (BPM) Framework used in the study, which consists of Six phases: first process identification, second process discovery, third process analysis, fourth process design, fifth process implementation, and sixth process monitoring.

The specific objectives of the research were met through the application of the Business Process Management (BPM) framework. Every element of the framework is directly related to the fulfillment of the specific objectives of the research noted below.

Process Identification. The researchers and the owner of the business laid out the objectives and analyzed the business's functioning in the current state by conducting interviews.

Process Discovery. The researchers interviewed the company's staff and documented KKBN's as-is processes in order to understand the business model in detail.

Process Analysis. After the researchers getting the clear picture of the existing processes at KKBN, they proceeded with the identification of all the issues and prioritizing the ones that required immediate attention.

Process Design. In this phase, the researchers created the configurations and added the specific Odoo modules that were required actively by KKBN, at the same time, this formed the to-be-processes.

Process Implementation. The researchers were at the phase where they configured and deployed the Odoo ERP system as well as conducted testing, the system was evaluated through ISO 25010 standards such as Performance Efficiency, Functional Suitability, Usability, Maintainability, Reliability, and Security.

Process Monitoring. The researchers conducted monthly monitoring of the system to analyze the outcomes on KKBN's business activities

3.2 Research Design

The research team that this study has combined qualitative and quantitative approaches through utilizing mixed-methods research design clearly anchors on both the general and specific objectives of the study. As Jane et al. [2024] pointed out, Mixed methods research combined qualitative and quantitative research methodologies to provide a more comprehensive understanding of research questions [2].

The quantitative part will mainly be focused on the collection and analysis of numerical data. According to Meissel et al. [2023], quantitative research used various instruments, for measurement of behavior and outcomes, which were psychometric [3].

On the other hand, the qualitative aspect will mainly be about collecting non-numerical data. Oranga et al. [2023] mentioned that qualitative research focused on individual experiences and group dynamics [4].

The researchers will conduct structured interviews and distribute survey questionnaires. The interviews with the KKBN owner and staff were helpful in analyzing the existing business workflows and determining which Odoo modules to be implemented effectively. The surveys will be distributed after the system deployment to the targeted users to assess the system based on the ISO 25010 quality standards. This type of research will not only contextualize both qualitatively but will also present the numerical aspect in the data collection. These findings will be a basis for the system enhancements and the satisfaction of both users and business requirements would be achieved.

3.3 Research Method

A mixed-methods approach was utilized in the study, which involved the combination of both qualitative and quantitative research methodologies in order to address the objectives associated with the Odoo ERP System implementation across KKBN Iriga and Nabua branches.

3.3.1 Data Sources

Primary data were gathered directly from KKBN stakeholders, including a business owner, secretary/cashier, and a sales lady. During interviews, they revealed their daily workflows, duties, and obstacles that were taken before and after ERP implementation. The use of an observation checklist provided documentation of behaviors and processes. All these efforts were attached to analyzing and assessing the performance of the implemented ERP system.

Sampling Technique

The respondents in this study will be ten individuals in total. Five of them will be from KKBN, including the owner, the secretary/cashier, and the sales personnel. These internal respondents will be selected based on their direct involvement in the business operations. Moreover, three IT experts will be invited in order to solicit their technical knowledge. Then, two ERP experts will be asked to assess the ERPs' suitability and efficiency. The complete distribution of the study's respondents is in Table 1 and Table 2.

Table 1: Distribution of Respondents for SOP 1

Respondents	No. Respondents	Percentage
Owner	1	20%
Secretary	1	20%
Staff	3	60%
Total	5	100%

Table 1 displayed the distribution of respondents for SOP 1. The five respondents were the owner, secretary/cashier, and three staff members. The direct involvement of these stakeholders in the daily operations was the reason why the data collected correctly represented the existing business processes, challenges, and user experiences. The interview is the first tools used for researcher to collect data from the owner because the purpose of the interviews was for the researchers to get an idea of how KKBN operates its business and where it can introduce changes. The observation checklist utilized by the researchers to observe the actual practices and workflows as well as the questionnaires that recorded the results of the respondents' perceptions and insights were the key methods of this research that presented a holistic view of the operational needs and readiness of KKBN for the Odoo ERP implementation. The combination of these tools made it possible to interject both the objective and subjective observations into the analysis.

Table 2: Distribution of Respondents for SOP 3

Respondents	No. Respondents	Percentage
Owner, secretary/cashier, and sales lady	5	50%
IT Experts	3	30%
ERP Experts	2	20%
Total	5	100%

Table 2 displayed the distribution of respondents for SOP 3, where the researchers will choose a total of 10 respondents for the study, including five users from KKBN (the owner, & secretary/cashier, sales personnel), along with two ERP experts and three IT experts, all of whom will meet the pre-decided inclusion criteria. The survey is the third tools will used the respondents to gather feedback and insight where the Odoo module improvement and assess the level of acceptability of the new ERP system based on ISO 25010 standards. The weighted mean and Likert scale that the researchers will use alongside the regular analysis of the collected data are powerful tools for the scientists to identify the trends and patterns existing in the data. These techniques would be adopted to appraise the feelings of the respondents as to the degree of the implementation of the Odoo ERP software at KKBN. The results of the ten respondents will be the first step in recognizing some of the areas where the system can be improved and at the same time in the evaluation of whether KKBN is prepared to implement the new ERP system.

3.3.2 Instrument

This subsection introduced the instrument used by the researcher, which helps ensure consistent and reliable data collection for the study.

Interview Guide. The interview guide will be one of the main instruments of data collection as this will contain a group of questions and discussion points so that the interviews will be maintained focused and will address the essential areas. The researchers will conduct an interview with KKBN owner using this guide which aims to understand KKBN existing business processes and identify the appropriate Odoo modules for digital transformation. This method will allow the researchers to gather detailed insights and directly address the first objective of the study, which is to examine the current workflows and system requirements of the business. The researchers will ask the owner about their goals to make sure the Odoo modules fit the business's needs.

Observation Checklist. The researchers will check off implements of the observation checklist to collect data from participants. In the predefined checklist format, specific actions, behaviors, or reactions that have to be seen are included, thus, enabling the data-gathering process to be structured and consistent. Even though the process will be manual, it will ensure reliable and organized data collection, forming a basis for a more systematic implementation of a tool like Odoo in the future.

Survey Questionnaire. The researchers will use this scale to evaluate the system's efficiency by means of the users' feedback.

3.3.3 Data Gathering Procedure

This section will be dedicated to the procedures that the researchers will follow to collect the necessary data for the study.

Interview Guide. The interview is the main method of ability to collect data from different sectors. The contributions of the interviewees in the discussion directly dealt with the first target of shipping. The purpose of the interviews was for the researchers to get an idea of how KKBN operates its business and where it can introduce changes.

Observation Checklist. The researchers will employ a checklist for observation to gather data from the participants. The checklist will provide the specific activities, behaviors, or reactions to be noticed that will guide the data-gatherers in a structured and consistent manner. Though the process will be manual, it will, however, ensure reliable and well-organized data collection, which will be the basis for the more systematic use of a tool like Odoo in the future and also to show what can be done to improve KKBN's current business processes.

Survey. The researchers will throw in the surveys and they will gather data from the owner of the business and also from the other people involved. The survey will reflect on the overall views of the people while at the same time will give the level of acceptance of the new ERP system that is based on the ISO 25010 guidelines. In particular, the researchers will look at the evaluation of these system characteristics: Performance Efficiency, Functional Suitability, Usability, Maintainability, Reliability, and Security.

3.3.4 Data Analysis Procedure

This study will applied both qualitative and quantitative data analysis techniques where to address all specific objectives (SO).

Thematic Analysis. Thematic analysis is a qualitative method of analyzing data that seeks to identify, explore, and interpret patterns in written or verbal data. The qualitative information that will be gathered from KKBN through interviews will be subjected to thematic analysis. To identify opportunities, challenges, and inefficiencies in KKBN's manual processes, the researchers will examine the current workflows and will determine the business needs essential process improvements and growth.

BPMN Analysis. Business Process Model and Notation (BPMN) analysis is a method of graphically showing and analyzing the sequence of tasks, decisions, and interactions in a business process. It uses prescribed symbols to present processes in clear terms so that it is easier to understand, improve, and communicate the way a business operates.

Descriptive statistics. Descriptive statistics is a branch of statistics concerned with summarizing, organizing, and visualizing data to make it meaningful. It uses quantitative measures and graphical methods to present the basic properties of a dataset. Data collected by means of the surveys will be analyzed using descriptive statistical tools. The responses gathered through a 5-point Likert scale will be computed to establish the mean, frequency, and percentage distribution for each quality attribute of the Odoo ERP system, according to ISO 25010 standards. The reached means will be interpreted using a predetermined scale to measure the level of acceptability which would range from "Highly Acceptable" to "Not Acceptable." This statistical analysis will give a way to the researchers to evaluate user satisfaction, system efficiency, and find out the points to be improved. The quantitative analysis such as the transformation of the respondents' feedback into numerical data aims to check for the success of the digital transformation initiative. The results will help identify areas for improvement and determine how effective the digital transformation initiative is.

Table 3: Likert Scale

Scale	Range	Level of Acceptability	Descriptions
5	4.21 – 5.00	Highly Acceptable	The design is outstanding and exceeds expectations. It fully meets or surpasses all requirements and standards, leaving little room for improvement. It is regarded as highly favorable and is strongly endorsed by users or stakeholders.
4	3.41 – 4.20	Moderately Acceptable	The design is above average and satisfies most expectations or standards. It is more than simply acceptable, with notable strengths that make it well-regarded. There may be a few minor areas for improvement, but overall, it performs well.
3	2.61 – 3.40	Acceptable	The design meets the minimum requirements and standards. While there may be some minor areas that could be improved, it is generally deemed Satisfactory and suitable for its intended purpose.
2	1.81 – 2.60	Slightly Not Acceptable	The design is close to meeting the required standards but still falls short in some important areas. There are clear deficiencies, though they are less severe than a fully unacceptable rating. Improvement is needed before it can be considered acceptable.
1	1.00 – 1.80	Not Acceptable	The design is completely unsatisfactory. It does not meet basic standards or expectations, and significant changes or improvements are necessary. It is strongly rejected by users or stakeholders.

Formula:

GV = Greate Value (Highest Point in the Likert Scale)

SV = Greatest Value (Lowest Point in the Likert Scale)

NP = Number of Points (total number of levels in the Likert scale)

Formula for Computing the Range: It was used to compute the range of Likert scale, below showed the formula for computing the range of 5-point Likert scale.

IL = (GV – SV) / NP

IL = (5 – 1) / 5

IL = 4 / 5

IL = 0.80

Table 3 presents the level of acceptability of the Odoo software base on the ISO 25010 metrics, utilizing a Five-Point Likert Scale to assess how well the software meets the specific criteria based on user feedback. This helps identify which areas of the system are performing well and which may need improvement.

The resource inventory of learning materials encompasses five (5) different classes that can be categorized according to their relevant score intervals. The uppermost class is "Highly Acceptable" (4.21 to 5.00), which directly implies that it exceeds the standard level; Scale 4, "Moderately Acceptable" (3.41 to 4.20) is a level typically associated with medium acceptance; Scale 3,"Acceptable" (2.61 to 3.40) is when the acceptability is average; Scale 2, "Slightly Not Acceptable" (1.81 to 2.60) is an indication of a small amount of negativity to the level of acceptability; and Scale 1, "Not Acceptable" (1.00 to 1.80) is a situation where the requirements are manifestly unacceptable. The whole range gives the scope to the general evaluation of the Odoo ERP software completely by the ISO 25010 standards which, in return, help the management and development processes. The formula for calculating the interval level is (5 - 1) / 5. The total interval scale is eight (0.80).

Statistical Test

The researchers used statistical tools to assess the data they had collected. The mean value was weighted to find the average of the responses provided by the respondents, and the mean value was also used to determine whether the respondents accepted or not tailored made and configured ERP software. The parameters utilized in the survey include functional suitability, performance efficacy, usability, maintainability, reliability, and security. Statistical software was used to support this computation as well. The formulas for the computation of the weighted mean are illustrated in the following part.

$$\boldsymbol{WM} = \frac{\Sigma\,(f_i \text{ x } w_i)}{N}$$

Where:

- f_i = frequency or the number of responses for each score
- w_i = weight or value of each score (from 1 to 5)
- $\boldsymbol{N}$ = total number of all responses

Notes

[1] Aal, S.I.A. 2025. Framework for Evaluating Business Processes Modeling Techniques under Neutrosophic Environment. *International Journal of Information Engineering and Electronic Business*. 17, 1 (Feb. 2025), 1–13. https://doi.org/10.5815/ijieeb.2025.01.01.

[2] Ahmed, A. and Pereira, L. 2024. *Mixed Methods Research: Combining both qualitative and quantitative approaches*.

[3] Meissel, K. and Brown, G.T.L. 2023. *Quantitative Research Methods*.

[4] Oranga, J. and Matere, A. 2023. Qualitative Research: Essence, Types and Advantages. *OALib*. 10, 12 (2023), 1–9. https://doi.org/10.4236/oalib.1111001.

CHAPTER 4
RESULTS AND DISCUSSION

This chapter shows the discussion of results according to identified specific objectives, integrated each phase in accordance with the selected framework, the Business Process Management Framework. Further, it covers different phases as illustrations to support the entire process of the Implemented Odoo ERP Software. It also covers the screen captured photos of output.

4.1 Process Identification

The researchers were efficient and strategic at this phase by conducting an interview with the owner of KKBN (Kayamanan Kalusugan Basta Natural). This part of the project was designed to obtain the details KKBN needed to analyze its ongoing activities. Process Identification is the process of recognizing the different tasks which are carried out in any business or organization. The researcher learns processes more clearly by recognizing them, such as how and who implements tasks and how they are all related. This is a significant step because it points out the process areas where enhancements could be done and sets the ground for a thorough business analysis.

The main processes that the researchers recognized, according to them, were the Inventory, Sales, Accounting, POS, Purchase, and Employee Management which they explained as the processes that the researchers identified are stated below.

4.1.1 Inventory

Proper inventory management plays a central part in the operation of companies to satisfy consumer demand for the products and raw materials. Errors in inventory management often causes problems such as shortages, excess stocks, or reduction of resources. Moreover, it helps the smooth running of the company and improves consumer satisfaction, thus, inventory management is an important process, especially to small companies, where the resources are limited. Disarray in the situation often leads to missed sales or financial damage for the business. The primary purpose of solid inventory control is to regulate the available stock and goods, decrease the rate of spoilage, and implement smart purchasing and selling practices. As stock is the main structural part, the business operates such ways like sales and customer support.

To sum up, inventory management was a very important process that was responsible for balancing, efficiency, and preparing the organization. Furthermore, it was helpful for businesses prevent problems, reduce losses, and enhance customer service. Minor changes were proven to lead to a significant improvement in the financial stability and growth of the business long-term.

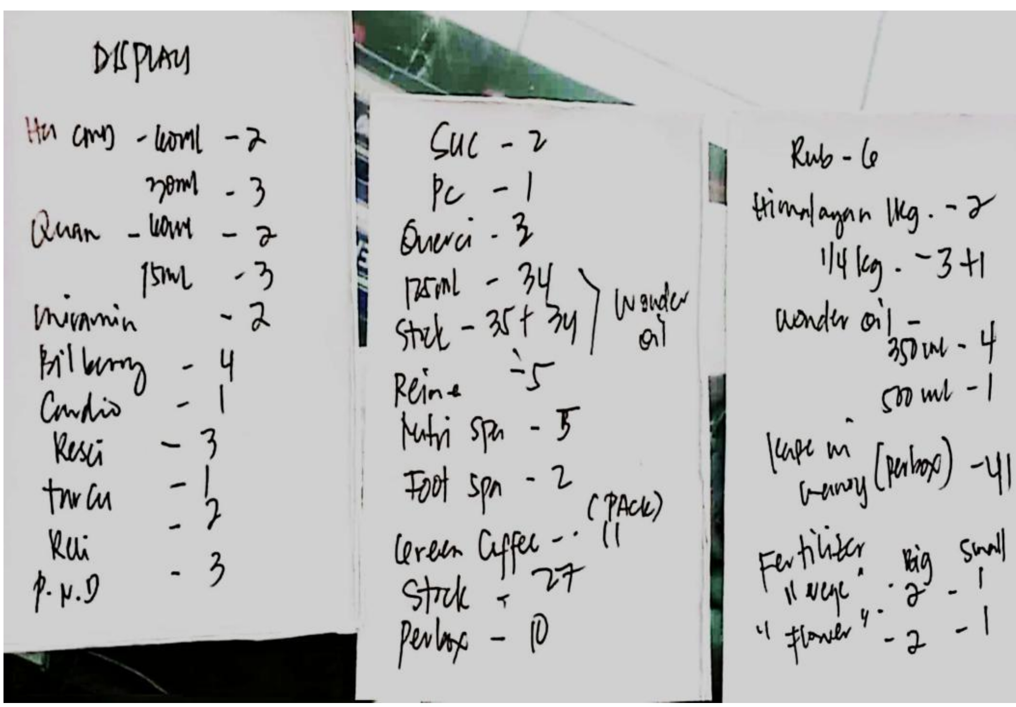


Figure 2. Manual Record of Product Stock

This record was handwritten on paper with names of products and their respective quantities and served as the primary reference in tracking the inventory. Though this is a process that has allowed the recording of product availability, it was very time-consuming and did a lot of manual checking and updating to make it inconsistent and erroneous. Handwritten stock records were also hard to read especially when they were kept by several staff with different types of handwriting. In some cases, no proper categorization was present thus it made the information about a specific product hard to find.

The manual method of stock checking product stock at KKBN was challenging to maintain accurate. Since stocks were mostly updated by hand, the risk of calculation errors, which could lead to delays in reporting, and the possibility of data loss due to the records being misplaced or damaged was very high. This made monitoring especially difficult when it came to report preparation or replenishment scheduling. Workers also had to physically check all the stock that was available on display and that which was in the stockroom before updating the records and this took a lot of valuable time [7].

Generally, the manual stocks recording or the monitoring method resulted in KKBN slow business operations and a great deal of time from the employees to realize the needed accuracy. It also limited the tracking of fast-moving items effectively; therefore, it was somewhat difficult to avoid stockouts.

4.1.2 Sales

Sales are acknowledged as one of the very essential elements in every business due to the reason that they are the main gate to company's goods reaching the end customers. In spite of that, in small businesses like KKBN (Kayaman Kalusugan Basta Natural), sales are even more crucial since they are the ones that directly impact the enterprise's growth and survival. The proprietors and the staff are often the ones present and they work directly with the customers. They make the sales process both personnel and essential.

In addition, sales are inextricably linked to other key business processes such as marketing, inventory control, and customer service. Overall, sales are more than just a transaction, they are a company's lifeblood. The process is all about trying to connect with clients, pay attention to their needs, and cultivate relationships that enable company growth and loyalty. Through research in the sales process, businesses can identify areas for improvement, streamline their processes, and deliver enhanced customer experiences.

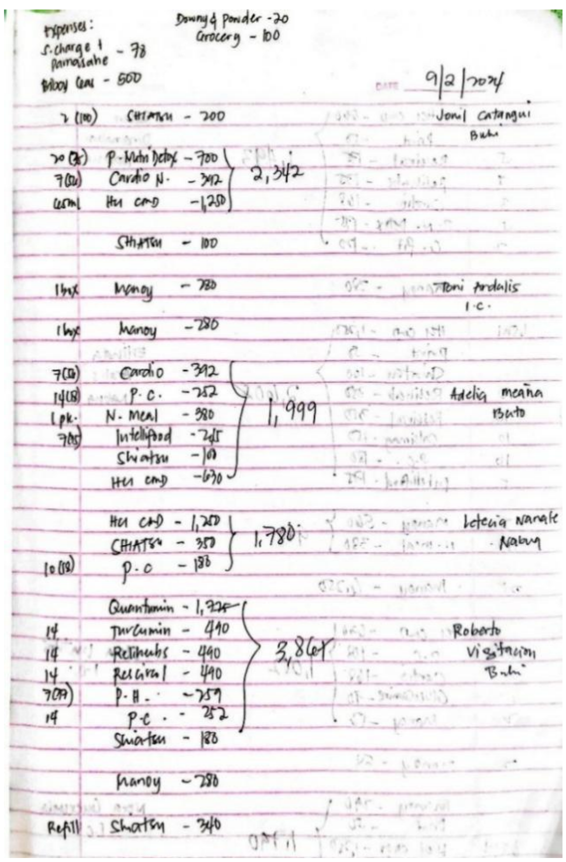


Figure 3. Manual Record of Everyday Sales Report

In the given figure, we can see the current manual record system of KKBN's daily sales report. KKBN's sales recording method of handwriting and manual material check was highly prone to mistakes and difficult to maintain over the long run. The notebook was filled with daily sales, making it twice as hard to track and organize information. Retrieving previous records from the notebook took a great deal of time and effort. The system was not equipped with backup solutions and hence was deemed to be volatile and at high risk. This manual operation took too much time, caused a delay in the workflow, and limited the time for service to the clients or for the development of the business.

These problems point out that despite the positive sides of the paper-based system, it has been one of the main reasons why KKBN could not expand, assess performance, and make good business decisions. Beyond this, employees had added difficulties, as the company was forced to manually record daily sales. Also, it was a hard and exhausting task to find out the total sales figures for the monthly sales report.

4.1.3 Accounting

The importance of accounting at KKBN cannot be undermined as it is through this that the company prevents financial problems and ensures the effective disposition of its limited means. The proper recording of sales, purchases, and expenditures is the very foundation of the budget and its forecasting. Well-planned and executed accounting methods had been responsible for things such as the avoidance of unnecessary expenditures, balance, and timely payments. Accurate and complete financial records help the management of taxes effectively, examine the performance, and keep trust with the stakeholders.

Through its role in assisting business to flourish, remain stable, and make sound decisions, accounting is a fundamental process too. It requires the establishment of systems, the provision of performance information that is useful, and the identification of areas of development. The financial management of the firm had to be in good order largely for the stability in the long term.

Purchasing was the readily discussed another important business process and required a well-thought-out approach: finding trustworthy vendors, securing the best deals, and fast delivery of the right products. Faulty purchasing sometimes led to production time being lost, receiving faulty materials, or incurring extra expenses.

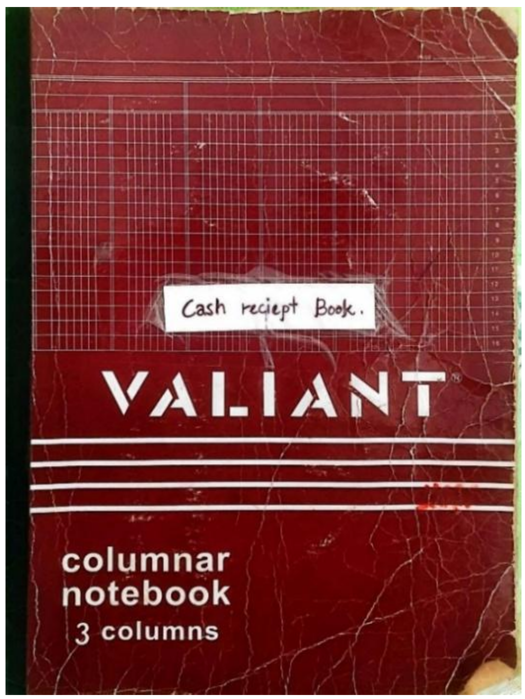


Figure 4. Manual Book Record of Monthly Sales Report

The manual cash receipt book that is utilized by KKBN for the purpose of recording monthly sales can be seen in figure 4. The book also acted as the main medium for documenting cash transactions and for one report each month, which included the consolidation of daily sales. It was possible for the business to depict its financial processes because the format was descriptive. All the pages in this booklet were used for recording the sales and income entries, which allowed the business to monitor its financial processes.

The manual format reflected a traditional method of bookkeeping, which has been prevalent among small businesses as a low-cost and readily available record-keeping tool.

From the other side, the use of this manual book had certain disadvantages. The entry of all data by hand resulted in calculation errors, missing entries, and inconsistencies, especially when summarizing totals at the end of the month. Folding the paper was also a problem; for the notebook, it was always crumpled or torn, thus posing a possibility of data loss in the long run. Nevertheless, despite the challenges these shortcomings posed, the manual cash receipt book was still a key player in KKBN’s financial documentation system and was used as the foundational tool for recording monthly sales.

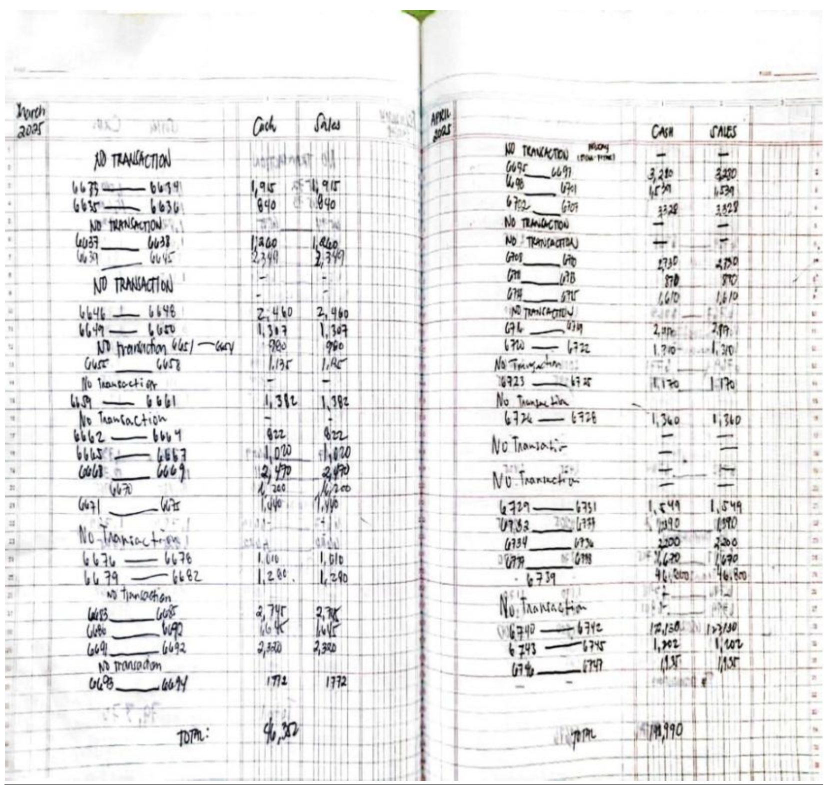


Figure 5. Manual Record of Monthly Sales Report

KKBN's Manual Record of the Monthly Sales Report was presented in Figure 5, where sales and cash transactions were documented in a logbook format as a summary of income for the month. "No Transaction" days were the periods when the records were carried over from the daily sales report. The manual record was the central method that brought together the daily reports to form a monthly overview. The handwritten entries were the major reason for such mistakes like miscalculations and missing entries. Moreover, the need to access or compare previous days' sales records came with heavy work of flipping through pages which was not only wasteful but also ineffective [8].

KKBN's record maintenance involved manual computations of total figures at the end of every period which were time-consuming and vulnerable to human mistakes. This also meant that retrieving and checking duplicating numbers from different dates required additional work during reports preparation for management or planning purposes. Although the monthly manual sales record mostly provided KKBN with the necessary transactional documents, the process remained deficient in terms of time, precision, and dependability thus not fitting for effective decision-making.

4.1.4 Point of Sale

The Point of Sale (POS) was a factor most businesses couldn't avoid as it facilitated consumer transactions and introduced a direct exchange between the company and the clients. Initially, it was a reference to a store's cash register but then included all such systems as were the hardware and software blend aimed at managing payment, sales tracking, and receipts.

The POS did not only allow a company to accept payments; it also was a master who controlled daily business operations. It was a prime process that helped to ensure accuracy, efficiency, and growth by enabling companies to assess performance, increase customer satisfaction, and lay a solid basis for decisions. The deployment of technology and incremental measures had a decisive effect on the success of the company and its sustained competitive edge.

In the business processes of KKBN, being without a POS system would have created a complication in daily transaction management. The absence of a POS, sales on the other hand, would need to be manually recorded which means a higher probability of mistakes like incorrect calculations or unaccounted sales. Inventory control also would be difficult because there wouldn't be an automated system that would see updated stock through real-time observations that would lead to overstock or shortage. These manual chores in KKBN could decrease customer transaction speed, delay peak times, and hurt consumer satisfaction.

4.1.5 Purchase

In any company, purchasing was an integral part which comprised of planning, selecting dependable suppliers, negotiating proper prices, and the most vital thing, timely delivery of products. Disorganized purchasing procedures usually resulted in delayed production, poor-quality supply, or costs that were unnecessary.

Small businesses like KKBN frequently dealt with tight budgets, strongly impacting performance and making purchasing a vital task. Paying too much or choosing the wrong supplier diminished revenue and damaged finances. Strategic purchasing aided in cash savings, as well as in supplier relationship building and securing reliable sources. Keeping stock levels stable and giving customer access also contributed to the inventory and sales processes.

Purchasing was an important process that interconnected different departments of the company such as finance, inventory, and sales through budgeting, cost tracking, and restocking. It had an indirect effect on employee productivity, customer satisfaction, and the growth of the company. Good control of the purchasing process lessened the risks, lowered the costs, and ensured proper resource acquisition.

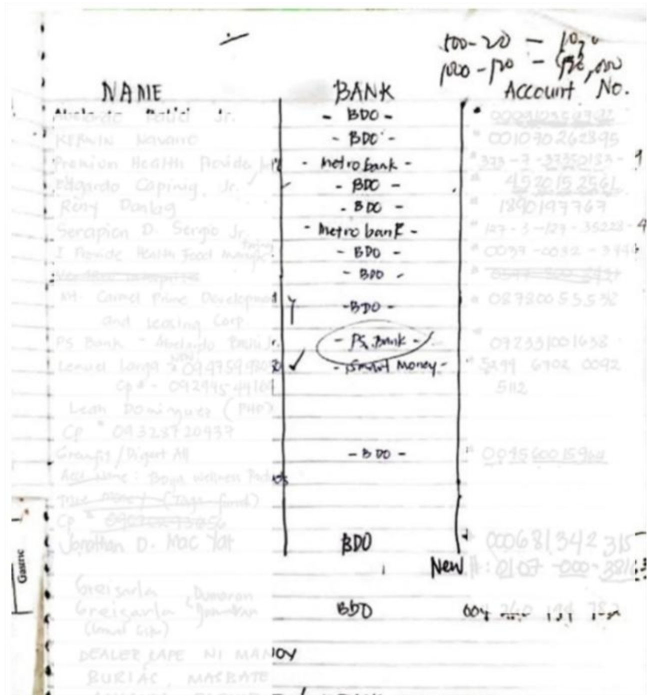


Figure 6. <u>Manual Record of Supplier Information</u>

The manual supplier information of KKBN is featured in Figure 6, which includes suppliers' names and their respective banks as well as account numbers. It was a reference book for payment processing, which was very efficient for the business to check whether the funds had been transferred to the right places. Furthermore, this record also carried written notes, phone numbers, and other information, which was the way the company documented supplier-related information in a very basic mode [1].

This document was especially important for the upkeep of supplier relations and the making of payments to the right bank. However, the fact of the supplier information record being handwritten created several challenges. Unreadable characters and wrong information often resulted in confusion or payment mistakes. Besides, the handwritten book itself being a physical object posed a threat to the security and reliability of supplier information because of the risk of damage or loss.

NUTRICARE HEALTH AND WELLNESS PRODUCTS TRADING
Door #3 Esplana Bldg. San Roque (Pob.), Iriga City Camarines Sur 4431 Philippines
JAEL A. RAÑOSA – Prop
Non-VAT Reg. TIN: 247-501-749-00000

SALES INVOICE No. 1136

SOLD to: ______ Date: ______
TIN: ______ Terms: ______
Address: San Roque, Iriga City OSCA/PWD ID No.: ______
Cardholder's Signature: ______
Business Style: ______

Qty.	Unit	ARTICLES	Unit Price	Amount
1	set		5,860.00	5,860.00
			Total Sales	5,860.00
			Less: SC/PWD Discount	
			Total Due	
			Less: Withholding	
			Total Amount Due	
			Sales Subj. to PT Exempt Sales	

[✓] Cash [ ] Check [ ] Credit

25 Bklts (50X2) 0001-1,250
BIR Authority to Print No. 066AU20220000002696
Date Issued: November 23, 2022
MODCOR, Printing Press Iriga City
TIN: 136-740-683-000 NV

By ______
Cashier/Authorized Representative
Printer's Accreditation No. 066MP20190000000001
Date Issued: 01-18-2019

"THIS DOCUMENT IS NOT VALID FOR CLAIMING INPUT TAXES"

Figure 7. <u>Purchase Invoice</u>

Figure 7 is a sample of a purchase invoice for Nutricare Health and Wellness Products Trading that is attached to a transaction between a buyer and a supplier. This was an official record of the sale made. Such invoices were necessary for suppliers and firms because they were the evidence of payment and the keys to maintaining the accuracy of financial records, which were needed for accountability and auditing. They were also instrumental in the trading of purchases, delivery confirmations, and the filing of accounting functions such as expense tracking and tax reporting.

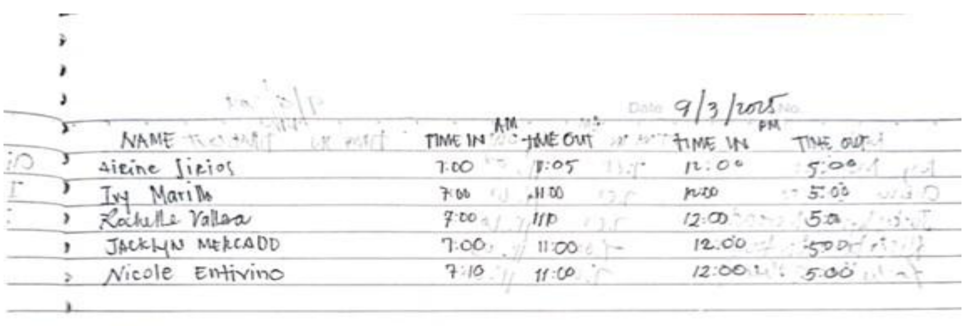
For KKBN, this document was a part of its manual purchase system, where all transactions were recorded on paper and stored physically. Therefore, it was exposed to certain risks such as loss, human errors, or damage to the physical paper records. However, it was still a crucial instrument for procurement activity and supplier transaction management.

4.1.6 Employee Management and Attendance

Employee Management was a key business function that involved recruitment, development, and directing employees to ensure that the organization shall be effective, employees will be satisfied, and the company will continue to grow. It was not just paperwork management, payroll, and everything but creating a productive workplace, where workers were motivated, and aligned with the company's objectives. Proper management of the employees made the organization have a strong team, involved the performance, and promoted the establish of the positive living organizational culture.

Firms were like the automatic bees: each employee is crucial to their job. Scheduling, participating in training, performance evaluations, and problem-solving effective HR strategies that guarantee the participation of workers and productivity. Also, HR supported customer contentment and profit margin by its effective help in other business operations like sales and inventory.

.


Date 9/3/2025

NAME	AM TIME IN	TIME OUT	PM TIME IN	TIME OUT
Aieine Jirios	7:00	11:05	12:00	5:00
Ivy Marillo	7:00	11:00	12:00	5:00
Rachelle Vallesa	7:00	11:0	12:00	5:0
JACKLYN MERCADO	7:00	11:00	12:00	5:00
Nicole Entivino	7:10	11:00	12:00	5:00

Figure 8. <u>Manual Record of Employee Attendance</u>

A manual record of employee attendance is presented in Figure 8 that registers the daily working hours of staff members in the business. This type of record was the backbone of tracking employee engagement, ensuring accountability, and providing daily detail on staff activity. It was indeed the most basic source for tracking employee attendance, but the paper-based process appears to disability to errors like inconsistent entries, poor handwriting, or missed logs. The physical logbooks are also in danger of wearing out, being stolen, destroyed, or erroneously read, thus affecting reliability and accuracy.

4.2 Process Discovery

The very first move towards sorting out how a particular company or organization works is through the process of discovery. The various transactions and activities that run day-to-day operations should be recognized, recorded, and visually represented first. Thus, firms become capable of having a clear view of how things are done, who is responsible for what, and the interrelation of each movement with others. This step is very important as it becomes the basis for further process analysis, problem detection, and presenting opportunities for improvement. Without the process of discovery, the existing system is not fully understood and hence, developing solutions or making recommendations for changes would be difficult.

The "As-Is" Model is a true representation of the performance and the way activities are done within an organization as it is a snapshot of the ongoing state of a business process. This is created in the initial stages of the process analysis without any adjustments or improvements to reflect the reality of the workflow, jobs, and assets. The main goal of the "As-Is" Model is to illustrate the accurate process flow and to highlight the strengths, weaknesses, and challenges associated with it. By doing the mapping, companies can observe the problems that the existing system might have, and that cannot be noticed in the daily operation of the system such as inefficiencies, redundancies, or gaps.

Based on this model, the "To-Be" Model is created which consists of the required designs and improvements to achieve better performance and efficiency. This form of structured approach made it possible for companies to set a clear path for the redesigning business processes. This form of structured approach guarantees that the changes are based on the best data, are specifically targeted, and are in alignment with the overall objectives of the organization.

4.2.1 Inventory

The Manual process of documenting transaction carries this inventory out by KKBN indicating they are recording their product inventory currently in a manual manner. Tracking the inventory of products flowing in and out of the system, adjusting stock levels, and filing entries are the most important features of this inventory management process. The system also details the staff to report on stock counts, and the final stage counts to be accurate. The new document on the current process refers to problems such as late inventory level updates, wrong inventory levels, or recognizing faster-moving materials.

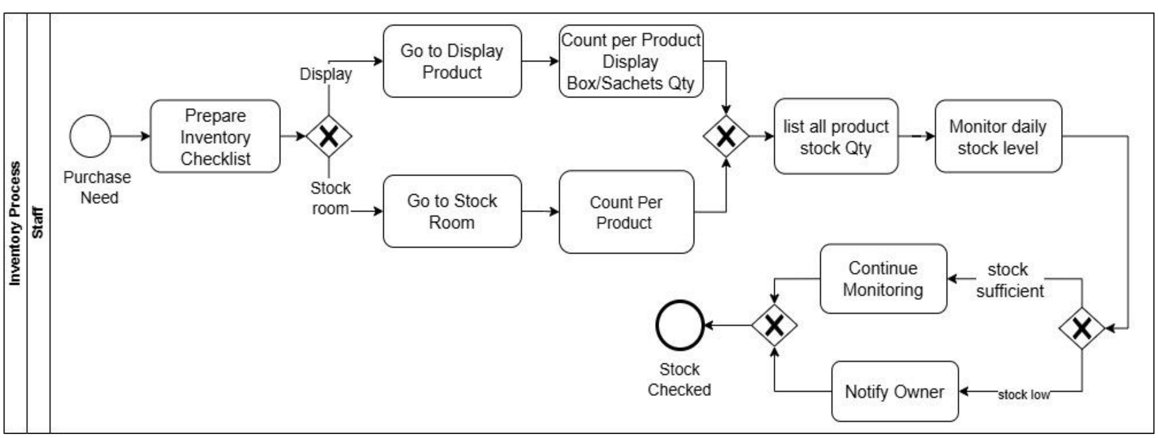


Figure 9. <u>Business Process Modeling Notation for Inventory</u>

Figure 9 shows a Business Process Modeling Notation representing the Inventory process of KKBN that employs a detailed manual method.

Prep List for Inventory. Before counting the stock. The staff will get the list of products they need to verify before the inventory counting process.

Go to Display Product. Then the staff will go to the product display area to check the items available on the shelves.

Count Per Product Display. The staff will count the different kinds of products on display separately in order to get the right quantities for sale.

Count Per Product Box Sachets Quantity. For example, if products are packed in boxes, the staff are also counting sachets to make sure it is correct in the inventory records. Go to Stock Room. After finishing the display check, the staff will head to stockroom which is the extra products room.

Count Per Product (Stockroom). While inside the stockroom, the staff will count the products per category to compare with the displayed count.

List of All Products Stock Qty. After the display and stockroom counts are finished, the staff will list all the product quantities in the inventory record. This documentation will provide an overview of product availability and help in stock tracking and management decisions accurately.

Monitor Daily Stock Level. The staff will observe these things; therefore, they will monitor the daily stock level with control over products stock movement and new product arrivals to ensure uninterrupted store operation and minimize shortages.

Continue Monitoring. The staff will keep up the monitoring routine in a commitment to be consistent.

Notify Owner (Low Stock). In the case of some products being raised above some level, the staff will inform the owner at once.

4.2.2 Sales

In this subsection, we will discuss about the identified sales process of KKBN, especially the sales transaction record and monitoring system which is done manually. The sales process workflow begins with the customer purchasing activities and ends with the creation of the daily and monthly sales reports. The sales process, as it was given, had some inefficiencies like manual calculations and unavailability of past sales data.

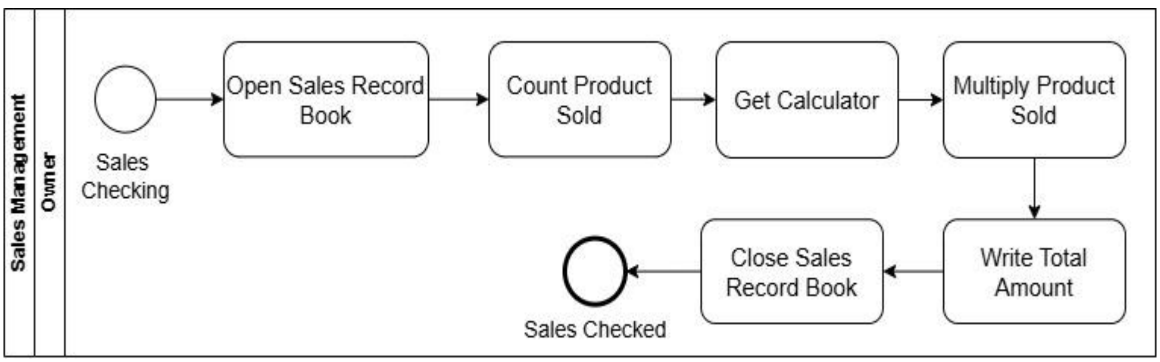


Figure 10. <u>Business Process Modeling Notation for Sales</u>

Figure 10 displayed the present sales manual process being used in the business. The steps in the process are:

Open Sales Record Book. The owner then opened it and reviewed the entries to find the list of products sold and the corresponding inventory.

Count Product Sold. After all the records of the sold products have come to an end, the owner selects the sold products for the verification of the accuracy of the transaction details.

Get Calculator. After that, the owner takes a calculator the calculation process precise. This is to avoid mistakes and to ensure that all the sales figures are correctly accounted for.

Multiply Product Sold. The owner respectively adds the number of products sold by their corresponding unit pricing to conclude the total sales amount per item.

Write Total Account. The sum resulting from the multiplication is then inscribed in the official sales logbook to create a written account of the financial transactions. This point is helpful for accounting purposes but also aids the owner in tracking daily profits.

Close Sales Record Book. At last, the owner closes the logbook thus indicating the end of the checking process and the activity with a simple.

4.2.3 Accounting

Accounting addresses the present situation of KKBN in terms of managing financial information like recording of revenues, expenditures, and transaction summaries. This subsection notes that the traditional method of recording and preparing the financial statements is manual and no real-time financial visibility is possible. The analysis of the current accounting process identifies various risks such as miscalculations, data inconsistency, and the inability to prepare accurate financial statements that are needed for the assessment of the business's performance.

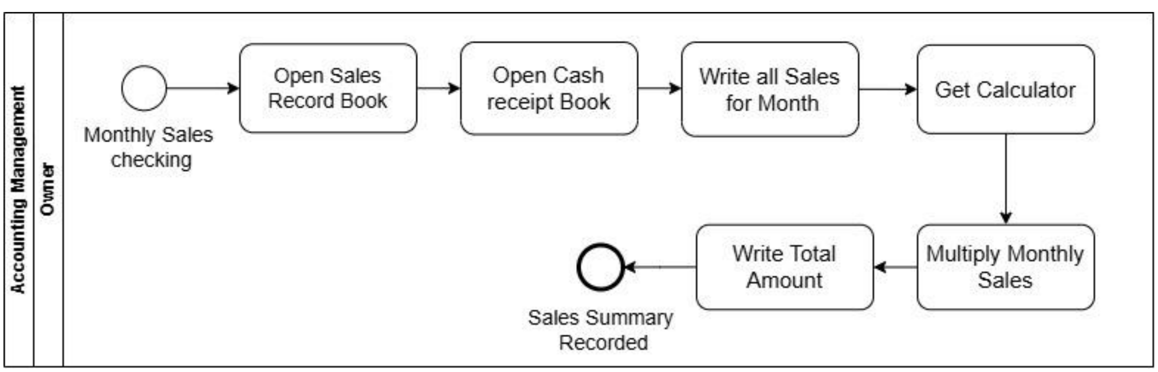


Figure 11. <u>Business Process Modeling Notation for Accounting</u>

Figure 11 depicts the manual accounting process currently run in the business, which deals with summarizing sales into monthly financial records first.

The method starts with the owner making the retrieval of the primary and secondary financial records:

Open Sales Record Book. The sales record book the owner opens it first, as it is the primary log of all daily sales allowance.

Open Cash Receipt Book. Additionally, the owner will also open the cash receipt book.; This book serves as a secondary record book of recorded daily sales, and it will help provide more accurate financial data.

Write all Sales for Month. The owner, based on the sales record book, manually records the total list of all daily sales transactions for the month in a single line.

Get Calculator. Now, the owner, after completing the monthly sales list, uses a calculator to make it easier to compute the monthly sales totals.

Multiply Monthly Sales. The total monthly sales are multiplied to calculate the overall sales.

Write Total Amount. After everything is calculated, the owner will write the total amount of the monthly sales in the cash receipt book, which is the official monthly record of income as well as for archival purposes.

4.2.4 Point of Sales

The POS operational procedure gives a broad understanding of how the company manages its selling and buying processes at the Point of Sale without a totally digital POS system. This particular section encompasses the specifics of the steps of the physical involved in processing customer purchases and generating their receipts. The as-is processes of the POS processes at KKBN are the real picture of the time-consuming manual admission of transactions, the non-integration of stock updates, and various issues affecting customer service and hence are inefficient.

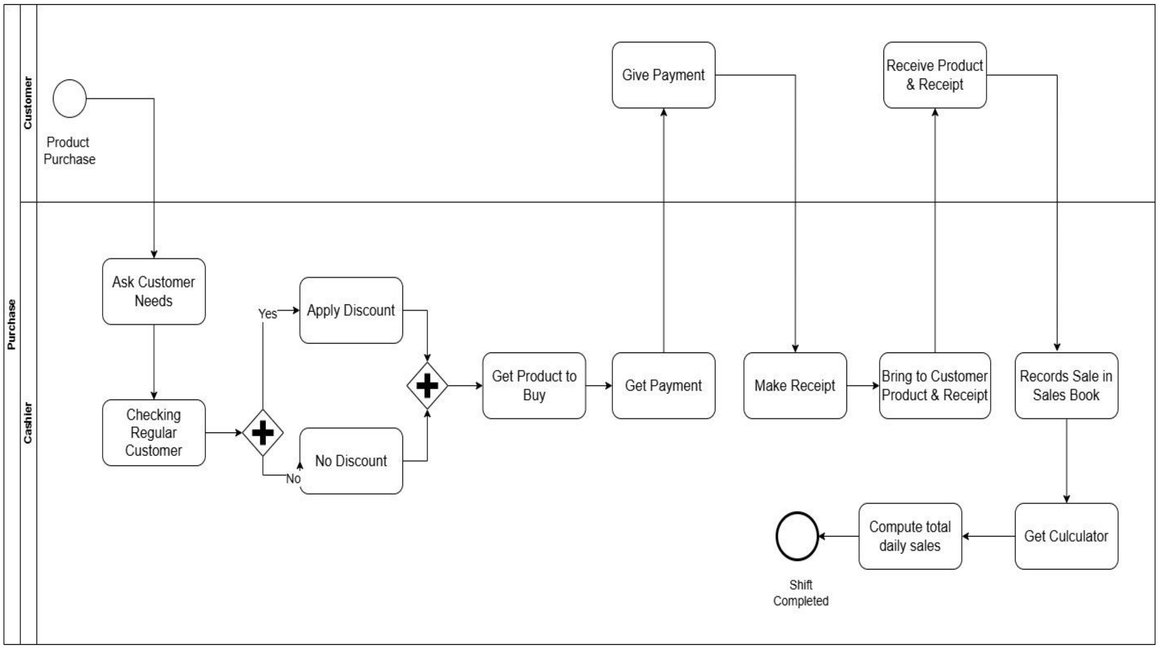


Figure 12. <u>Business Process Modeling Notation for Point of Sales</u>

Figure 12. represented by the manual Point of Sale (POS) process being followed in the business.

Ask Customer Needs. At the start of the workday, the cashier prepares for customer transactions. When a customer approaches the cashier, the cashier formally questions about the products that the customer needs to purchase.

Checking Regular Customer. The cashier verifies if the customer is an existing customer. This step is vital in recognizing the eligibility of the special discount for the regular customers.

Ask Customer Needs. The cashier asks formally about the products that the customer needs to purchase.

Checking Regular Customer (Decision Point). The cashier verifies if the customer is an existing customer. This step is vital in recognizing the eligibility of the special discount for the regular customers.

Applied Discount (Yes). If the customer is recognized as a regular customer, the cashier applies the proper discount on the product.

No Discount (No). In case the customer is not a regular customer, no discount is given. The product is sold at its usual price or standard price.

Get Product to Buy. After the confirmation of the customer's choices, the cashier or a store attendant will take the items from the display shelves.

Get Payment (Customer Action). The customer then completes the transaction by providing payment through the available mode of payment.

Give Payment (Cashier Action). The cashier collects the payment from the customer. The transaction is documented properly and acknowledged.

Make Receipt. The cashier either writes the transaction details on the receipt or in the transaction log. This including the name of the product, quantity sold, price, and amount paid for documentation issues.

Bring Customer Product & Receipt (Cashier Action). In addition to the purchased product and change, the cashier also gives the receipt to the customer.

Receive Product & Receipt (Customer Action). The customer gets the products, checks the items and the receipt, therefore, confirming the whole transaction.

Record Sales in Sales Book. The details of the transaction are handwritten in the sales book with an enumeration of the product sold. This practice ensures proper documentation, helps with the inventory count, and provides a reference for sales and financial reporting.

Get Calculator. The cashier utilizes a calculator after writing the daily sales list to ease the computation of the total sales. This step is important for minimizing the possibility of mistakes and for ensuring that the sales are accounted for correctly.

Compute total Daily Sales. At the end of the day, the cashier reviews all the recorded sales, multiplies the quantities by unit prices, and totals them to determine the daily sales. This process summarizes the day and prepares the management for the financial report and switching shifts.

4.2.5 Purchase

This Purchase subsection presents the identified buying process of KKBN, including how supplies and products are ordered from suppliers. It describes manual processes for detecting stock shortages, communicating with suppliers to order goods, recording the purchase transaction, and filing of purchase invoices. The as-is purchase process pinpoint's problems such as delays in procurement, very poor coordination with the stock records system, and minimal tracking of suppliers' transactional data.

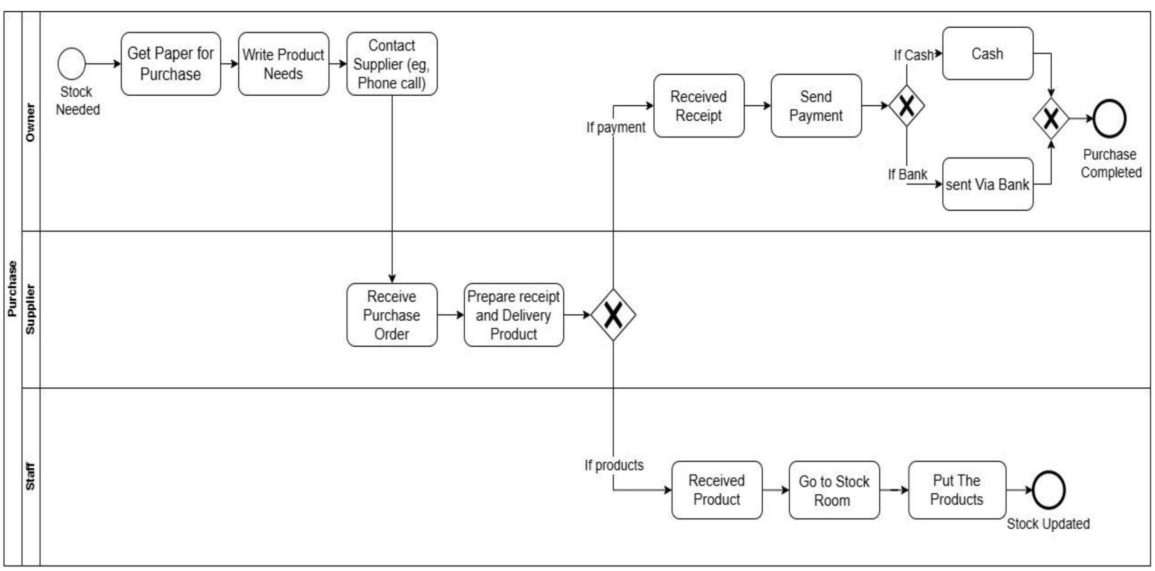


Figure 13. Business Process Modeling Notation for Purchase

Figure 13 documented the manual purchase process in the business.

The sequence of events starts when the boss recognizes the need to add products to the inventory.

Stock Paper for Purchase. The owner at the start of the purchasing process feels it is necessary to buy new stock because the current product is low. After confirming the need, the owner takes a piece of paper to make a purchase order.

Write Product Needs. The owner, by hand, makes a note of the products needed: this includes the specific items and quantities on the paper.

Contact Supplier. After the owner prepares the list, he/she typically places an order by phone call to the supplier, who is the person contacted.

Receive Purchase Order List (Supplier Action). The supplier then accepts the order from the owner *and confirms it.*

Prepare Receipt and Delivery Product (Supplier Action). The supplier collects together the products that were ordered and arranges for them to be delivered to KKBN.

Get Receipt. Along with the delivery of the products, the owner also obtains a receipt that serves as proof of the transaction from the supplier.

Send Payment. The owner goes on to send payment as per the total amount indicated on the receipt. The payment process can be completed based on the method of payment.

Cash. If the owner chooses to go cash, the payment is handed to the supplier directly while the product is getting delivered.

Via Bank. If the owner goes for a bank payment option, the quantity is sent to the supplier's bank account.

Done Purchase. The purchase from the owner's end is seen as complete after the payment is sent out and confirmed.

Received Product (Staff Action). After the vendor arrives, the staff is responsible for taking the products from the vendor to check them against the invoice.

Go to Stockroom (Staff Action). The staff, thereafter, takes the items directly to the stockroom. When they arrive, they orderly put the items in their respective storage areas for tracking purposes.

Put The Products (Staff Action). In the stockroom, the staff arranges and places the products correctly in their respective places. The stock levels are regarded as updated, since new items have been added to the inventory.

4.2.6 Employee Management and Attendance

On the subject of employee attendance and management, KKBN is currently dealing with the practice of keeping manual employee and attendance records manually. This part explains KKBN's method of dealing with the attendance of employees and management of employee data. It was further observed that the current process has certain inefficiencies, which in turn, create problems in employee management.

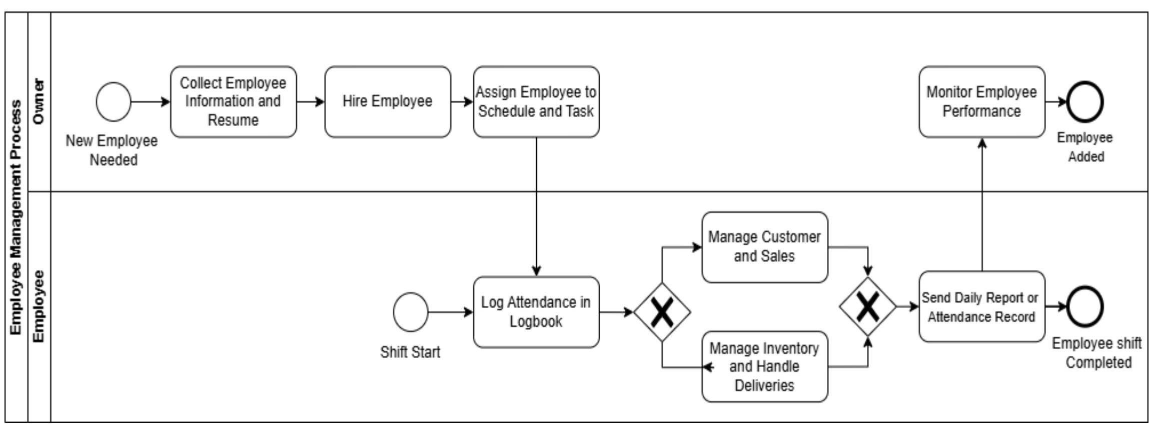


Figure 14. <u>Business Process Modeling Notation for Employee Management/HR</u>

The current workflow of the employee management and attendance process being deployed in the business is shown in figure 14.

Collect Employee Information Resume. The owner needs more employees for business operation. The start processes owner collects the personal information resumes.

Hire Employee. In the following step, the owner looks through the resumes and identifies the most appropriate one among them. The owner finally chooses and hires a new employee to fill the position he required.

Assign Employee to Duties. Then, the owner specifies different tasks and gives out some management responsibilities to the newly recruited employee.

Assigned Work Schedule and Tasks. Besides, the owner allocates the employee's work schedule for the whole week. That information serves for the employee daily responsibilities.

Log Attendance in Logbook (Employee Action). At the beginning of the shift, employees use the logbook to document their time-in and their time-out as proof of attendance.

Manage Customer & Sale (Employee Action). The employee whose to work takes care of the customer, processes sales, and makes sure that all the sales are recorded correctly.

Manage Inventory and Handle Delivery (Employee Action). The employee keeping track of the existing inventory assists in product deliveries and ensures that the proper documentation is finished for any item inputted or taken out of the stock.

Send Daily Report or Attendance Record (Employee Action). At the end of the rotation or business day, an employee typically composes and then gives the owner a daily report to fill him in on the day's activities.

Monitory Employee Performance (Owner Action). The owner observes the performance of the employee on a regular basis, to make sure that he/she is doing the work efficiently.

4.3 Process Analysis

The process analysis involved the careful examination of the corporate processes that were carried out to find out how effective, how they operated and which were the areas to be improved [8]. It was not just a matter of enumerating the tasks but a comprehensive investigation of the interconnections, the time it took, the resources needed, and the possible problems. The principal aim of the process analysis was to find out the inefficiencies, bottlenecks, and/or redundancies that were adversely affecting the performance. The businesses were able to have a better decision-making, increase their productivity, and enhance their overall performance through the learning gained from the process analysis.

Identified Problems

Process analysis was an essential factor in finding as well as solving issues that stood in the way of the company's productivity. These included long, complicated procedures, ambiguity about who is responsible, mistakes in recording and tracking data, and customer service delays due to which they had to deal with unwanted issues as well. Manual recording of sales and inaccurate inventory checks were the identified weak points because of the analysis of the current procedures. Awareness of these problems was the beginning point for taking further steps and establishing solutions, as well as recommendations for further progress.

KKBN was experiencing multiple operational hurdles, such as the over-reliance on manual workloads, the lack of integration among sales, inventory, purchasing, accounting, and errors in data. Employees have experienced a lot of time taking sales records, tracking purchases, and verifying inventory that is tiresome and time-consuming. This manual procedure increased the risks of human errors, for instance, miscalculations in the sales amount, and incorrect inventory figures. No integration also caused double work and discrepancies between the actual and reported data. Slow decision-making, lack of monitoring activities, and communication segments persisted as a result of employee reports that had been forwarded to the owner.

Table 4: Identified Problem

Tasks	Identified Problems	Descriptions
Inventory	Inaccurate and time-consuming stock monitoring	Physical inventory counts have the potential to cause mistakes, stock inconsistencies, and a delay in identifying shortages.
Sales	Manual recording of sales transactions	Writing sales by hand slows down the process and increases the risk of errors, missing receipts, and trouble keeping track of daily totals.
Accounting	Errors in financial recording and reporting	Because financial reports rely on manual entries, there may be errors, records that are missing, or trouble producing accurate reports.
Point of Sales	No automated sales system	Real-time sales tracking is difficult, and transactions take longer when done manually without a point-of-sale system.
Purchase	Lack of systematic purchase tracking	Purchases are manually entered, which frequently results in lost or duplicate orders and delays in stock updates.
Employee Management and Attendance	Limited monitoring of employee performance and tasks	The lack of automation in employee records and performance monitoring makes it challenging to efficiently manage staff activities, track progress, and documentation.

The 4 table above capture the various problems identified in the existing processes of KKBN across all functional departments of the business. In sales, the use of a manual recording system was the main problem that slowed down the transaction process and raised the possibility of errors, lost receipts, and daily totals that were difficult to track. The purchasing process was also tough, as the absence of systematic tracking often resulted in misplaced or repeated orders, in addition to delayed reporting on inventory levels. Likewise, inventory management was heavily dependent on physical counts which not only took a significant amount of time but also caused stock discrepancies and delayed stock shortage recognition.

With regard to point of sales, the absence of an automatic system substantially increased transaction times and did not allow for the business to track the real-time sales information that was crucial for instant decisions. The accounting department was also hit by manual recording, thus the financial statements became error-prone, incomplete, and inconsistent; this led to a lack of accuracy and reliability.

Finally, in human resources, as employee data and performance monitoring was not computer-based, management found it difficult to keep a proper track of attendance, productivity, and task assignments. In general, the table conveyed the message of how manual and non-integrated processes in different areas had created inefficiencies. Very strong emphasis, therefore, was placed on the need for a more systematic and automated process that would significantly reduce human errors, improve data accessibility, promote better employee management, and make possible a higher level of accuracy, efficiency, and decision-making at KKBN.

Proposed Solutions

To enhance the daily operations of the firm, the researchers came up with various recommendations that involve streamlining of routine jobs, lowering of mistakes, and improvement of data tracking in such areas as inventory, sales, and purchases. The execution of these measures saved time, led to increased productivity, and created a much more orderly system of staff development that in turn, supported better decision-making, and the long-term growth of the firm. These solutions assist in staff coordination improvement through the provision of correct and information for daily activities.

Table 5: Proposed Solution

Tasks	Identified Problems	Descriptions
Inventory	Inaccurate and time-consuming stock monitoring	Utilize the Odoo Inventory module to manage stock levels, automate stock moves, and generate real-time inventory reports without relying on barcodes.
Sales	Manual recording of sales transactions	Implementing Odoo Sales Module to create and track orders quickly and easily.
Accounting	Errors in financial recording and reporting	Use the Odoo Accounting module to automate journal entries, manage invoices, perform bank reconciliation, and generate financial reports.
Point of Sales	No automated sales system	Deploy the Odoo Point of Sale (POS) module, which allows for quick transaction handling and automatic syncing with sales and inventory data.
Purchase	Lack of systematic purchase tracking	Implement the Odoo Purchase module to manage suppliers, automate purchase orders, and track purchases easily.
Employee Management and Attendance	Limited monitoring of employee performance and tasks	Implement Odoo employee modules (Employees, Attendance, Time Off) to efficiently track employee activities, performance, and records.

The above table 5 included the capstone project solutions proposed for KKBN that needed to be improved through the Application of an open-source ERP solution Odoo. The table clearly listed the key business areas: Inventory, Sales, Accounting, Point of Sales, Purchase, and Employee Management and attendance, with their respective existing problems.

In Inventory, the solution was to use the Odoo Inventory module for stock level management, automatic movements, and real-time inventory report generation. The manual recording of transactions in Sales was resolved by implementing the Odoo Sales module which allowed for the fast creation and tracking of sales orders. With regard to the Accounting, the solution proposed was the automation of journal entries, invoice management, the performance of bank reconciliation, and generation of precise financial reports. The Point of Sale system was automated which led to the fast transaction processing and the seamless synchronization of sales and inventory data.

In Purchasing, the Odoo Purchase module made supplier management more efficient by automating purchasing orders and monitoring relevant purchasing activities. Lastly, the Employee Management and Attendance section were updated with the integration of Odoo Employees, Attendance, and Time Off, which together recorded and tracked the progress of employee tasks very efficiently.

4.4 Process Redesign

The process redesign phase was dedicated to the creation of the To-Be model, which projected improvements that would be made to the existing As-Is model. This phase included the comprehensive design of the processes, which was possible due to the ideas obtained from the analysis of the As-Is model. The objective was to define the operation of the processes, prioritizing the introduction of the new technologies as a means of solving the issues that were made evident. The design framework intended the creation of procedures that were more efficient, effective, and accurate and that was in line with the purpose of the research.

The To-Be Model presented the scenario in which the organization, in order to increase customer satisfaction, productivity, and efficiency, achieved transformation of operations, technology, and procedures. It was a guiding document for turning the current operations into far more efficient and effective systems that namely, concentrated on automation, scalability, and user-centric approaches. To achieve operational excellence and comply with the company's long-term objectives.

4.4.1 Inventory

The process redesign phase was dedicated to the creation of the To-Be model, which projected improvements that would be made to the existing As-Is model. This phase included the comprehensive design of the processes, which was possible due to the ideas obtained from the analysis of the As-Is model. The objective was to define the operation of the processes, prioritizing the introduction of the new technologies as a means of solving the issues that were made evident.

The To-Be Model presented the scenario in which the organization, in order to increase customer satisfaction, productivity, and efficiency, achieved transformation of operations, technology, and procedures. It was a guiding document for turning the current operations into far more efficient and effective systems that namely, concentrated on automation, scalability, and user-centric approaches. To achieve operational excellence

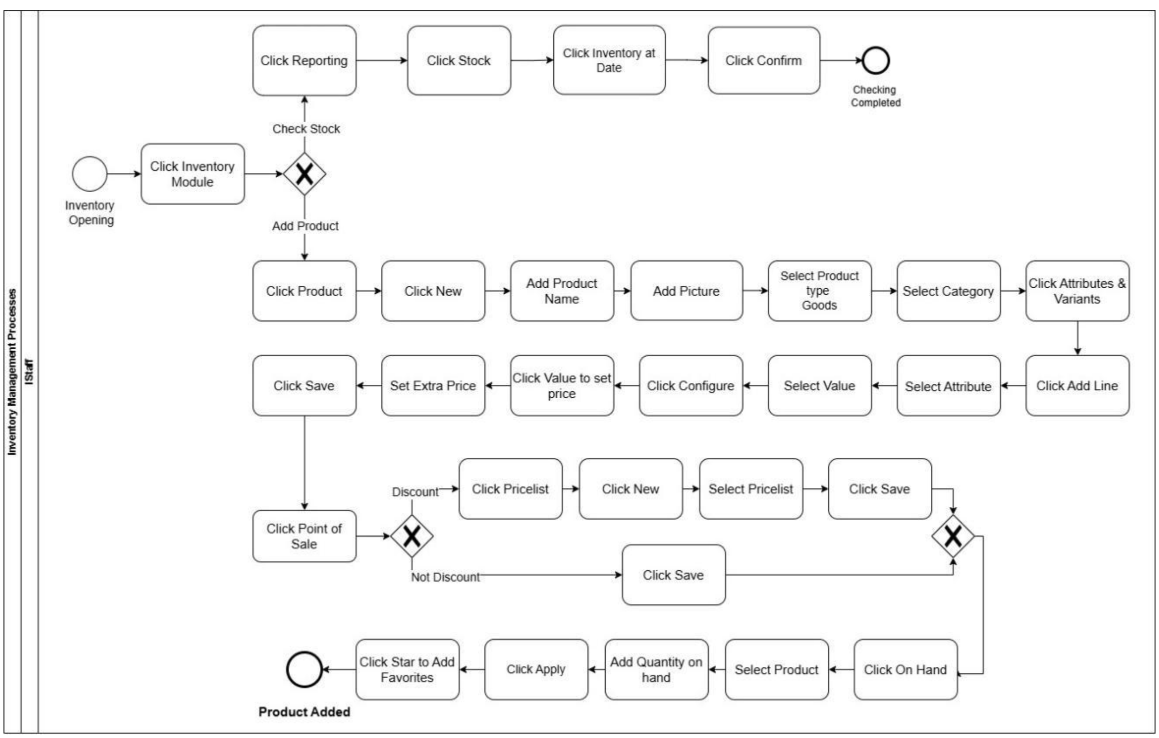


Figure 15. Business Process Modeling Notation for Inventory

Figure 15 showed the new process flow of Inventory in KKBN using Odoo The process starts when the inventory personnel open the Inventory module. Here, they can perform two primary activities: either inquire about the available stock or enter a new product into the system. When inquiring about stock, the personnel just create a report, chooses the record of stock, sees the inventory on a given date, and then verifies it. This ensures that the present stock levels are updated and correct.

When introducing a new product, the procedure becomes more specific. The team makes a new product record, enters the name of the product, uploads an image, and selects the appropriate product type and category. They can even include personal attributes, variations, and prices to fill in the product details. Once this is done, the product is saved within the system and is ready for use. When the item is sold on a discount first, the product list is created and attached to the item in question. This mechanism guarantees that all product promotions and any price changes are made exclusively in the sales module. After the prices get fixed, the employees modify the available quantity, fill up the full product specifications, and finally, save the required data.

The product is then included in the process is through the Point of Sales adding it. In addition to that, the staff can mark it as a favorite for quick access later. With this workflow complete, Odoo not only makes sure that the products are correctly posted to the inventory, but they are also easily accessible for sales transactions.

4.4.2 Sales

The newly designed sales process, as outlined in the Odoo Sales Module, is expected to significantly improve the accuracy and efficiency of transaction processing. Invoicing, bills, and sales have been mapped directly to the warehouse management and accounting module in the process that would be. Consequently, with the help of this tool, KKBN now has the possibility to track the daily sales and act. With the new process, the sales information is regularly updated, and accurate reporting is achieved to assist in informed management decisions. Overall, the process reduces manual work and minimizes errors in sales manual recording.

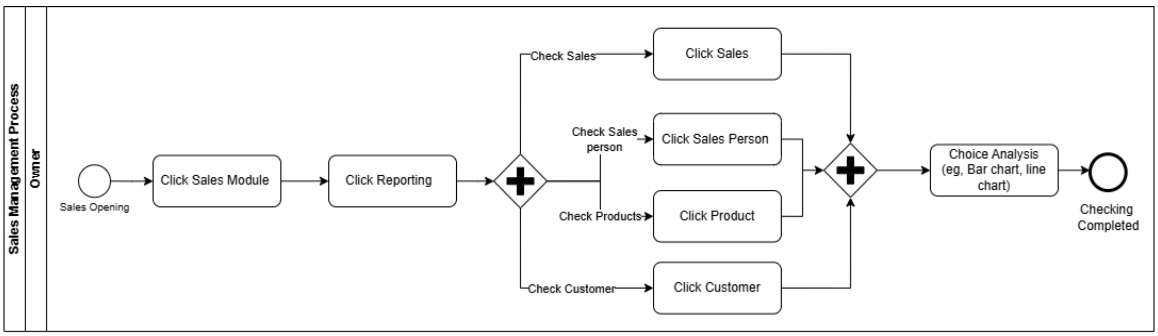


Figure 16. <u>Business Process Modeling Notation for Sales</u>

Figure 16 showed the new process flow of Sales in KKBN using Odoo The process starts with the Owner logging in to the Sales module. After entering, they are able to run reports for viewing sales-related data. The user is then able to select various areas to view depending on what they are looking for: total sales, individual salespeople, products, or customer data. Having this level of flexibility allows it to be simpler to track various elements of the sales process. If the admin wants to view the overall sales number, they simply need to click on the Sales option. If the aim is looking at sales by salesperson, the system enables them to do so. Administrators sometimes had reasons to see the sales depending on the product or to check the customer info. The discussed abilities were the primary means to acquire precise knowledge of the company's performance.

Following the selection of the region, the programmer can give the user the bar chart or the line chart as a tool to analyze the data. It becomes simple to find the patterns, to make the comparisons, and to draw the conclusions through the facts. To wind up, the process is finalized with the sales report signed, this way, the data is well organized and ready for all decision-making.

4.2.3 Accounting

Besides the KKBN manual financial reporting, the transformation of the accounting procedure also includes the Odoo Accounting Module as its main tool for improvement. In the desired state, financial transactions are automatically recorded in the system based on the supporting evidence of sales, purchases, and point-of-sale activities. This procedural change initiates the automatic recording of financial events instead of manual entry which is prone to errors and ensures the accuracy and up-to-dateless of the financial data. The new financial reporting method gives KKBN the opportunity to track financial revenues and expenses and directly creates the true financial reports for better management of the finances.

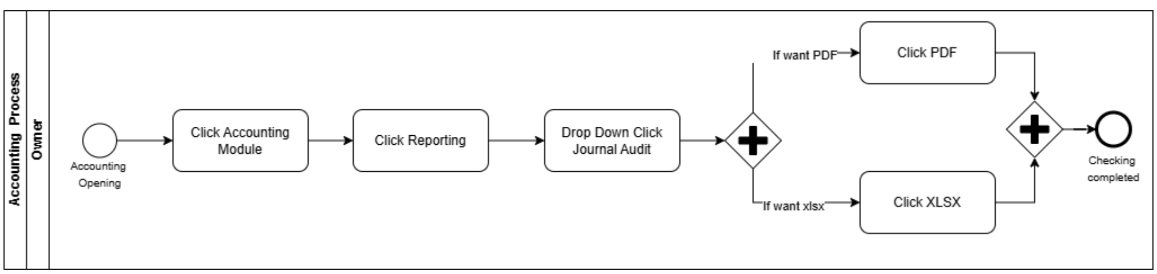


Figure 17. <u>Business Process Modeling Notation for Accounting</u>

Figure 17 showed the implementation of Odoo in KKBN required a new process flow for Accounting. From there, the user accessed the Reporting section and selected the Journal Audit option. This feature allowed them to be more comprehensive in their financial observations regarding the organization. At this step, the user got the option of either PDF or XLSX (Excel) to generate the report. For the selection of PDF, they pressed the button PDF; and for the selection of Excel, they pressed the XLSX option. After they had selected the wished-for format, and after the report was formulated, the process was finished by logging out of the accounting module. The flow was thus: the financial data was not only available easily but also was in a format suitable for presentation, audit, or analysis and thus could be exported.

Odoo Accounting has served as a platform that allowed KKBN to generate financial reports more conveniently, reduce incidences of manual errors, and boost the overall efficiency of managing accounting information. The middle decision point was a demonstration of the flexibility in using reports based on the needs of the users.

4.2.4 Point of Sales

The Point of Sale (POS) process that is new and automatic greatly improves the transaction speed and the Odoo POS module purity. In the To-Be stage, the sales process is managed rapidly and the inventory and accounting management are strict according to the schedule. With this action, the inventory and accounting systems will be automatically changed for every sale transaction. The new POS process that is installed at the KKBN's retail outlets will speed up the sales transaction processing and will let the customers have a better shopping experience

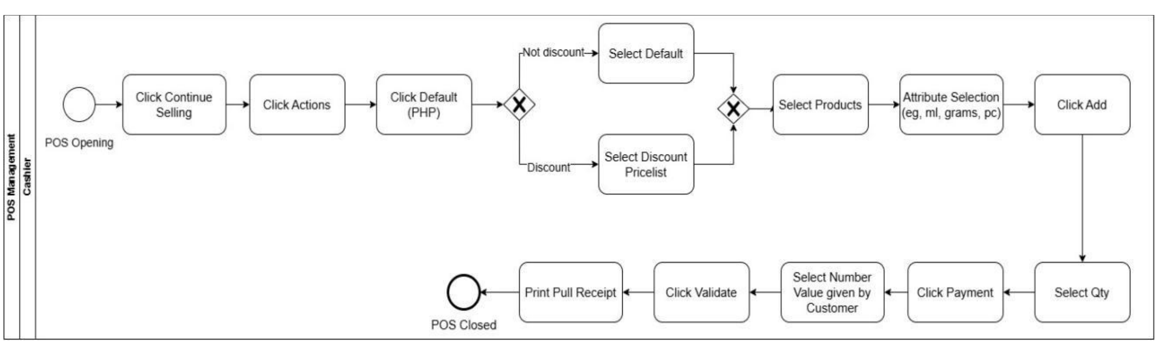


Figure 18. Business Process Modeling Notation for Point of Sales

KKBN has adopted a completely new Point of Sale process flow which is depicted in Figure 18. The cashier started with the POS module entry and immediately transferred to the sale transaction to add a transaction. Next, she selected Actions and opted for the Default (PHP) option. The system confirmed a discount on the transaction or no discount and provided the appropriate answer. The transaction was without a discount, so the cashier continued with the Default option; in contrast, if the item was discounted, the Discount Pricelist would be selected. The next step was for the cashier to click on the products he wants to purchase and then specify their quantities such as grams, or pieces. Having agreed on these first details, the cashier clicked on Add, entered the quantity that the customer wanted, and proceeded to the Payment stage. During payment, the cashier mentioned the amount received from the customer and pressed the Validate button to complete the transaction with it. Therefore, the cashier had a physical printed document showing the proof of a successful sale. The cashier then signed off the POS system after having accomplished all the tasks, thus, he had proved to be a good employee. This transaction flow was a typical and hassle-free process of selling items either at a discount or at a regular price that was correctly registering in the product selecting, quantity and payment.

4.2.5 Purchase

The acquisition of the new purchase process is to be implemented in order to improve the procurement process of KKBN by the use of the Odoo Purchase feature. According to the To-Be Model, the acts of purchase request, supplier information, and purchase order are done with the help of the system. Consequently, this enables KKBN to have the correct purchase process, to keep track of the transactions made by the supplier with precision, and to coordinate the process with the inventory.

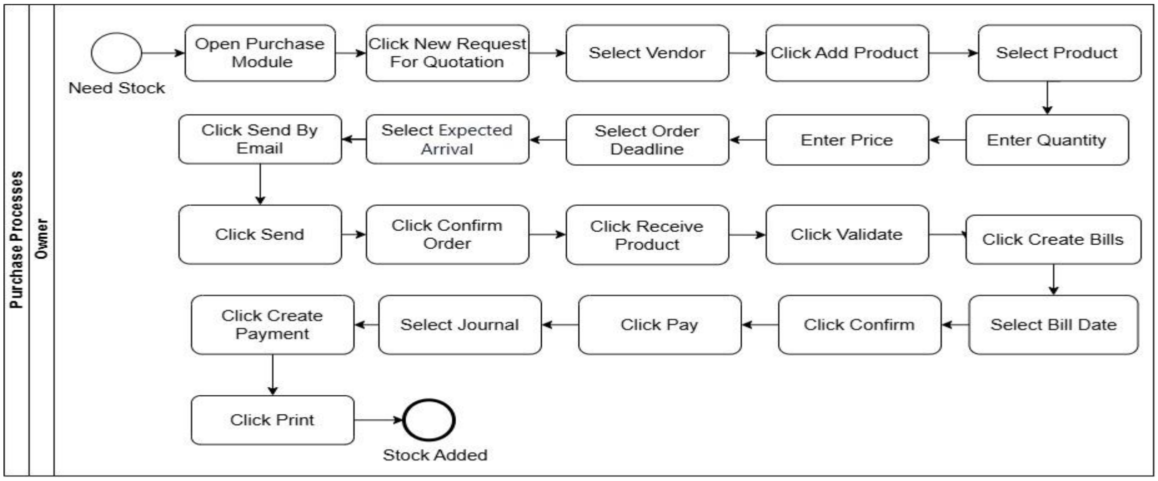


Figure 19. <u>Business Process Modeling Notation for Purchase</u>

The Purchase process flow in KKBN, carried out via Odoo, is projected in Figure 19. This is the sequential procedure starting with the opening of the Purchase Module and the creation of a New Request for Quotation (RFQ). The administrator chose the relevant vendor, selected the required products, and added information such as product type, quantity, price, order deadline, and estimated arrival date. All the required information was included in the request which was electronically sent to the vendor through the email system.

Once the confirmation was received, the purchase order was placed, and the organization was waiting for the delivery. The products came, the admin clicked on Receive Product, and the delivery was confirmed to ensure that everything was alright. The system then moved to the billing section where the admin made bills, set the invoice date, and made the payout by checking the right journal, confirming the business transaction, generating the payment ledger, and printing the logs for the documentation. The stock was added correctly so a forementioned system was finally updated. The description of the process was the correct way to do the procurement process which was in a methodical way and was documented well. This article said it was the reduction of errors, the positive impact of vendor communication, and the right flow of inventory in the system.

4.2.6 Employee Management and Attendance

The designed process for managing employees and taking attendance relies on the Odoo Employee and Attendance modules. The process is fully automated if you work with such modules. In the described process, employee data, in addition to marking attendances, are done through the digital modes, which makes the attendance sheets obsolete. This results in efficient human resource management for the organization, KKBN. It also improves accuracy and allows management to monitor employee attendance more easily.

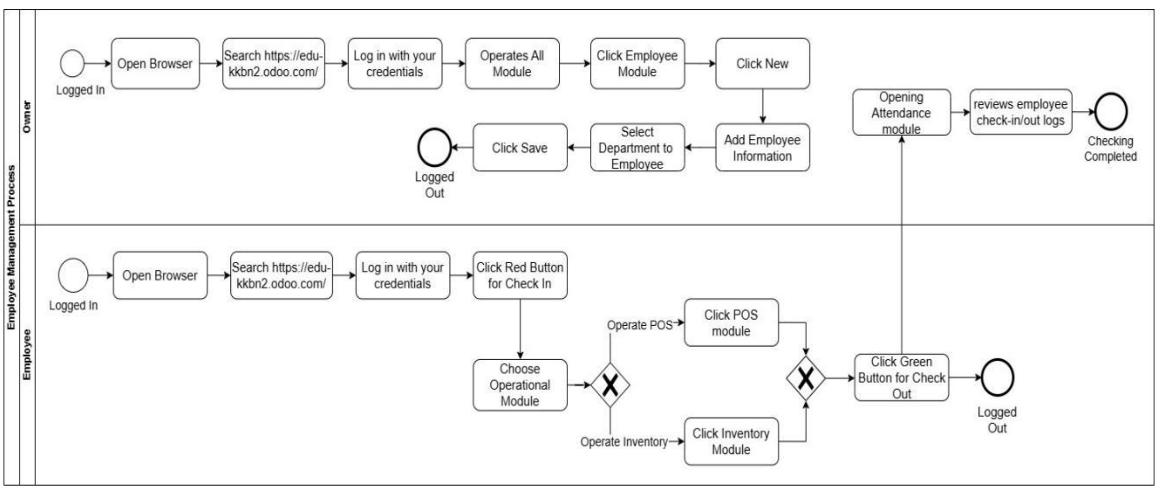


Figure 20. <u>Business Process Modeling Notation for Employee Management</u>

Figure 20 showed the new process flow of Employee Management in KKBN using Odoo.

The process began with system login through a web browser by visiting the firm's Odoo portal and providing login details. After the admin entered the Employee Module, he was able to see all modules and directly go to this section. By clicking on the New. The owner generated a new employee record, filled out the necessary employee info and assigned the employee to a specific department. When he saw the data was correct, the admin saved it and logged out, thus making sure that the employee records were accurate.

For employees, it was a simpler process. They also entered the system to their personal documents, but they pressed the red button to check in, they were not granted access to the full module. But they could only use certain modules that correspond to their responsibilities. For instance, they could work with the POS Module to sell goods or the Inventory Module for stock processes. When they finished their work, they clicked on the green button for check out and were then logged out of the system.

This policy ensured that the duties of the employees and the Owner were properly exercised while the workers were the ones to operate the system. As a result, the performance was countered by the reliability and, through more effective security.

4.5 Process Implementation

The implementation phase of BPMN focused on shifting from the To-Be Model to actual process execution. This included translating the designed improvements into practical steps and applying them in real-world operations. The purpose was to ensure that the new procedures were well-integrated and efficiently implemented while addressing the concerns and challenges identified during the analysis. This phase helped validate whether the redesigned processes could deliver the expected improvements and ensured that the procedures were aligned with the objectives of the study.

Hardware and Software Tools

The software and hardware tools needed to implement ODOO in KKBN business. This includes reliable computer, server and network system for smooth operation. together with the ODOO software and supporting applications.

Table 6: Hardware Tools

Hardware Tools	Specifications
Desktop Computer/Laptop	2.4 GHz processor
	5 GB of free storage
	4 GB of RAM
Internet Connection	Minimum of 1 Mbp/s

The table 6 shows the hardware specifications required to efficiently operate ODOO in the KKBN company. For the system to run effectively, the use of either a desktop or laptop with a minimum of 2.4 GHz processor, 4 GB of RAM, and 5 GB of free space is advisable. With these requirements, the system will be able to process data and retain effective performance without slowdown. Also needed is a stable internet connection at minimum speed of 1 Mbps to facilitate real-time access to ODOO's capabilities. These pieces of hardware create the basis for the successful running of ODOO so that the business can shift from manual operations to a digital and automated system without encountering technical glitches.

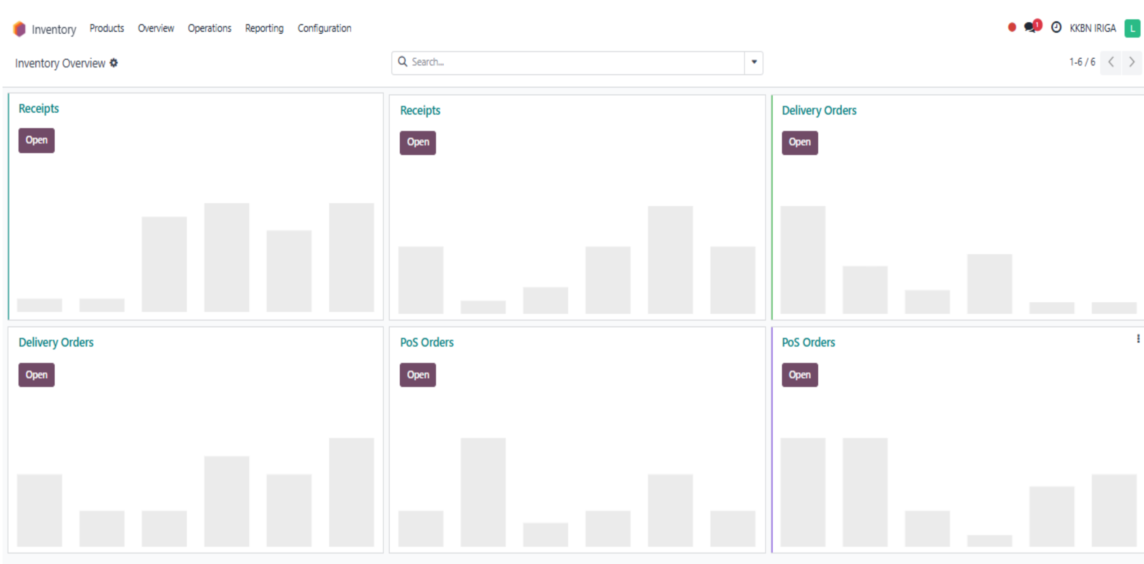


Figure 21. Main Dashboard of Inventory Module in Odoo

Figure 21 showed the Inventory Overview dashboard of Odoo, which was used for managing products and stock operations. The system permits the user to visit such areas as Products, Overview, Operations, Reporting, and Configuration. The arrangement displayed was with the three primary sections namely Receipts, Delivery Orders, and POS Orders, each revealing their status as Open. The area was used to display the various stages of the inventory flow: receiving goods into the warehouse, delivering products to customers, and processing point-of-sale orders.

The visible name of the company was KKBN IRIGA. Thus, the total arrangement gave the users the chances to oversee the stock functions and inventory In real time and speed up the workflow and be efficient.

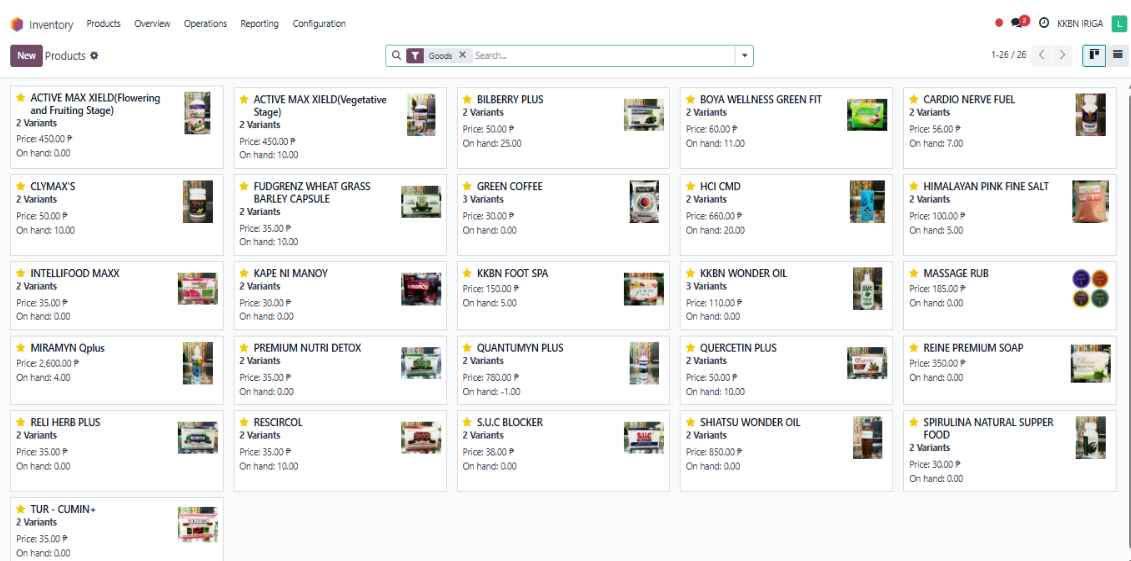
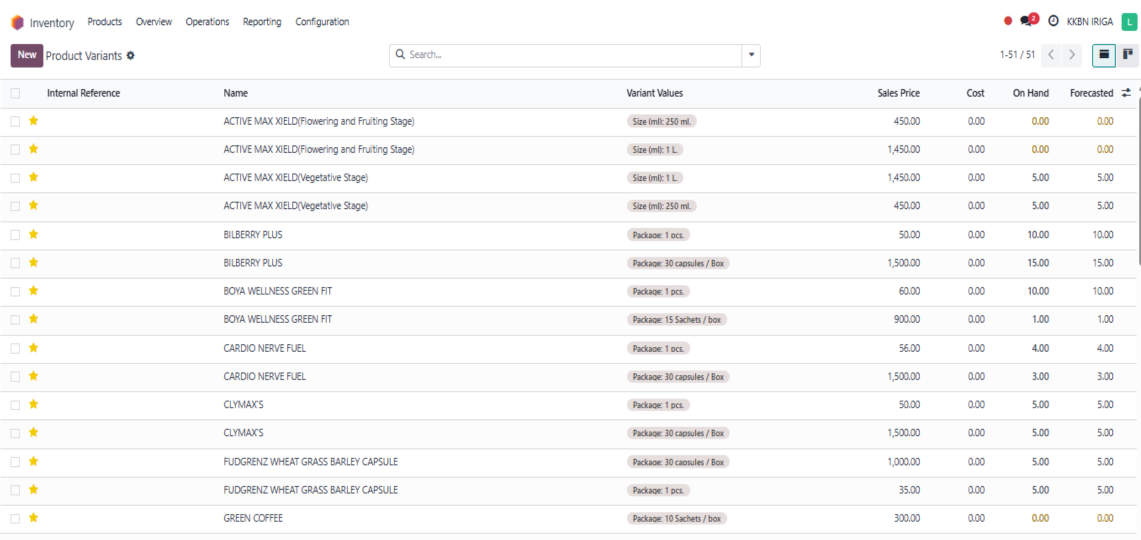


Figure 22. Products

The Odoo Product module in Figure 22 acted as a digital catalog of all the products that the company has in stock. The information provided for each product included the name, image, price, available quantity on hand, and a number of variants. The product cards for each product also had a thumbnail image that made it easy for users to find the product. Active MAX XIELD which had two priced choices was one of the products that costs ₱450.00, and CARDIO NERVE FUEL which was tagged at ₱650.00 and had a stock amount of seven units. Thus, the businesses were capable of instantaneously viewing the number of items that were available in stock, what products were low on stock, and what items had none which was denoted as On Hand: 0.00. The function was mainly for planning inventory and stockout avoidance.

The image also included the search bar and filter options that made it easy to order specific items within a short period. The module was a centralized platform through which users can modify product specifics, stock levels, and price making corrections thus ensuring stock uniformity in all the sales. Briefly, the setup served to control on the product to the companies, lessen mistakes, and therefore, smoothen the sales operations.



Internal Reference	Name	Variant Values	Sales Price	Cost	On Hand	Forecasted
	ACTIVE MAX XIELD(Flowering and Fruiting Stage)	Size (ml): 250 ml.	450.00	0.00	0.00	0.00
	ACTIVE MAX XIELD(Flowering and Fruiting Stage)	Size (ml): 1 L.	1,450.00	0.00	0.00	0.00
	ACTIVE MAX XIELD(Vegetative Stage)	Size (ml): 1 L.	1,450.00	0.00	5.00	5.00
	ACTIVE MAX XIELD(Vegetative Stage)	Size (ml): 250 ml.	450.00	0.00	5.00	5.00
	BILBERRY PLUS	Package: 1 pcs.	50.00	0.00	10.00	10.00
	BILBERRY PLUS	Package: 30 capsules / Box	1,500.00	0.00	15.00	15.00
	BOYA WELLNESS GREEN FIT	Package: 1 pcs.	60.00	0.00	10.00	10.00
	BOYA WELLNESS GREEN FIT	Package: 15 Sachets / box	900.00	0.00	1.00	1.00
	CARDIO NERVE FUEL	Package: 1 pcs.	56.00	0.00	4.00	4.00
	CARDIO NERVE FUEL	Package: 30 capsules / Box	1,500.00	0.00	3.00	3.00
	CLYMAX'S	Package: 1 pcs.	50.00	0.00	5.00	5.00
	CLYMAX'S	Package: 30 capsules / Box	1,500.00	0.00	5.00	5.00
	FUDGRENZ WHEAT GRASS BARLEY CAPSULE	Package: 30 capsules / Box	1,000.00	0.00	5.00	5.00
	FUDGRENZ WHEAT GRASS BARLEY CAPSULE	Package: 1 pcs.	35.00	0.00	5.00	5.00
	GREEN COFFEE	Package: 10 Sachets / box	300.00	0.00	0.00	0.00
	GREEN COFFEE	Package: 1 pcs.	30.00	0.00	0.00	0.00

Figure 23. Products Variants

Figure 23 represents the Product Variants module, which comprised the essential overview needed for inventory and sales management. It included a list of product variants that was required to comprehend the contents such as Internal Reference, Name, Variant

Values, Sales Price, Cost, On Hand, and Forecasted Stock. The listed items were basically health supplements and wellness products such as ACTIVE MAX XIELD, and GREEN COFFEE. All the items had different package formats; active experiences such as the package type (Box of 30 Capsules or 1 Bottle) or package size (250 ml or 1 L) available, so the system could know all the versions of the same item. This method was an incredible feat, for Product Line Management and it facilitated the right stock for each variant.

The table also had the Sales Price column, which was a significant part of the table that onlookers could see retail prices-of course directly-a customer sales and sales forecasts very important factor. On the opposite side, the Cost column indicated that there were zero costs for all items under. In addition, the On Hand and the Forecasted Stock volumes were the two that indicated a need for the reconciliation of physical and projected stock (one of them or both) by cashing out any discrepancies.

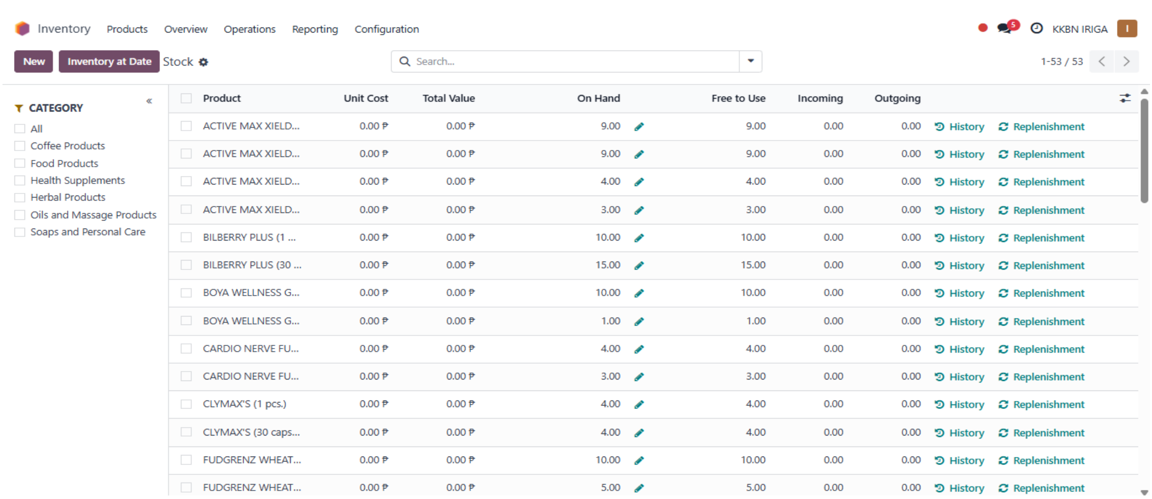


Figure 24. Products Stock

The product stock tracking module showed in Figure 24 was the one that displayed the products in accordance with their categories like Coffee Products, Food Products, Health Supplements, Herbal Supplement, oil and Massage Products, and Soaps and

Personal Care. The rows had product-specific info such as Unit Cost, Total Value, On Hand, Incoming and Outgoing stock movements. The products on the display included ACTIVE MAX XIELD, and GREEN COFFEE in various packaging or size differences. Besides that, it was easy to take quick action such as checking stock history and creating replenishment orders that the interface provided and this made it much simpler for the users to cope with restocking tasks directly from that page.

The application of this function delivered that the inventory was not only managed by the amount of merchandise but also through the classification of products, which increased the correctness of tracking and decision-making. Moreover, the On Hand and Free to Use columns, which displayed additional items in actual stock less sold, that were still available for customer orders, were another critical component in the figure. The very same function then was useful in preventing overselling and made sure the customer was served correctly at the entitled time and with the right goods. In a broader view, the incorporation of this change promoted better planning with the procurement of goods, and the distribution of resources, thus aligning the business with the demands of the market.

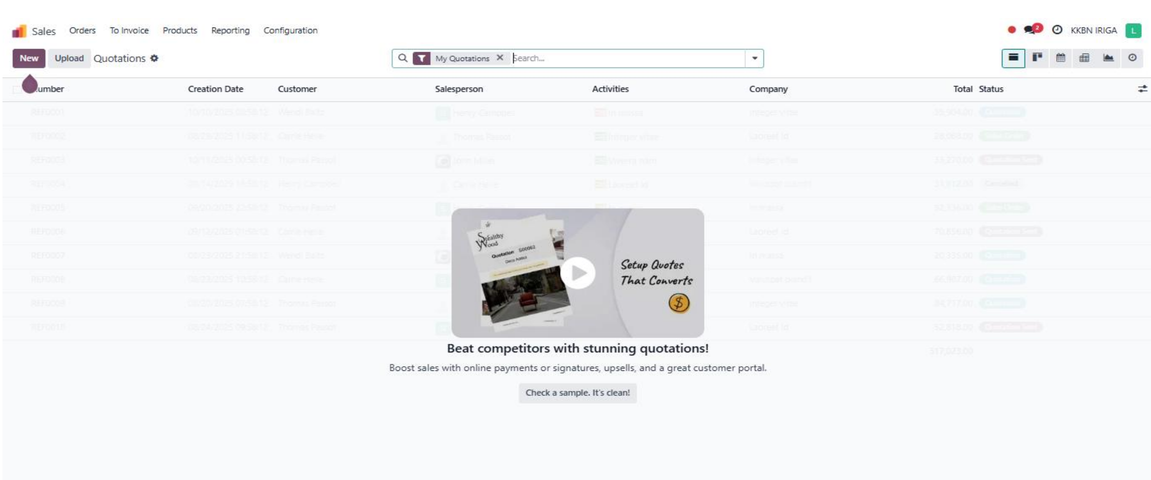
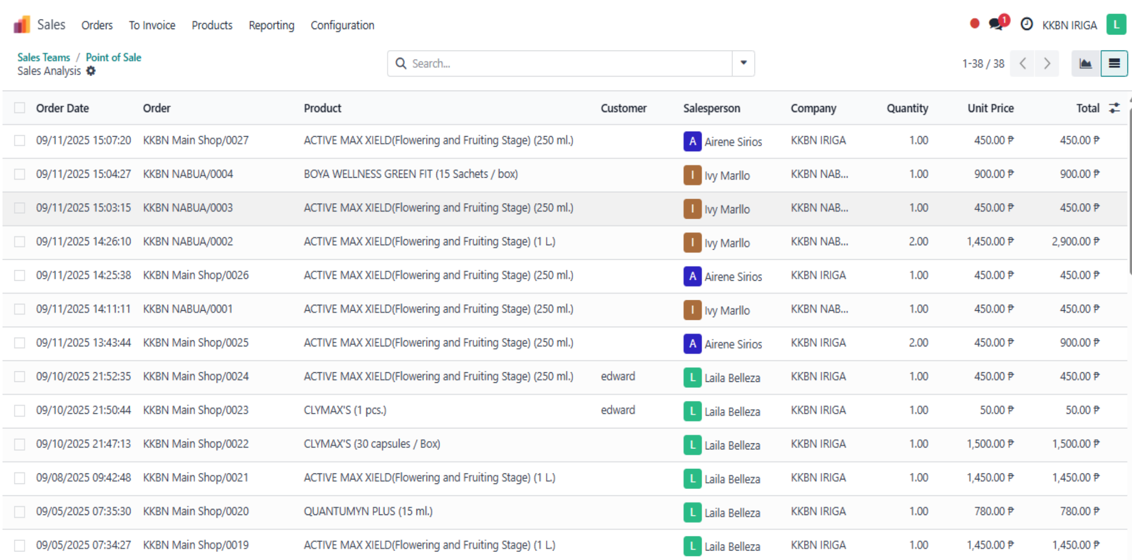


Figure 25. <u>Main Dashboard of Sales Module in Odoo</u>

Quotations module in the ERP sales management system, most likely Odoo, is represented in Figure 25. The buttons like Sales, Orders, To Invoice, Products, Reporting, and Configuration at the top show that the user could access these options, thus implying that the module is part of the overall sales workflow.

This design highlighted the efficiency of the system in fully automating the quote process and increasing the rate of conversions. A button for testing a sample quotation was available on the interface, which was a tutorial and an exhibit in one. All in all, this particular view showed the dual aim of the system: to give operational capabilities for handling real sales quotations and at the same time promoting best practices to increase efficiency and competition in selling.


Sales Orders To Invoice Products Reporting Configuration
KKBN IRIGA
Sales Teams / Point of Sale
Sales Analysis
Search...
1-38 / 38

Order Date	Order	Product	Customer	Salesperson	Company	Quantity	Unit Price	Total
09/11/2025 15:07:20	KKBN Main Shop/0027	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (250 ml.)		Airene Sirios	KKBN IRIGA	1.00	450.00 ₱	450.00 ₱
09/11/2025 15:04:27	KKBN NABUA/0004	BOYA WELLNESS GREEN FIT (15 Sachets / box)		Ivy Marllo	KKBN NAB...	1.00	900.00 ₱	900.00 ₱
09/11/2025 15:03:15	KKBN NABUA/0003	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (250 ml.)		Ivy Marllo	KKBN NAB...	1.00	450.00 ₱	450.00 ₱
09/11/2025 14:26:10	KKBN NABUA/0002	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (1 L)		Ivy Marllo	KKBN NAB...	2.00	1,450.00 ₱	2,900.00 ₱
09/11/2025 14:25:38	KKBN Main Shop/0026	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (250 ml.)		Airene Sirios	KKBN IRIGA	1.00	450.00 ₱	450.00 ₱
09/11/2025 14:11:11	KKBN NABUA/0001	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (250 ml.)		Ivy Marllo	KKBN NAB...	1.00	450.00 ₱	450.00 ₱
09/11/2025 13:43:44	KKBN Main Shop/0025	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (250 ml.)		Airene Sirios	KKBN IRIGA	2.00	450.00 ₱	900.00 ₱
09/10/2025 21:52:35	KKBN Main Shop/0024	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (250 ml.)	edward	Laila Belleza	KKBN IRIGA	1.00	450.00 ₱	450.00 ₱
09/10/2025 21:50:44	KKBN Main Shop/0023	CLYMAX'S (1 pcs.)	edward	Laila Belleza	KKBN IRIGA	1.00	50.00 ₱	50.00 ₱
09/10/2025 21:47:13	KKBN Main Shop/0022	CLYMAX'S (30 capsules / Box)		Laila Belleza	KKBN IRIGA	1.00	1,500.00 ₱	1,500.00 ₱
09/08/2025 09:42:48	KKBN Main Shop/0021	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (1 L)		Laila Belleza	KKBN IRIGA	1.00	1,450.00 ₱	1,450.00 ₱
09/05/2025 07:35:30	KKBN Main Shop/0020	QUANTUMYN PLUS (15 ml.)		Laila Belleza	KKBN IRIGA	1.00	780.00 ₱	780.00 ₱
09/05/2025 07:34:27	KKBN Main Shop/0019	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (1 L)		Laila Belleza	KKBN IRIGA	1.00	1,450.00 ₱	1,450.00 ₱

Figure 26. <u>Sales Teams of Point of Sale</u>

The sales teams feature of the POS module in Odoo ERP captured and tracked the sales transactions for KKBN which was shown in figure 26. The table presented data such as Order Date, Order Number, Product, Customer, Salesperson, Company, Quantity, Unit Price, and Total Amount Sold.

This module has the performance ratio of sales in a clear way so you can monitor which products are selling and which salespeople have the highest sales. The gathered information accounted for the accountability of the sales staff and enabled management to monitor customer activity. The program also did not require the manual summing of sales totals which made it a time saver, decreased the number of errors and delivered the correct documents. Apart from that, the promptness of the sales information made it possible to respond more quickly to sales patterns and customer needs. The Point-of-Sale module, by centralizing this data, improved the decision-making process in areas such as inventory management, employee performance evaluation, and sales strategy development.

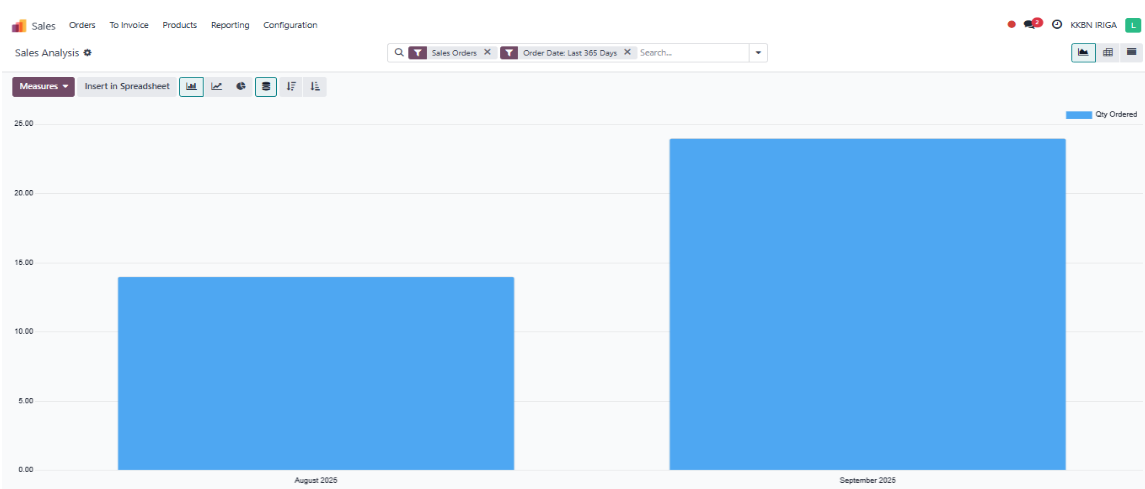


Figure 27. <u>Sales Analysis</u>

The graph that is seen in Figure 27 could have been generated with a business management system like Odoo in a sales analysis dashboard. The x-axis depicted time (August 2025 and September 2025) while the y-axis illustrated the Qty Ordered metric. The chart utterly informed on sales units sold in September 2025 in total in contrast to August 2025. The chart indicated that the sales in the month of September were better than the previous month. The figure informed that in August, the company received approximately 13 orders, and in the month of September, the company achieved 23 orders. It denoted a rise in sales activities which signify growth. The rise was the indication that the recent promotional strategies worked and thus sales could continue climbing.

This information can be used for the company decision making. Marketing, customer engagement or the product that has the highest demand in September, the three options that could have been provided for the positive sales volume were attributed to. The need for more frequent monitoring and evaluation of business processes to detect changes in the business environment and objectives is the other main point they made. Comparing these metrics regularly allowed companies to produce sales forecasts, allocate resources, and draw up development plans more efficiently than ever before.

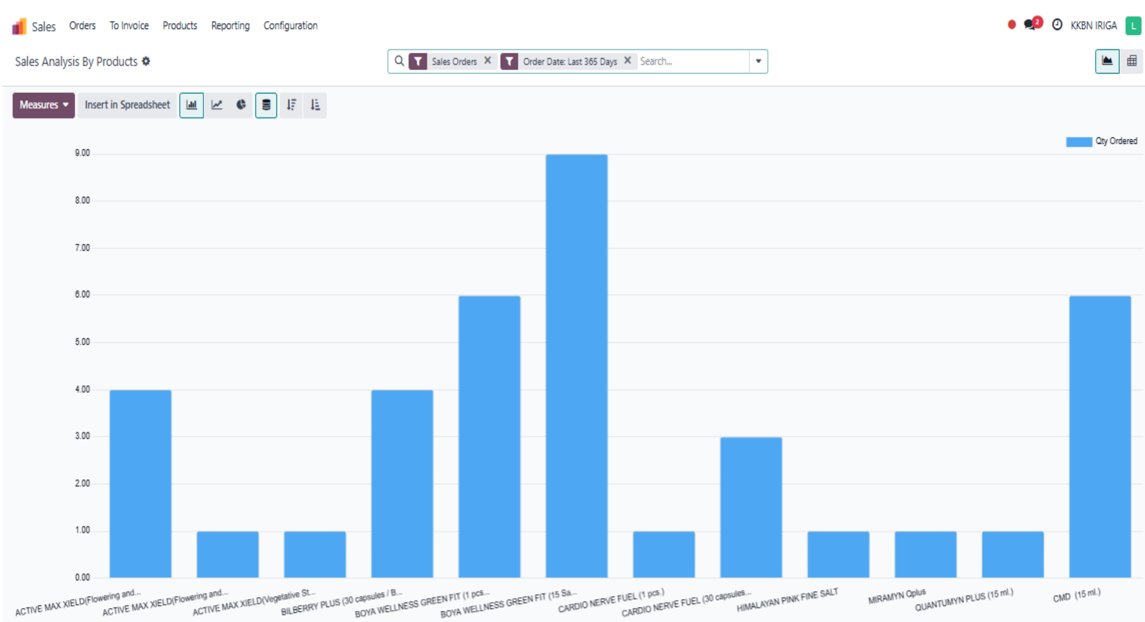


Figure 28. <u>Sales Analysis by Products</u>

Figure 28 was a bar chart that showed the sales of specific products and it also the purchase of different products for the last 365 days. Cardio Nerve Fuel 15 softgels, for instance, the highest amount of orders that were made to the shop. Then came Boya

Wellness Green Fit 15 softgels. Other products thus recorded almost no sales, like, for example, Bilberry Plus 30 capsules and Mirantin Qplus.

This diagram proved to be a powerful instrument in examining the market and the various customer preferences as well as the demand diversions of the different products in the range. Cardio Nerve Fuel and Boya Wellness Green Fit were the product lines that were showing to be the high-volume sellers as it seems that those sold the most and in that way were the ones adding the most profit too. Meanwhile, some products which were not ordered frequently by customers had to be investigated to understand if they required to be marketed differently, repositioned in the marketplace, or taken out of circulation altogether. Realizing these factors gave the companies the chance to prioritize the fast-selling items and as a result, permitted an enhancement in the sales techniques.

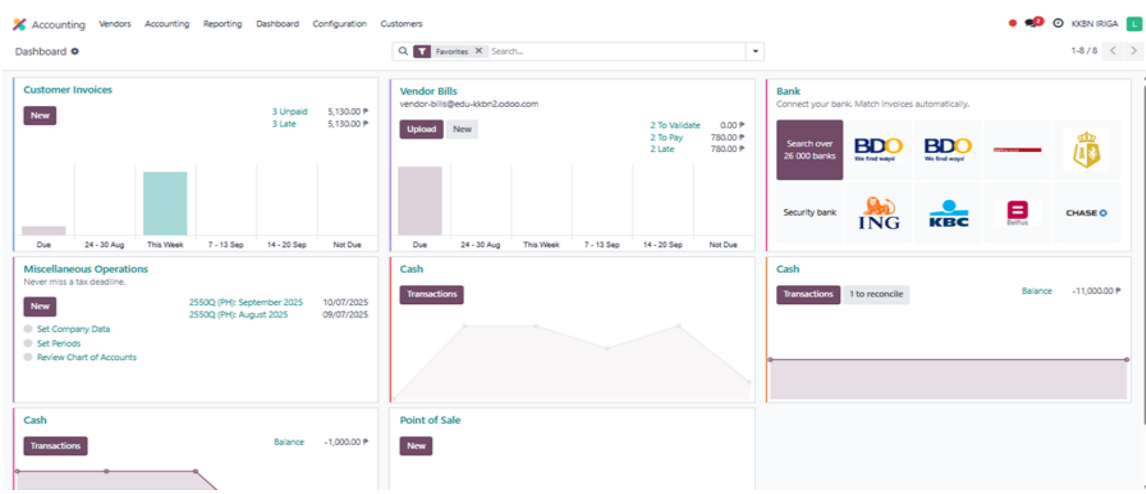


Figure 29. <u>Main Dashboard of Accounting Module in Odoo</u>

The accounting module for all the payments transactions is considered as the displayed item on the window of the Figure 29. It integrated important information such as customer invoices, vendor bills, cash flow, bank accounts, miscellaneous operations, and point-of-sale records to present an overview. The module combines the possibilities of handling the user like Vendors, Accounting, Reporting, Dashboard, Configuration, and Customers. The proprietor was able to see the total financial condition of the company directly with the application without the need to wait for detailed reports. Additional dashboard features were the shortcuts for loading bills, putting down transactions, reconciling accounts, and setting financial periods, which turned accounting into an easier daily task. The picture illustrated the dashboard among which sections as Customer Invoices were tracking outstanding and overdue invoices, and Vendor Bills were listing pending and late supplier payments. The Bank panel was an integrating connection with the banks that would automatically enable the comparison of invoices and payments on the other hand. The Point of Sale and Cash panels showed available transactions and balances, thereby facilitating liquidity tracking. Miscellaneous Operations informed users on important accounting activities, like company data setup, the chart of accounts review, and tax due date tracking.

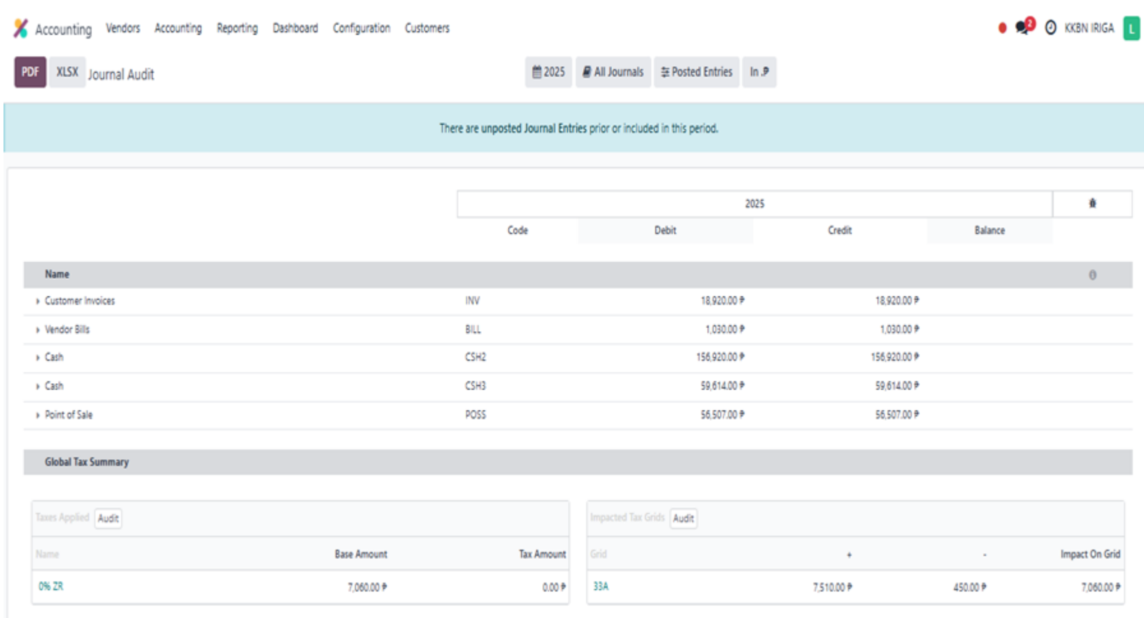


Figure 30. Journal Audit

Audited Journal report included the most recent report by an accounting software package which was displayed presumably Odoo ERP. The image provided journal entries for the year 2025 that were filtered to show all journals and posted entries only. The heading at the top pointed out that there were some unposted journal entries which might have affected the accuracy of the report. Below, the list of Netherlands journals such as Customer Invoices, Vendor Bills, Cash, and Point of Sale were mentioned which all contained both debit and credit balances. For example, customer invoices, and vendor bills both showed equal debit and credit amounts, which indicated balanced entries, while cash and point of sale journals also displayed the same steady financial activity.

The Global Tax Summary was presented as per the tax base amounts along with related Tax effects. In this case, there was a 0% ZR (Zero-Rated) tax applied to a base of ₱7,060.00, with no result in tax amount as it was zero. The tax grid affected (33A) was characterized by a credit of ₱7,510.00, a debit of ₱450.00, and a net effect of ₱7,060.00. Office of the account or the API showed the accountants and managers an overview of all financial operations carried out in the various journals accompanied by the summary of the way taxes affected reported transactions

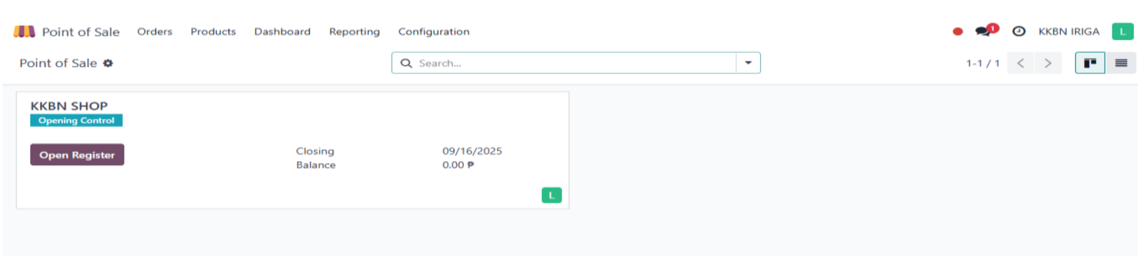


Figure 31. Main Dashboard of Point of Sale in Odoo

Figure 32 represents the Point of Sale (POS) module section in an accounting or ERP system, which is suspected to be Odoo. It exhibited the POS dashboard for KKBN

Main Shop where the register can be managed. It showed the closing date as 09/05/2025 with a total closing balance of ₱0.00. This referred to the fact that no sales slip had been entered or completed in the POS system for the day and the register was (already) closed but it could be reopened to accept new transactions if the Open Register button was pressed.

This interface was the part of retail operations where this was used a lot, thanks to cashiers could start and finish daily sales sessions. Register opening was the first step in which sales transactions were counted, observed, and finally, at the end of the working day, the amount was corrected. The equal amount of ₱0.00 would be due to either no open transactions being available or all previous transactions before were taken care of. This is a simple but good way to both control and monitor sales in a store's POS system.

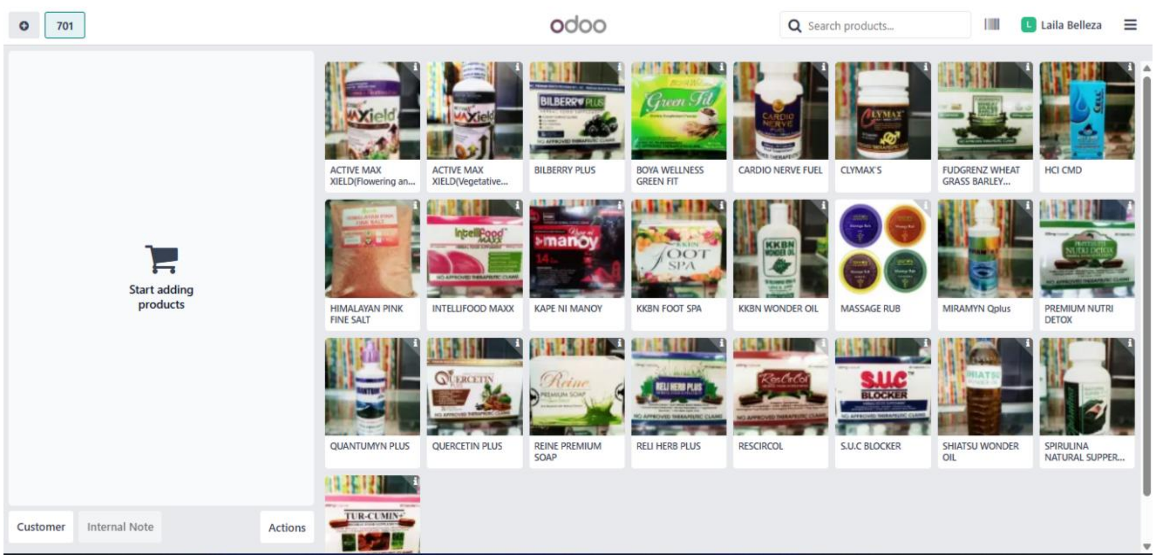


Figure 32. Point of Sale in Products

As presented in Figure 33, Odoo Point of Sale (POS) interface which depicted a grid of products that could be selected was viewed be the reader. This was a specific product of Odoo which had products arranged in a grid format. With the possibility of viewing the image that accompanies each product and the label that accompanies each product, adding items to a customer's order was facilitated for cashiers. Products in the catalog mostly terminated in the category of health and wellness such as Cardio Nerve Fuel, Rene Premium Soap, and Massage Rub were shown. The structured and systematic way of displaying the products made transactions faster as the staff members were able to find the items straight away instead of searching through lengthy text lists.

The board was showing "Start adding products" to the user, a place where you can list items that you have selected when you are taking an order. Chosen items appeared on the order list. Below this, Customer, Internal Note, and Actions allowed the assignment of the purchase to a customer, the addition of remarks, or the completion and administration of the order.

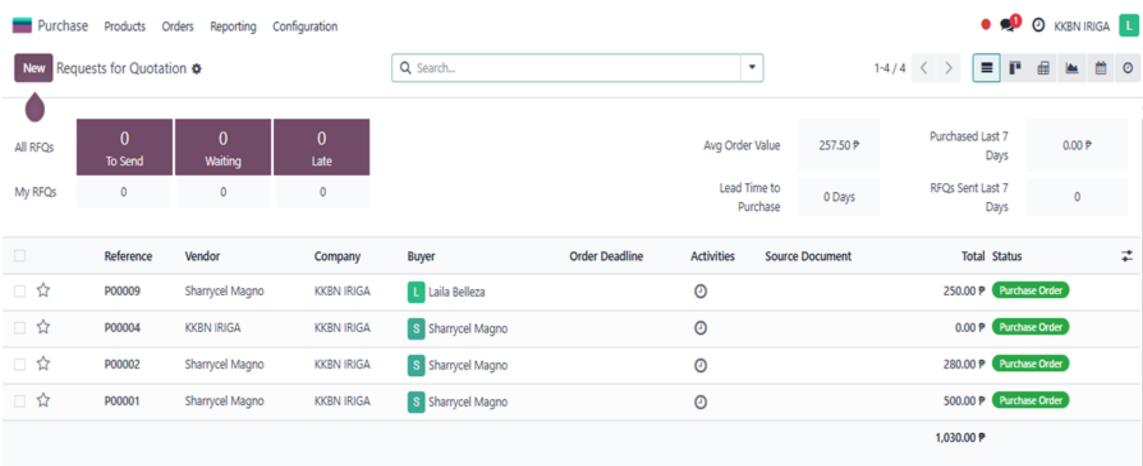


Figure 33. Purchase Module List of Requests for Quotation

In Figure 34 was the Display Purchase Order Management window which contains the system through which all purchase transactions are tracked and monitored with suppliers. The lines were arranged in such a way that each line represented a purchase order record with certain details like Reference Number, Confirmation Date, Vendor, and Buyer which made it easy to find and follow individual transactions. The Activities column showed pending or active activities related to the purchase order, on the other hand, the Source Document column was the source of references to documents connected for traceability. Transparency was mainly done by the Total column which was showing the purchase cost and the Billing Status column displayed the payment, which was signified as Fully Billed, which means that the vendor invoice was processed and paid. The Expected Arrival column was the one that showed the predicted delivery date and time which were the basis that the company would prepare the stock availability and inventory management.

The New button at the top left side of the interface was the main way of creating a new purchase order which in turn will speed up procurement and make it more efficient. The Filter button and the search facility at the top of the interface potently assisted users to easily locate specific purchase orders based on the vendor, date, or status. This figure presented a centralized purchasing order management tool, which enhanced accountability and ensured that procurement were well monitored from order making to delivery.

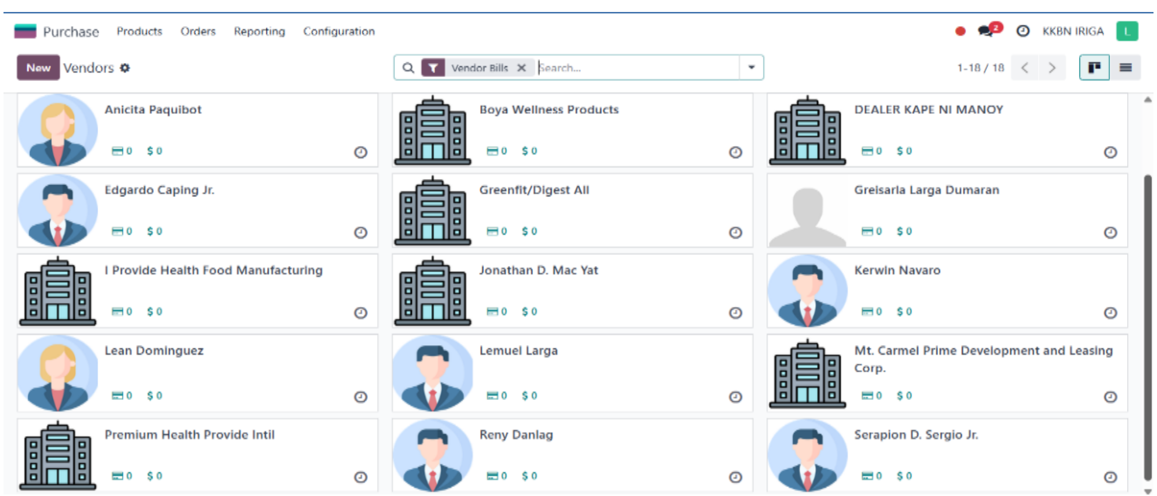


Figure 34. Vendor Lists

Figure 35 showed the Vendors list dashboard from a purchasing or procurement system interface. Alongside each vendors identified are detailed such as Name example

Anicita Paquibot and company name Boya Wellness Products and outstanding amount all of which showed $0 indicating the lack of present payables. Each vendors have respective Bank Account example BDO to ensures that financial transaction, such as payment for purchase can be processed seamlessly without needing to re-enter details manually. These systems allow businesses to efficiently manage and monitor their supplier by centralizing vendor information in one accessible view. The dashboard also enhances the supplier management and managing financial transaction effectively and better vendor relationship and decision making through offering a complete view of all active vendors within the system.

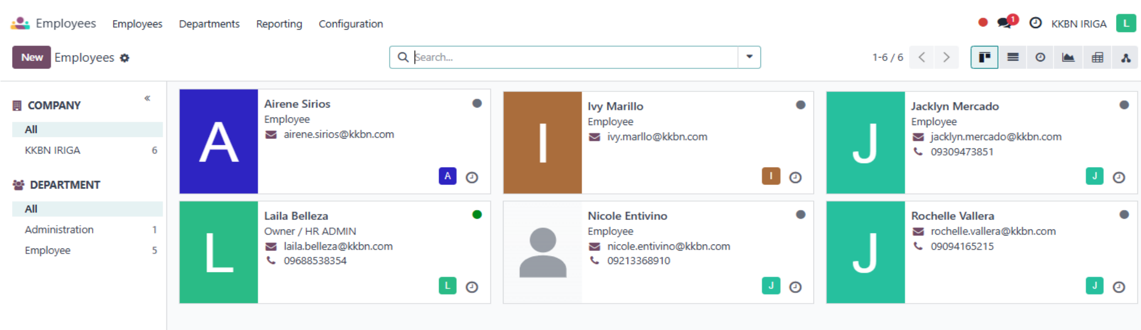


Figure 35. <u>Employee Module in Odoo</u>

The Employee Management Dashboard is a web-based management platform to monitor and manage employee data that the employee management hardware dashboard centralises. Employee profiles were listed on the dashboard in a card-style format, with each card containing the necessary details like full name, job title, contact email, and phone number. The card format made it possible for employees to find and access the information even quicker-minimum searching through records would be done. On the left side, the system offered a department filter which categorized the workers into Administration, Employee, and the number of members in each group.

This feature contributed to better employee distribution, so that managers could easily identify and organize personnel according to their units. Staff categorization, along with the way it was done, made it easier to mention the employees by their respective departments which was one of the crucial preconditions for correct resource allocation and effective interdepartmental communication. Up at the screen of the dashboard, there were also some other options such as adding a new worker, searching for certain profiles, and toggling between different diagram modes to enhance data visualization. These suggested features significantly helped the process, making it easier for administrators to maintain the records updated, staff tracking, and real-time accuracy of their personal database.

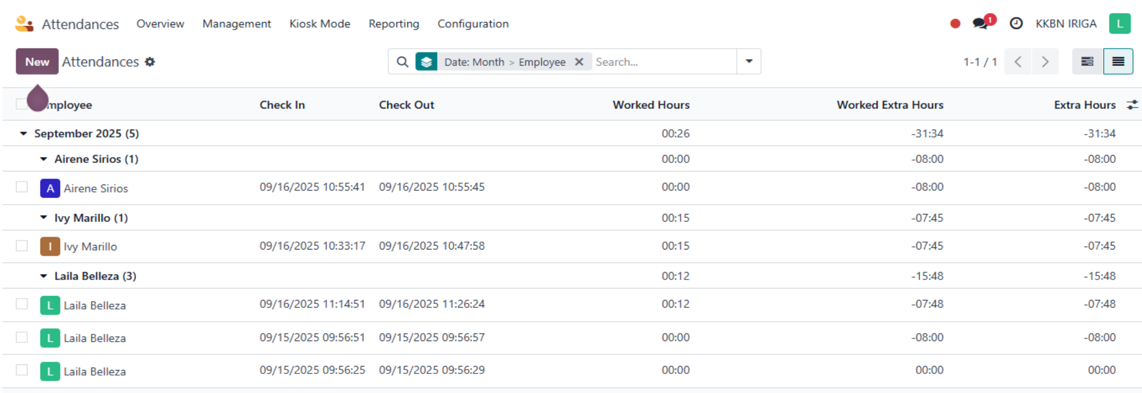


Figure 36. Attendance Module

The Attendance Module feature of Odoo ERP that was chosen in Figure 37 was the representation of the employee attendance records. The table occupied various specifics such as the Employee Name, Check-In, Check-Out, Worked Hours, Worked Extra Hours, and Extra Hours. On the opposite hand, every row that was attached to this table was, in fact, the record of one certain employee's attendance of the day. The principal purpose of the module was to supervise the timekeeping of employees, the productivity and the observance of work schedules. KKBN's owner got a simple and comprehensive track of the attendance records. This centralization was a combination of the record-keeping process which was improved due to the monitoring errors that were made manually and employee attendance data was correctly reported and performance assessments.

4.6 Process Monitoring

Within this stage, the team of researchers integrated the system into KKBN as a solution to the problems it had been facing with its daily transactions. By the ISO 25010 standard, the researchers did evaluate the customized and configured ERP software according to the following characteristics: functional suitability, performance efficiency, usability, maintainability, reliability, and security [2].

Functional Suitability

The table below showcases the systems functional Suitability and also the overall system having to be assessed, along with the required data generated in terms of functional suitability.

Table 7: Level of Acceptability of Functional Suitability

Descriptions	Weighted Mean	Remarks
The degree to which all necessary business operations and functions are covered by the ERP solution.	4.9	Highly Acceptable
The ERP solution's capacity to generate dependable and accurate outputs from the input data.	4.9	Highly Acceptable
How well the ERP system satisfies the user's particular functional needs.	4.8	Highly Acceptable
The ERP system's capacity to automate repetitive processes and operations.	4.6	Highly Acceptable
The ERP solution's capacity is to easily share and integrate data across many module	4.7	Highly Acceptable
Average	**4.78**	**Highly Acceptable**

The results reveal that the Odoo ERP system largely has a very high level of functional suitability with a weighted mean of 4.78, which can be termed Highly Acceptable. This implies that the system is mostly accepted by users since it can carry out necessary business activities, provide accurate and reliable output, and meet specific user needs, automate routine activities, and share data easily and efficiently between modules. This assumes that the system can carry out its intended duties efficiently within the organization.

This finding is also found in some studies based on the ISO 25010 model in recent years. The importance of functional suitability is not to be achieved through aligning functions with business processes. Sarwosri et al. [2023] clarified that ERP systems are expected to achieve high functional suitability if there is stronger alignment between the functions and tasks. This aspect leads to closing functional gaps and satisfying the users [7]. The functional suitability is found to be high in this study. This implies that the implemented Odoo ERP system can fulfill the activities KKBN performs daily. This is considering that the needs are met and the system is functional. This implies that the employees can undertake their duties efficiently.

The limitation is though the outcome is outstandingly positive, the analysis of this system has been made through user perception at the time of analysis. As usage of the system increases and/or as business requirements evolve, certain other functional challenges may emerge.

The findings have a wide range of implications; the primary one being proving the capabilities and effectiveness of the Odoo ERP system in the actual business process execution support. This has its very specific significance as the functional suitability level is so high that this indicates that the solution is not just a technical instrument that is used, but it relates to the kind of process which is performed in the company. In general, this improves the relevance of the research's findings and conclusions, proving the effectiveness of a customized solution for improving the efficiency of the process while making all necessary decisions. The Implementation of ERP system is highly recommende. This also supports the relevance and effectiveness of the ISO 25010 model in analyzing the efficiency of the solution.

Performance Efficiency

The table below presents the system's level of acceptability in terms of performance efficiency.

Table 8: Level of Acceptability of Performance Efficiency

Descriptions	Weighted Mean	Remarks
The ERP solution eliminates the need for conventional manual paperwork, enabling users to accelerate their document processes and reduce their reliance on hard-copy forms.	4.8	Highly Acceptable
The ERP solution can respond to user activities rapidly and efficiently, ensuring smooth and uninterrupted operations while systems utilize.	4.7	Highly Acceptable
The ERP solution reduces transaction delays, enhances user experience, and accelerates routine operations.	4.8	Highly Acceptable
The ERP solution increases the system's efficiency compared to the previous one and allows the users to notice apparent boosts in the productivity of operations.	4.7	Highly Acceptable
The ERP solution can maintain a high level continuously when facing different levels of workload and always ensure dependable operation.	4.7	Highly Acceptable
Average	**4.74**	**Highly Acceptable**

Concerning the performance efficiency use of the Odoo Enterprise Resource Planning system, the weighted mean obtained was 4.74. The classification of the obtained mean is termed as Highly Acceptable. This shows that the users obtained a high response rate from the system and a lower reliance on paperwork. Current ERP research emphasizes that performance efficiency can be increased by systems with less manual and more optimized transaction processing. Al-Shboul et al. [2021] emphasized that contemporary ERP systems increase efficiency by optimizing system response times and subsequently minimizing processing times. This corroborates with the current research, which indicated that the system supports efficiency and faster completion of work [2].

The result of this research is indicate that Odoo ERP has the potential to enhance multiple work processes with the help of the software, which in turn, facilitates faster employee execution of the tasks and sometimes makes handling of the tasks by employees with no interruptions. Therefore, the software is, on the one hand, a workflow optimization method and, on the other hand, a tool for productivity and operation regularity.

The limitation is evaluation was the user experience. This meant that the testing process did not evaluate the system under intense workload conditions or at its peak. Future studies should consider testing the system under higher workloads to determine its performance limits and scalability.

These results are important as they prove that the Odoo ERP system has the capability of supporting organizational operations. The fact that there is a high level of functional suitability shows that the system not only has the potential to exist to support organizational activities, but it also enhances activities in a way that fits the organizational environment. This further enhances the findings of the study that a customized system has the capability of supporting organizational activities in a way that ensures efficiency in activities as well as decision-making. This further confirms that ISO 25010 has the capability of supporting the process of evaluating organizational activities.

Reliability

The table below presents the system's level of acceptability in terms of reliability of the system.

Table 9: Level of Acceptability of Reliability

Descriptions	Weighted Mean	Remarks
The degree to which the ERP solution keeps working even when software defects occur, enabling users to continue being productive despite unforeseen problems.	4.7	Highly Acceptable
To ensure integrity of data and reduce potential loss, the ERP solution can recover and restore data after disruptions or failures.	4.7	Highly Acceptable
The ERP is always available to the users to enable access for critical processes on a continuous basis.	4.7	Highly Acceptable
The operations of the ERP system are continuously executed without unexpected deviation to provide users with providing a reliable and steady experience.	4.9	Highly Acceptable
The capability of the ERP solution to perform regular maintenance and upgrades without disturbing the current operations.	4.9	Highly Acceptable
Average	**4.78**	**Highly Acceptable**

Reliability The weighted mean received by the Odoo ERP system is 4.78, which fell within a Highly Acceptable level, meaning it is generally available when needed; it can recover data in case of disruptions, and it operates without frequent, or unexpected, interruptions. The recent research highlights the importance of reliability as a main factor affecting the ERP success rate due to the fact that these systems basically cater to the core business processes. Sarwosri et al. [2023] have put forth the argument that the trustworthy ERP systems are the ones that build up the user confidence and offer the operational continuity, particularly those systems, with the help of recovery and fault-handling mechanisms, that are supported. Thus, this highly agrees with the high ratings of reliability, which are found in the present study [6].

The results indicate that the ERP system is able to enhance the continuous flow of business operations with only some minor technical problems. This is made possible by the high level of reliability that allows workers to carry out crucial tasks without having to worry about the system becoming offline. This enhances productivity. In this case, management is also able to make decisions on reliable information. Trust among stakeholders is also created through this system's reliability.

The limitation is the system is not designed to address is severe system failures, as these were not seen during the evaluation period, meaning that extreme system performance is not assessed. How well the system will fare during large failures or when heavily used is not known. Its ability to recover from normal difficulties is all that its ability to recover from outages is known to be capable of handling. In fact, these results may not generalize to other organizations.

The significance of this finding is evident since it confirms that the Odoo ERP system can support business activities in an appropriate manner. The degree of functional suitability indicated by this finding demonstrates that the ERP system in question is not only available in the market but also has a closer relationship with business activities in the organization. This finding generally supports the conclusion drawn from this study that a well-designed ERP system can effectively support business activities. This finding also supports the significance of this study in using the ISO 25010 standard for selecting ERP systems.

Usability

The table below presents the system's level of acceptability in terms of usability of the system.

Table 10: Level of Acceptability of Usability

Descriptions	Weighted Mean	Remarks
Because of the ERP solution's ease of use, users may effectively do daily activities with little effort.	4.9	Highly Acceptable
The ERP solution is user-friendly and simple to use and comprehend without requiring a lot of training.	4.7	Highly Acceptable
Users can accomplish their goals effectively and satisfactorily with the help of the ERP solution, enabling productive results of the work.	4.9	Highly Acceptable
The ERP system's capacity to recognize and fix mistakes on its own, reducing the need for users to perform manual troubleshooting	4.1	Highly Acceptable
The user may easily learn the ERP solution object to have more comprehension and result in faster adaptation and more effective daily use.	4.8	Highly Acceptable
Average	**4.68**	**Highly Acceptable**

The usability of the Odoo Enterprise Resource Planning system scored Highly Acceptable, which means that the users found the system very easy to understand, learn, and use. The overall response of most of the users revealed that the menu items in the system are well-structured, and the tasks can be performed without any confusion. It is an indication that the users can navigate the system very smoothly, which reduces errors, saves time, and increases efficiency.

Recently, it was highlighted that usability plays an essential role in ERP system acceptance and its success. According to Lusiani & Princes [2024] ergonomic and friendly design of ERP system interfaces increases user satisfaction and minimizes resistance to system implementation. This was supported by this study, where the users had a positive perception of ease of usability and understanding of the ERP system. High usability means that the ERP system will easily enable employees to adopt it and work with it to complete their tasks, and this will enable the organization to fully utilize the system. leading to improved productivity, efficiency, and overall operational performance [2].

The limitation is the usability test was based on the perceptions of users during the test period. Differences in levels of user experiences and extensive system usage can bring to light other usability problems not identified by this study. Additionally, it might not be an ideal evaluation since it took a short period. Users might face different problems based on their acquaintance with advanced features within the ERP system. In future studies, perhaps a more extended period of usage could be considered, with a larger number of respondents.

Results of the usability test are important as they show that, apart from being functional, the ERP system is also user centric. A user-centric system promotes adoption, increases user confidence, and minimizes errors. The results show that users are efficiently able to navigate the system and perform their duties without much difficulty in the system. It also results in increased productivity. A high level of usability will help in the easy adaptation of workers in the ERP system. Thus, successful implementation of the ERP system relies on ensuring that the entire organizational members are at ease with the system and can exploit its functionalities to their benefit.

Maintainability

The table below presents the system's level of acceptability in terms of maintainability of the system.

Table 11: Level of Acceptability of Maintainability

Descriptions	Weighted Mean	Remarks
Solution's flexibility as business demands changes and its simplicity of customization and configuration to match business requirements.	4.9	Highly Acceptable
The ERP system's capacity to reduce the amount of time spent on error correction by enabling users to swiftly view and fix any errors.	4.8	Highly Acceptable
The ERP solution's ability to run without hiccups and guarantees dependable performance while in use.	4.7	Highly Acceptable
Users may manage the ERP solution with ease.	4.6	Highly Acceptable
The ERP solution reduces interruptions to corporate operations and is made for rapid and effective maintenance or updates.	4.9	Highly Acceptable
Average	**4.78**	**Highly Acceptable**

The maintainability of the Odoo Enterprise Resource Planning system was found Highly Acceptable since there was a perception that it was flexible, stable, and easy to manage. It also shows that the system can be maintained without disrupting normal business activities. It also comprises the acknowledgment by the users that there will be efficient handling of errors and system updates that facilitate smooth business activities.

Recent studies carried out around software quality have proven that in the case of ERP systems, delivered very often due to changing demands in businesses, the criterion of maintainability becomes an important one. Additionally, concerning the definition of maintainability itself, Behutiye et al. [2020] affirms that a system with good maintainability is less costly in the long term and supports continuous improvement since changes and updates can be easily performed. It appears that the adjustments of the ERP system are efficient and thus help the organization be flexible and responsive to change. This ensures that the organization has less system downtime and enables the organization to maintain system performance [6].

The limitation is the assessment was that it was largely perception-based while it could have relied on technical maintenance information for a more refined assessment of maintainability.

The results for maintainability are important as they indicate the capability of the ERP system not only to perform effectively in the current state but to support the organization in the long run as well and without error encountered. A system that is maintainable helps the organization to make changes, modify, and upgrade the ERP system without incurring a lot of cost and effort in the long run. The Implementation of ERP system at KKBN based on the respondents is highly recommended. This is important as a system that is highly maintainable is always a good and long-term investment for the organization.

Security

The table below presents the system's level of acceptability in terms of security of the system.

Table 12: Level of Acceptability of Security

Descriptions	Weighted Mean	Remarks
Sensitive data is shielded from unwanted access while being kept secure and private within the ERP system.	4.9	Highly Acceptable
The ERP solution provides a secure environment for all users while protecting the data from a variety of dangers.	4.8	Highly Acceptable
The ERP system is strengthened to stop data loss while giving users assurance about the security of their data.	4.7	Highly Acceptable
The ERP system can stop unapproved changes to data and regulate access.	4.6	Highly Acceptable
The ERP solution's database is safe and protected from unwanted access.	4.9	Highly Acceptable
Average	**4.78**	**Highly Acceptable**

The weighted average of 4.78 was already attained by security measures within the Odoo ERP system. This translates to Highly Acceptable. This means that they strongly believe in their capability to provide security for confidential business data, handle user access, and secure against unauthorized modification of business data. The respondents were satisfied with the areas involving user authentication, management of user access by means of role-based security, and confidentiality of data within the system because of its capability to create a secure setting for doing business.

Recent research carried out to evaluate the security of the ERP system suggests that the most critical attributes of a trustworthy system are access control mechanisms and data protection. Regarding the above-mentioned topic, Alenezi et al. [2023] clarified that in order to keep a high trust level of the users at a maximal value in the modern-day ERP system, the system ought to provide its users with a guarantee of integrity, confidentiality, and availability of the data. The current information provides a basis to understand the high security score obtained in the research study and suggests the compliance of the Odoo's ERP system with general security standards [6].

The findings have demonstrated that the organization is assured of the secure processing of its sensitive data by implementing the ERP system. The security of the organization is strengthened due to proper measures developed in the system to mitigate risks associated with security threats, unlawful access, and data loss.

However, the limitation in this case is that the testing carried out did not involve advanced techniques for penetration tests as well as vulnerability tests. There might have existed other ways of testing the level of security that would have fared better security evaluation methods unfortunately now.

Very essential, however, are the findings related to security, which state that it is safe to use the system within the business context, particularly regarding maintaining the confidentiality of data. This is particularly essential, given the fact that this system appears to work best, with high security and the facility to accomplish all tasks with no hassles whatsoever. The Implementation ERP system high security measures would thus guarantee a successful installation process.

Table 13: Summary of Results

Descriptions	Weighted Mean	Remarks
Functional Suitability	4.78	Highly Acceptable
Performance Efficiency	4.74	Highly Acceptable
Reliability	4.78	Highly Acceptable
Usability	4.68	Highly Acceptable
Maintainability	4.74	Highly Acceptable
Security	4.78	Highly Acceptable
Average	**4.75**	**Highly Acceptable**

Table 13 below shows an overall summary of the respondent evaluation of the performance of the ERP system, focusing on its effectiveness in terms of six broad aspects of software quality that include functional adequacy, performance efficiency, usability, maintainability, dependability, and security. The results indicate that all aspects recorded Highly Acceptable levels of satisfaction, ranging and an overall average of 4.75, measured using the weighted means of 4.68, 4.78, and 4.75, respectively. These results indicate an overall high level of satisfaction such that the Customized Odoo ERP system is effective in performing its operations and maintaining overall stability and security.

The overall high ratings imply that the users are satisfied with the effectiveness of the ERP system in satisfying their operation requirements, its efficiency in executing their tasks, its reliability in system operations and execution of tasks, its usability and maintainability as well as its ability to safeguard their confidential business data.

This result is indeed in concurrence with the ISO/IEC 25010 software quality model which outlines the four main factors that influence the quality of software: Functional Suitability, Performance Efficiency, Usability, and other features about software like Reliability, Maintainability, and security [4]. The factors listed above are essential in the implementation of the ERP system Peters et al. [2020].

Even though the result is limited to user preferences during the scarce time left during the review, the answer of the reviewer is still encouraging; the situation where really big problems of performance and scalability occur is not mentioned at all.

It is worth mentioning that these results are very important because they show that the ERP system proposed is not only an effective and reliable system but also it has a positive impact on the organizational processes and a high level of user satisfaction and support. Accepting the system at a high level means that it can further improve productivity and decision-making and also lay a good foundation for other steps and the organization's expansion

Notes

[1] Bharadia, R. et al. 2021. Purchase Request Management System. *2021 International Conference on Computational Performance Evaluation, ComPE 2021* (2021), 636641.

[2] Lusiani, S. and Princes, E. 2024. Evaluating the Effectiveness of Mobile JKN Application in Indonesia: A User-Centric Approach Using the ISO 25010 Quality Model. Journal of Logistics, Informatics and Service Science. 11, 10 (2024), 485–500. https://doi.org/10.33168/JLISS.2024.1028.

[3] Peters, E. and Aggrey, G.K. 2020. An ISO 25010 based quality model for ERP systems. Advances in Science, Technology and Engineering Systems. 5, 2 (Apr. 2020), 578–583. https://doi.org/10.25046/aj050272.

[4] Ramiah, S. and Razali, N.F. 2023. Inventory Management Systems (IMS)

[5] Rochimah, S. et al. 2023. INTERNATIONAL JOURNAL ON INFORMATICS VISUALIZATION Software Quality Measurement for Functional Suitability, Performance Efficiency, and Reliability Characteristics Using Analytical Hierarchy Process.

[6] Rojas, H. et al. 2025. INTERNATIONAL JOURNAL ON INFORMATICS VISUALIZATION journal homepage : www.joiv.org/index.php/joiv INTERNATIONAL JOURNAL ON INFORMATICS VISUALIZATION Understanding Performance Efficiency in ISO/IEC 25000: A Systematic Literature Review.

[7] Teixeira, A.R. et al. 2024. Optimization of Business Processes Through BPM Methodology: A Case Study on Data Analysis and Performance Improvement.

Information (Switzerland). 15, 11 (Nov. 2024). https://doi.org/10.3390/info15110724.

[8] Yuniarto, A.A. and Siregar, I.W. 2025. From Manual to Digital: Modernizing the Sales Recording System for Growth at MSME Rumah Abon. Journal of Scientific Insights. 2, 5 (Sep. 2025), 510–525. https://doi.org/10.69930/jsi.v2i5.537.

CHAPTER 5
SUMMARY, FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

This chapter, regarded as the key findings, drew the most important conclusions, and recommended it practically based on the complete implementation and evaluation of an integrated Odoo ERP system at KKBN. The research demonstrated that the expert evaluation and end-users' experiences had a difference which was a way to highlight both the technical advantages and practical difficulties of the system. The findings brought out the possible role of Odoo to revolutionize the way of working and the facts pointing to critical problems with application and hardware during peak times. Thus, the proposals were comprised of the addition of system training to enhance understanding, and modules updates for the purpose of long-term sustainability of ERP and future development.

5.1 Summary

This study illustrates the intervention of centralized Odoo ERP system at KKBN Iriga and Nabua branches to improve operations, precision, and decision-making. It dwells on the configuration, start-up, and evaluation of a single Odoo ERP system designed primarily for the businesses' needs. The major goals are mapping the existing business processes, installing the suggested modules, validating the system based on ISO 25010 criteria, and operating in the physical work environments to demonstrate the way integrated ERP can wipe out manual tasks, faults, and help in the productivity of the business.

The research focuses on six main modules: inventory, sales, accounting, point-of-sale, purchase, and employee management and attendance. It did not examine other components like CRM or marketing, and it was limited to two branches under one ownership, affecting generalizability.

A mixed-methods approach combined qualitative and quantitative methods based on interpretivist and post-positivist philosophies. Interviews, observations, and surveys using Likert scale questionnaires were used to study workflows, system performance, and user satisfaction.

The approach used systematic Business Process Management through process identification, discovery, analysis, redesign, implementation, and tracking. BPMN was used to represent workflows. Internal participants included the owner, employees, and customers, on the other hand, external ERP experts provided technical verification. Interviews, surveys, and Realtime observations using interview guides, questionnaires, checklists, and audio/video recordings were included in the data collection. For the data analysis, thematic interpretation was used to understand the qualitative data, while in weighted mean calculations were used for the quantitative results that were aligned with ISO 25010 and rated using a five-point Likert scale.

The findings show clear differences between how experts and end-users rated the system. IT professionals rated the system high in terms of functionality, performance, compatibility, and ease of use. But on the other hand, the owner and staff are still experiencing issues with the system's reliability during peak hours, data integration, ease of use, and their overall trust in the system.

The showcasing of the Odoo ERP's full potential in managing the functions and offering services to many businesses while also bringing up the issue of obstacles such as those related to user acceptance and system performance is the aim of this research study. To realize the first-mover advantage and maintain the competitive edge, research results state the necessity of the ongoing user training, system optimization, and performance.

5.2 Findings

Based on the data collected and examination in alignment with the study's, Specific Objective. The following findings are presented:

1. The findings of the research conducted by the team indicated that KKBN is still relying on manual techniques and paper-based systems in the areas of Inventory, Sales, Accounting, Point of Sale, Purchase, Employee Management/HR, and Attendance. The recorded data were in the form of papers and were then modeled using Business Process Modeling Notation (BPMN) to show the actual state of the business processes. The BPMN didactic models depicted stepwise workflows. With the help of this inquiry, the weaknesses were recognized such as improper and prolonged stock control procedures, manual sales records that were slow but full of errors, financial statements which were wrong in accounting and inconsistent, a Point of Sale system was those who had not implemented an automated system or human error and longer transaction time, lack of a proper purchase system which ended up with misplaced or duplicate orders, and tracking the performance of employees was limited as the attendance logs were paper based. This was due to the manual processes that were prone to mistakes and delay in the operations.
2. The ERP Odoo Modules configurator encompassed a very fast track of installation and structured personalization that strictly adhered to the business demands. On the contrary, it led to the birth of six various modules which are: Inventory, Sales, Accounting, Point of Sale, Purchase, and Employee Management/HR and Attendance. Every single module was intensely tailored in such a way that they maximize effectiveness and at the same time, they fulfill the special needs of KKBN business

processes. To mention some of the new tools that were integrated into the system were real-time inventory tracking, sales and accounting transaction records' summary, seamless POS transactions, automatic procurement orders, and the functionalities of employee management which were simplified to include attendance monitoring, time tracking, performance tracking, and centralized storage of employee records. The researchers harnessed the full potential of the Odoo 18 Community Edition during the customization and configuration activities.

3. The evaluation of the Odoo ERP system based on the ISO 25010 standard was rated by 10 respondents, which included the owner, employees, 3 IT experts, and 2 ERP experts. The results showed Highly Acceptable with a weighted mean of 4.75 for functional suitability, 4.78 for performance efficiency, 4.74 for reliability, 4.78 for usability, 4.68 for maintainability, and 4.78 for security.

5.3 Conclusions

This conclusion of the study on the implementation of an integrated Odoo ERP system at KKBN the following findings are presented

1. The findings emphasize that to increase operational efficiency, KKBN must think about process automation. inventory, sales, accounting, Point of Sales, purchasing, Employee management/HR and attendance. operations could be monitored in real time, workflows could be streamlined, and errors could be decreased by putting in place an integrated system like Odoo ERP.
2. This study aims to design, and implement, and evaluate an integrated Odoo ERP system to address the problems that KKBN faced in their daily operations that caused by manual processes. The researchers analyzed the existing workflows, and configured

the necessary Odoo ERP modules, and assessed their performance using ISO 25010 standards. Based on the results of the implementation and evaluation of the system, the following conclusions

2.1 Inventory Module:

Using the inventory module, we observed that KKBN was able to monitor stocks in real time, which solved the common issue of shortage and overstocking. The system automatically updated stock levels whenever sales or purchases were made, which removed the need for manual checking. We also noted that the system will be seen immediately when the products are running low, which allowed their business to be restored on time. However, during peak business hours, some employees have had difficulty using the module and there is still a need for proper user training.

2.2 Sales Module:

As for the sales module has been evaluated, the researchers found out that it has simply revolutionized the process of recording transactions by the automatic sales workflow. Previously, transactions were recorded manually, which led to delays and errors. But now with Odoo, the sales can be entered directly in the system, and the reports will be generated automatically, this will allow the owner to have accurate and updated information regarding the sales performance. Our observation revealed that the communication with customers has also been improved because the orders were processed more quickly and reliably. Even though there are these benefits, some of the employees still found the digital systems hard to adjust. Overall, this indicates that the sales module can deliver considerable efficiency gains, but user adaptation is also an important factor in the success of the overall system.

2.3 Accounting Module:

According to what the researchers have discovered, the KKBN accounting module has improved the financial grip of KKBN due to its effect on the process of transaction automation and generating precise financial reports. The problem of manual accounting sometimes finding mismatches is practically difficult to explain or admitting loss of records, and this problem with the Odoo system can be easily neglected. This software not only properly shows cash flows and financial results but also is a helper to the entrepreneur to make a wise decision. However, we witnessed that the efficiency of the accounting module was largely depending on the exactness of the data that is being fed into it. If mistakes were made in the sales or in the purchase entries, they would directly influence the financial statements. This implies that, even though the module is reached, it still needs the users to do regular checking and be disciplined with the data.

2.4 Point of Sales Module:

The POS module is proven to be the main driver for the increase in Customers' transactions. It was the ability to pay using more than one channel, having digital receipts and real-time sales and inventory updates that facilitated this. The module promoted the fast service of digital receipts and reducing the time that customers had to wait in line. Hence, we not only think but are sure that it also raised the level of customer satisfaction. The module addressed the manual receipt issue and also unrecorded deductions in the inventory that KKBN was dealing with previously. However, encountered during the employees' simultaneous operation, system response went down. This represented the issue of under-staffing that can be solved by proper training to assure reliability particularly at peak times.

2.5 Purchase Module:

The researcher according to the analysis the purchase module was the main reason for the procurement to be simplified by the automating of purchase requests, vendor management, and order tracking. It was also the inventory module which makes it easier to find out whether the stock replenishment was done accurately and in a timely manner. This version simplified the manual work of KKBN people dealing with delayed or forgot purchase orders. Moreover, the system also provided a better understanding of how supplier transactions were managed. This enhanced the relationship with the suppliers. On the other hand, we found that when the volume of transactions was high the system also slowed down and extra customization was needed. This means that while the module made the purchasing process much better, the system would be more reliable if it was optimized to a greater extent and if a stronger infrastructure was built.

2.6 Employee Management/HR and Attendance Module:

In the employees and attendance module, the researchers noticed that KKBN has successfully centralized the employees' data and tracked their attendance automatically which in turn has greatly reduced manual timekeeping errors making the records more reliable for management. Along with this, the system has helped the owner to view employee schedules and evaluate their work performance more clearly. In spite of these enhancements, there are still some employees who find it difficult to adapt the new system, simply because of their long-term acquaintance with the old manual process. The module is quite useful for handling employees, but ongoing orientation, and training play a significant role in helping the personnel conform to the new system.

3. The findings bring out the fact that ERP systems should be assessed not only for their characteristics but also for user participation and their durability over time. The main reason is that these factors have a direct impact on the process and thus, on the installation of "green" technology.

5.4 Recommendations

Based on the findings and conclusions, the following suggestions were proposed to enhance the existing system and guide future research:

1. KKBN is required to schedule evaluations of all operational processes on a quarterly basis to maintain synchronization with actual workflow requirements and also to identify new bottlenecks or inefficiencies. This will be the guarantor of system efficiency as the business develops.
2. KKBN should make improvements on their Odoo ERP system to solve the problem of customers being able to order products that are not in stock. This bug renders the system incapable of showing realistic records in the Point-of-Sale (POS) system with the relevance of inventory hence losing them money. The retail store should implement a more stringent inventory tracking system and notify users automatically when an item goes out of stock.
3. KKBN should guarantee reliable internet connection for the Odoo ERP system since it is purely an online platform. They must also think about backup solutions or offline features to keep the operations going even during internet outages; this will ensure business continuity.
4. KKBN should think of the integration of another essential Odoo module that can solve the problem and also, a Customer Relationship Management (CRM) module for the

management of the clients more efficiently and the eCommerce/Online Sales module to add operations. Additionally, according to the assessment results, KKBN needs to generally update the workflows, remedy bugs, improve loading speeds, and increase module reliability this to achieve high performance in their systems based on the ISO 25010 standards.

5. The POS receipt could not be edited as it was using the default configuration template. It is suggested to change receipt templates to Official Receipts (OR) for financial and regulatory compliance.
6. Employees should be required to undertake deeper training so that they will know how to operate the system correctly and avoid making mistakes that will possibly lead to wrong records. Other than this, KKBN could also try to add other useful Odoo modules, like Customer Relationship Management (CRM) and Online Sales, so their operations would be more organized.
7. Based on the assessment results, KKBN has to regularly update workflows, fix errors, optimize load times, and enhance the reliability of the modules to be able to achieve high performance in the system as described in ISO 25010 standards.

BIBLIOGRAPHY

Book Sources

Ahmed, A. & Pereira, L. (2024) Mixed Methods Research: Combining both qualitative and quantitative approaches.

Kc, D. and Hagen, T. (2021) ERP software system comparison between Odoo and Microsoft Dynamics NAV Is Odoo able to replace the existing Microsoft Dynamics NAV core functionalities?

Meissel, K. and Brown, G.T.L. (2023) *Quantitative Research Methods*.

Momani, F., et al. (2022) ERP Implementation Process Improvement: A Methodological Proposals.

Penelitian, J., et al. (2023). Challenges and Strategies for Inventory Management in Small and Medium-Sized Cosmetic Enterprises: A Review.

Ramiah, S. & Razali, N. F. (2023). Inventory Management Systems (IMS).

Sauvant, K. P., et al. (2023) An Inventory of Concrete Measures to Facilitate the Flow of Sustainable FDI: What? Why? How? (Third edition).

Utami, R. S. D. Dyah. (2024) Implementation of Odoo Based on Enterprise Resource Planning System with Sales and Purchasing Module Using Rapid Application Development.

Journal Article

Aal, S. I. A. 2025. Framework for Evaluating Business Processes Modeling Techniques under Neutrosophic Environment. International Journal of Information Engineering and Electronic Business, 17(1), 1–13.

Alaskari, O., et al. 2021. Framework for implementation of enterprise resource planning (ERP) systems in small and medium enterprises (SMEs): A case study. Procedia Manufacturing, 424–430.

Anaya, L. & Qutaishat, F. 2022. ERP systems drive businesses towards growth and sustainability. Procedia Computer Science, 854–861. https://doi.org/10.1016/j.procs.2022.08.103

Almutairi, S. & Shaiti, H. 2020. The impact of enterprise resource planning on business performance: With the discussion on its relationship with open innovation. Journal of Open Innovation: Technology, Market, and Complexity, 6(3). https://doi.org/10.3390/joitmc6030087

Alhayek, W. Y. & Odeh, R. A. A. 2024. Exploring the Impact of ERP Systems on Business Efficiency

Arvianto, A., et al. 2022. ERP ODOO Implementation in Small Retailers. International Journal of Applied Science and Engineering Review, 03(06), 66–85. https://doi.org/10.52267/ijaser.2022.3605

Bharadia, R., et al. 2021. Purchase Request Management System. 2021 International Conference on Computational Performance Evaluation (ComPE 2021), 636–641. DOI: 10.1109/ComPE53109.2021.9751825

Fatoni Efendi, H. & Aditya, A. 2022. Business Process Analysis and Implementation of Odoo Open Source ERP System in Inventory, Purchasing and Sales Activities (Case Study: Captain Gadget Store).

Ibrahim, M. F., et al. 2024. A literature review on ERP implementation: Methodologies, module, software, and policy. AIP Conference Proceedings. https://doi.org/10.1063/5.0192339

Ian Pabustan, S. D., et al. 2021. Analyzing the Impact of ERP Application to the Business Operations in SIA Engineering (Philippines) Corporation.

Ivanović, T. & Marić, M. 2021). Application of modern Enterprise Resource Planning (ERP) systems in the era of digital transformation. Strategic Management, 26(4), 28–36. DOI: 10.5937/StraMan2104028I

Lubis, M., et al. (2021). Enterprise Resource Planning (ERP) System Customization with the Inventory Management Module: Case Study of Rumah Yatim. ACM International Conference Proceeding Series, 7–12. https://doi.org/10.1145/3462676.3462678

Lusiani, S. and Princes, E. 2024. Evaluating the Effectiveness of Mobile JKN Application in Indonesia: A User-Centric Approach Using the ISO 25010 Quality Model. Journal of Logistics, Informatics and Service Science. 11, 10 (2024), 485–500.

Mansour, K., et al. (2022). Novel Integrated Framework for ERP Selection and Implementation. Journal Europeen des Systemes Automatises, 55(6), 787–792. https://doi.org/10.18280/jesa.550611.

M. Teja Sri, et al. (2024). Integration of ERP in Optimizing Business Process. International Research Journal on Advanced Engineering and Management (2(05), 1584–1587. DOI: 10.47392/IRJAEM.2024.0216

Setiawan, D., et al. (2024). A Proposed Framework for ERP System Implementation in SMEs. International Journal of Artificial Intelligence Research, https://doi.org/10.29099/ijair.v7i2.1102.

Pandey, A. K., et al. (2022). Examining the Role of Enterprise Resource Planning (ERP) in Improving Business Operations in Companies. ECS Transactions, 107(1), 2681–2696.

Peters, E. and Aggrey, G.K. 2020. An ISO 25010 based quality model for ERP systems. Advances in Science, Technology and Engineering Systems. 5, 2 (Apr. 2020), 578–583. https://doi.org/10.25046/aj050272.

Panduwiyasa, H., et al. (2021). Accounting and Smart System: Functional Evaluation of ISO/IEC 25010:2011 Quality Model (a Case Study). IOP Conference Series: Materials Science and Engineering, 1092(1), 012065.

Peters, E. & Aggrey, G. K. (2020). An ISO 25010 based quality model for ERP systems. Advances in Science, Technology and Engineering Systems, 5(2), 578–583. https://doi.org/10.25046/aj050272.

Putra, D. G., et al. (2021). The Influence of ERP Implementation System on Company Performance Mediated by Organizational Capabilities. Journal of Accounting and Investment, 22(2), 221–241.

Qowindra, M. R. G. & Wiratama, J. (2023). Development of ERP using RAD Method for the Garment Industry in Indonesia. G-Tech: Jurnal Teknologi Terapan, 7(2), 504–513. https://doi.org/10.33379/gtech.v7i2.2296.

Oranga, J. & Matere, A. (2023). Qualitative Research: Essence, Types and Advantages. OALib, 10(12), 1–9. https://doi.org/10.4236/oalib.1111001.

Rojas, H. et al. 2025. INTERNATIONAL JOURNAL ON INFORMATICS VISUALIZATION Understanding Performance Efficiency in ISO/IEC 25000: A Systematic Literature Review.
Rochimah, S. et al. 2023. VISUALIZATION INTERNATIONAL JOURNAL ON INFORMATICS VISUALIZATION Software Quality Measurement for Functional Suitability, Performance Efficiency, and Reliability Characteristics Using Analytical Hierarchy Process.

Syamsuddin, S., et al. (2023). Implementation of an Enterprise Resource Planning (ERP)System and its Impact on Manufacturing Company Operational Efficiency. Global International Journal of Innovative Research, 1(2), 194–199.

Teixeira, A. R., et al. (2024). Optimization of Business Processes Through BPM Methodology: A Case Study on Data Analysis and Performance Improvement. Information (Switzerland), 15(11). https://doi.org/10.3390/info15110724.

Tongsuksai, S., et al. (2021). Critical success factors and challenges for cloud ERP system implementations in SMEs: A vendors' perspective. IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE 2021).

Utami, M. P. & Kharisma, I. L. Implementation of Enterprise Resource Planning (ERP) Based Information System Using Odoo Software. The Eastasouth Journal of Information System and Computer Science, 1(02), 55–64.

Wu, J. Y. & Chen, L. T. (2020). Odoo ERP with Business Intelligence Tool for a Small Medium Enterprise: A Scenario Case Study. ACM International Conference Proceeding Series, 323–327.

Yoo, B. K. & Kim, S. H. (2021). A Suggestion for ERP Software Customization Model Using Module Modification Factors. Lecture Notes in Electrical Engineering, 563–568.

Yosevine, P., et al. (2021). Enterprise Resource Planning (ERP) Evaluation and Implementation: A Case Study. Journal of Multidisciplinary Issues, 1(1), 49–66.

Yuniarto, A. A. & Siregar, I. W. (2025). From Manual to Digital: Modernizing the Sales Recording System for Growth at MSME Rumah Abon. Journal of Scientific Insights, 2(5), 510–525.

Zahra, S. L. Z., et al. (2023). Implementation of Odoo-Based ERP in The Case Study of Micro, Small, and Medium Enterprises “Woody Moody Jakarta.” Instamatics, 3(2), 68–77.

APPENDICES

APPENDIX A

EVALUATION TOOL

INTERVIEW GUIDE

Dear: Ma'am/Sir

Good day! We hope this message finds you well. We are **Team Aport**, a third-year Colleges Student under the program of **Bachelor of Science in Information System.** We are currently conducting a study entitled "Digital Transformation of KKBN Business Process using ODOO: A Capstone Project".

This interview aims to gather insights about Kalusugan, Kayamanan, and Basta Natural (KKBN) current business processes to identify areas that can benefit from Customization and configure using Odoo ERP. All information you share will be kept confidential and used solely for academic purposes. With your permission, we may take notes and, if allowed, record this interview for accuracy. Your participation is highly appreciated and will play a crucial role in the success of our project.

Respectfully yours,

TEAM APORT

John Edward Hontalba

Sharrycel Magno

Gerald Buergo

INTERVIEW QUESTIONS

GENERAL QUESTIONS:

1. What are the process under the Kalusugan, Kayaman, Basta Natural (KKBN) product? Kindly explain each below.
2. Which operational inefficiencies at Kalusugan, Kayamanan, and Batsa Natural Product could be resolved with the use of Odoo modules?
3. Are there documented process flows or procedures currently in place in (Kalusugan,Kayamanan,Basta Natural) Product? If yes, could you provide a copy of these documents? If not, describe the process flows either verbally or in written form.

CLOSING:

Thank you much for your time and valuable insights. Your contribution is important to the success of our project. We assure you that all information gathered during this interview will remain confidential and used only for academic purposes. Before we end, may we kindly ask:

Would you be open to answering follow-up questions if we need additional clarification?

Again, thank you for your cooperation and support.

OBSERVATION CHECKLIST

Observation Checklist for Employee Management, Sales, Inventory, and Purchase

Location: KKBN Iriga Branch

Date of Observation: _____________

Observer: ______________________

Section A: Employee Management Module

Purpose: To evaluate the effectiveness of the Odoo Employee Management module in managing employee data and processes.

Observation Indicator	Yes	No	Not Applicable	Remark
Accurate employee data maintained				
Efficient management of employee leaves and absences				
Effective performance management tools available				
Streamlined recruitment and onboarding processes				
Easy access to employee information for managers				

Section B: Sales Module

Purpose: To assess the efficiency and accuracy of the Odoo Sales module in managing sales processes.

Observation Indicator	Yes	No	Not Applicable	Remark

Accurate recording of sales orders and invoices				
Efficient management of sales opportunities				
Timely generation of sales reports				
Integration with other modules (e.g., Inventory)				
Accurate tracking of sales performance (KPIs)				

Section C: Inventory Module

Purpose: To assess the accuracy and efficiency of inventory management using the Odoo Inventory module.

Observation Indicator	Yes	No	Not Applicable	Remark
Accurate stock levels maintained				
Efficient stock management and tracking				
Timely generation of inventory reports				
Integration with sales and purchasing modules				
Effective management of low-stock alerts				

Section D: Purchase Module

Purpose: To evaluate the efficiency and accuracy of the Odoo Purchasing module in managing procurement processes.

Observation Indicator	Yes	No	Not Applicable	Remark
Efficient management of purchase orders				
Accurate tracking of purchase invoices				
Timely generation of purchasing reports				
Integration with other modules (e.g., Inventory)				
Effective management of supplier information				

Section E: General Notes / Other Observations

Use this space to document any other observations relevant to the study that are not covered in the checklist.

____

____

____

SURVEY QUESTIONS

Dear Respondents,

We are Team Aport, a 4th year Colleges Student under the program of Bachelor of Science in Information System. We are currently conducting a study entitled "Digital Transformation of KKBN Business Process using ODOO: A Capstone Project".

This survey aims to gather feedback on the effectiveness of the newly implemented Odoo ERP system at Kalusugan, Kayamanan, at Basta Natural (KKBN), to evaluate improvements in business processes and identify any areas that may require further customization or configuration. All information you share will be kept confidential and used solely for academic purposes.

Your participation is highly appreciated and will play a crucial role in the success of our project.

Sincerely Yours,

Hontalba, John Edward

Magno, Sharrycel

Buergo, Gerald

Researchers

I. Respondent Profile. Please provide the requested details below.

Name: ________________________ Sex: Male ☐ Female ☐

Age: __________ Position: ________________________

5-POINT LIKERT SCALE

Scale	Level of Acceptability	Description
5	Highly Acceptable	The design is outstanding and exceeds expectations. It fully meets or surpasses all requirements and standard, leaving little room for improvement. It is regarded as highly favorable and is strongly endorsed by users or stakeholders.
4	Moderately Acceptable	The design is above average and satisfies most expectations or standards. It is more than simply acceptable, with notable strengths that make it well-regarded. There may be a few minor areas for improvement, but overall, it performs well.
3	Acceptable	The design meets the minimum requirements and standards. While there may be some minor areas that could be improved, it is generally deemed satisfactory and suitable for its intended purpose.
2	Slightly Not Acceptable	The design is close to meeting the required standards but still falls short in some important areas. There are clear deficiencies, though they are less severe than a fully unacceptable rating. Improvement is needed before it can be considered acceptable.
1	Not Acceptable	The design is completely unsatisfactory. It does not meet basic standards or expectations, and significant changes or improvements are necessary. It is strongly rejected by users or stakeholders.

Functional Suitability	5	4	3	2	1
The degree to which all necessary business operations and functions are covered by the ERP solution.					
The ERP solution's capacity to generate dependable and accurate outputs from the input data. 3. how well the ERP system satisfies the user's particular functional needs.					
How well the ERP system satisfies the user's particular functional needs.					
the ERP system's capacity to automate repetitive processes and operations.					

The ERP solution's capacity to easily share and integrate data across many modules.					
Performance Efficiency	**5**	**4**	**3**	**2**	**1**
The ERP solution eliminates the need for conventional manual paperwork, enabling users to accelerate their document processes and reduce their reliance on hard-copy forms.					
The ERP solution can respond to user activities rapidly and efficiently, ensuring smooth and uninterrupted operations while systems utilize.					
The ERP solution reduces transaction delays, enhances user experience, and accelerates routine operations as a whole.					
The ERP solution increases the system's efficiency compared to the previous one and allows the users to notice apparent boosts in the productivity of operations.					
The ERP solution can maintain a high level continuously when facing different levels of workload and assure dependable operation at all times.					
Reliability	**5**	**4**	**3**	**2**	**1**
The degree to which the ERP solution keeps working even when software defects occur, enabling users to continue being productive in spite of unforeseen problems.					
To ensure integrity of data and reduce potential loss, the ERP solution can recover and restore data after disruptions or failures.					
The ERP is available at all times to the users to enable access for critical processes on a continuous basis.					
The operations of the ERP system are continuously executed without unexpected deviation to provide users with providing a reliable and steady experience.					
The capability of the ERP solution to perform regular maintenance and upgrades without disturbing the current operations.					

Usability	**5**	**4**	**3**	**2**	**1**
Because of the ERP solution's ease of use, users may effectively do daily activities with little effort.					
The ERP solution is user-friendly and simple to use and comprehend without requiring a lot of training.					
Users can accomplish their goals effectively and satisfactorily with the help of the ERP solution, enabling productive results of the work.					
The ERP system's capacity to recognize and fix mistakes on its own, reducing the need for users to perform manual troubleshooting					
The user may easily learn the ERP solution object to have more comprehension.					
Maintainability	**5**	**4**	**3**	**2**	**1**
The ERP solution's flexibility as business demands changes and its simplicity of customization and configuration to match particular business requirements.					
The ERP system's capacity to reduce the amount of time spent on error correction by enabling users to swiftly view and fix any errors.					
The ERP solution's ability to run without hiccups and guarantees dependable performance while in use.					
Users may manage the ERP solution with ease.					
The ERP solution reduces interruptions to corporate operations and is made for rapid and effective maintenance or updates.					
Security	**5**	**4**	**3**	**2**	**1**
Sensitive data is shielded from unwanted access while being kept secure and private within the ERP system.					
The ERP solution provides a secure environment for all users while protecting the data from a variety of dangers.					

The ERP system is strengthened to stop data loss while giving users assurance about the security of their data.					
The ERP system can stop unapproved changes to data and regulate access.					
The ERP solution's database is safe and protected from unwanted access.					

Comments/ Suggestion:

____

____

THANK YOU!!

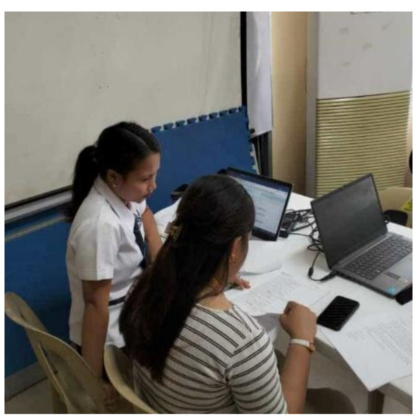
APPENDIX B

DOCUMENTATIONS

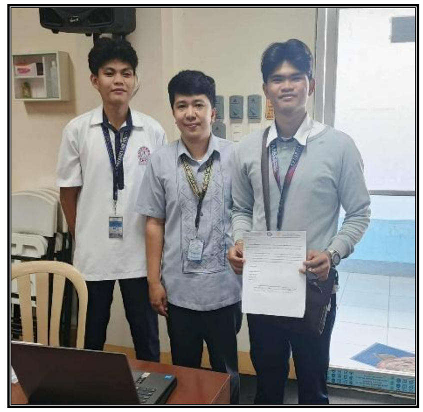
The Image captured during the interview conducted by the researchers to the Owner, using interview guide and observation checklist to thoroughly identify the objectives of this study and examine the current state of KKBN Business Processes.

The image was taken during the data gathering process with the owner, secretary/cashier, and saleslady to collect all necessary data such as inventory records, product photos, product sales reports, financial records, and owner and employee information. This activity was conducted to ensure the accuracy and completeness of the information needed for system development.

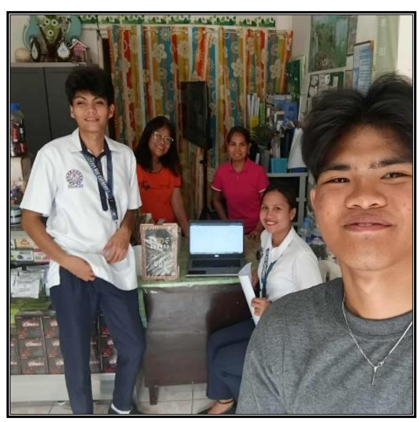
These images were taken during the demonstration of the new system like Odoo ERP and to assess the new system in terms of ISO 25010 with six components to the owner, secretary/cashier, and sales lady in KKBN.



The image above captured during the demonstration of the Odoo modules customized and configured by the researchers to the 3 IT Experts and assess the survey questionnaire prepared by the researchers.



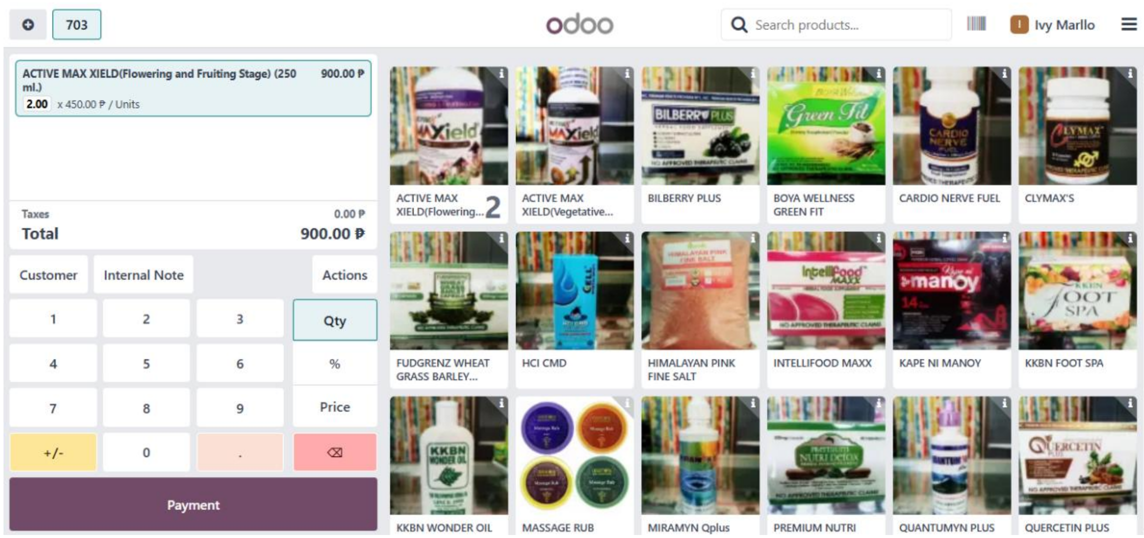
These images taken during the of the Odoo modules customized and configured by the researchers to the 2 ERP Experts.

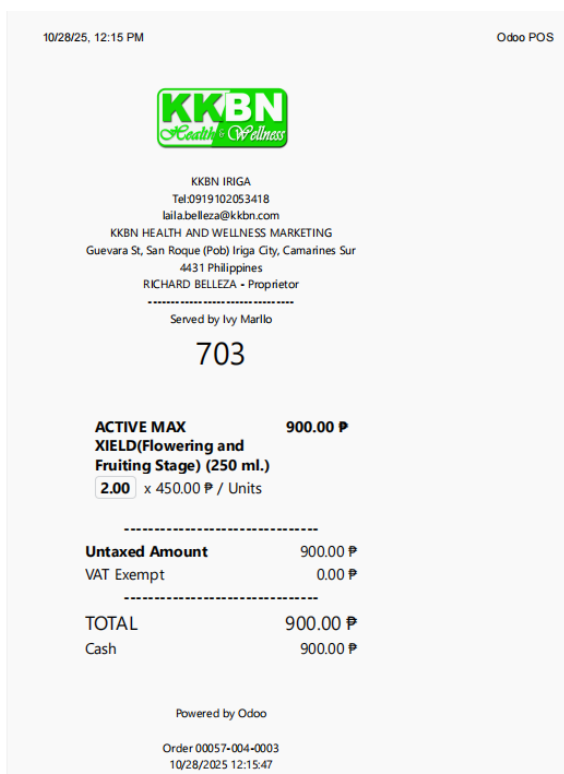


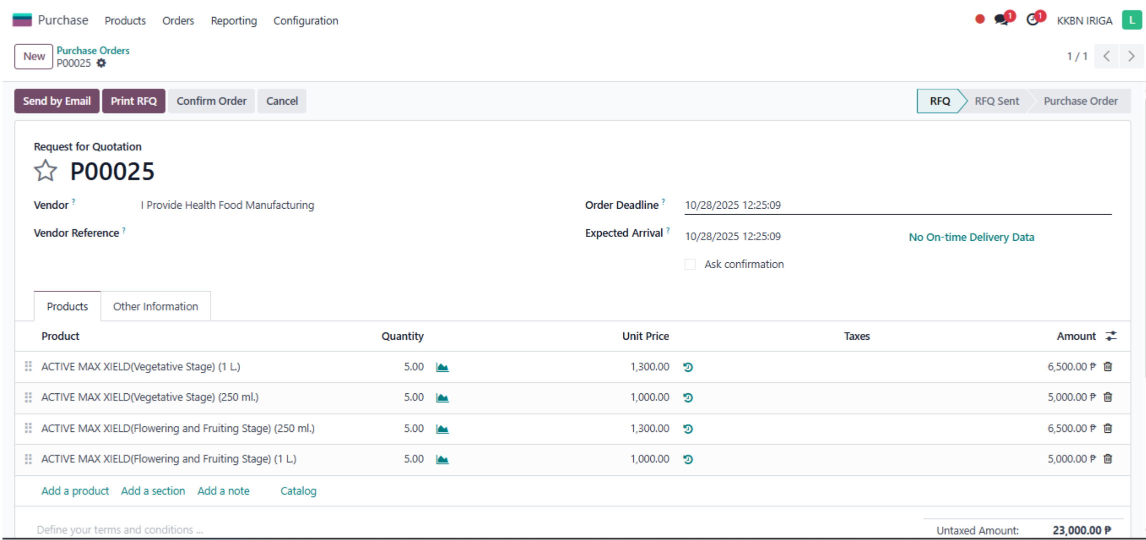
The image above is the implementation of the Odoo ERP Software in KKBN.

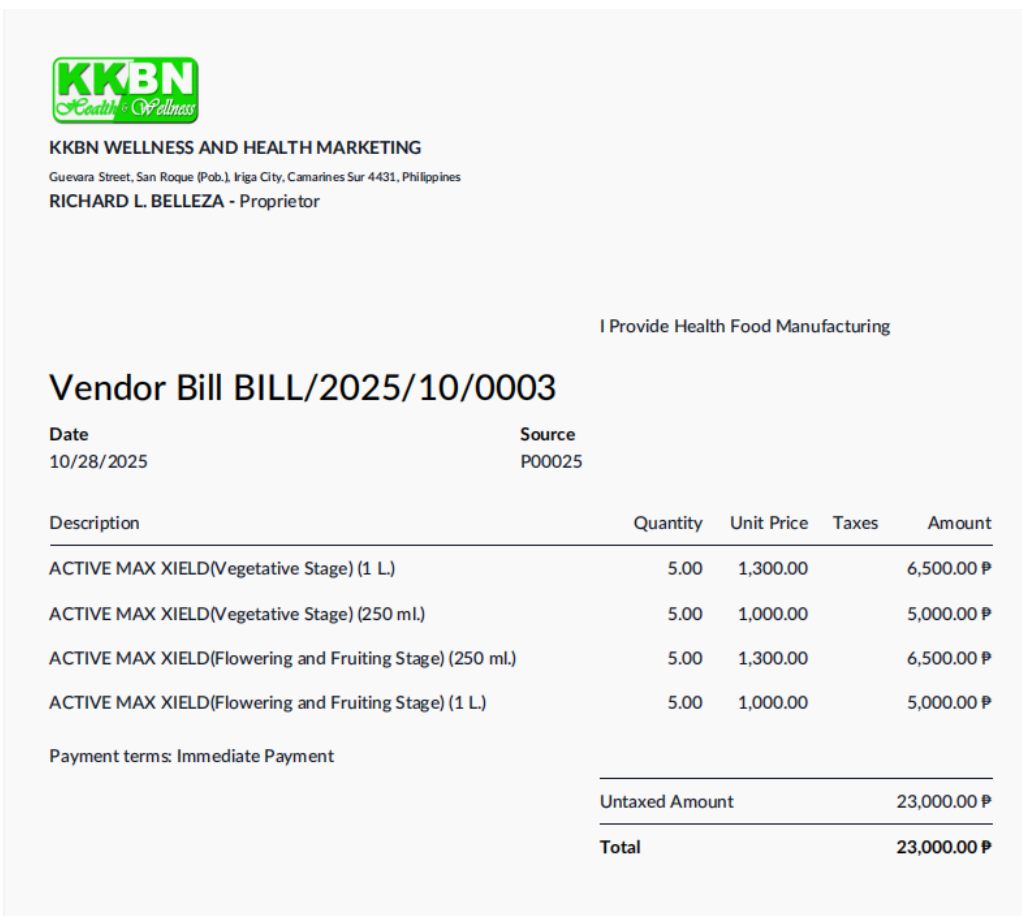
APPENDIX C

SAMPLE INPUT/OUTPUT REPORT









KKBN WELLNESS AND HEALTH MARKETING
Guevara Street, San Roque (Pob.), Iriga City, Camarines Sur 4431, Philippines
RICHARD L. BELLEZA - Proprietor

I Provide Health Food Manufacturing

Vendor Bill BILL/2025/10/0003

Date	Source
10/28/2025	P00025

Description	Quantity	Unit Price	Taxes	Amount
ACTIVE MAX XIELD(Vegetative Stage) (1 L.)	5.00	1,300.00		6,500.00 ₱
ACTIVE MAX XIELD(Vegetative Stage) (250 ml.)	5.00	1,000.00		5,000.00 ₱
ACTIVE MAX XIELD(Flowering and Fruiting Stage) (250 ml.)	5.00	1,300.00		6,500.00 ₱
ACTIVE MAX XIELD(Flowering and Fruiting Stage) (1 L.)	5.00	1,000.00		5,000.00 ₱

Payment terms: Immediate Payment

Untaxed Amount	23,000.00 ₱
Total	**23,000.00 ₱**

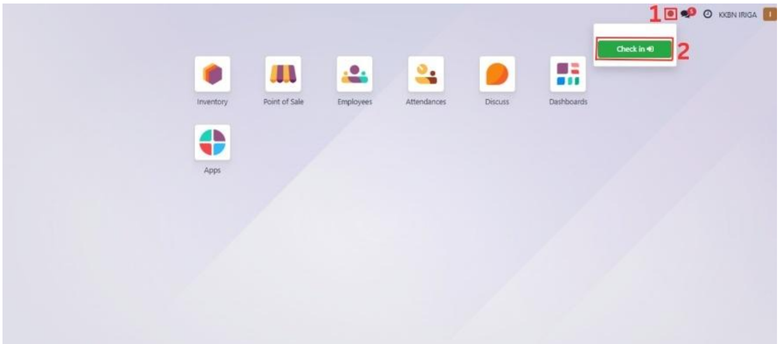
APPENDIX D

USER GUIDE

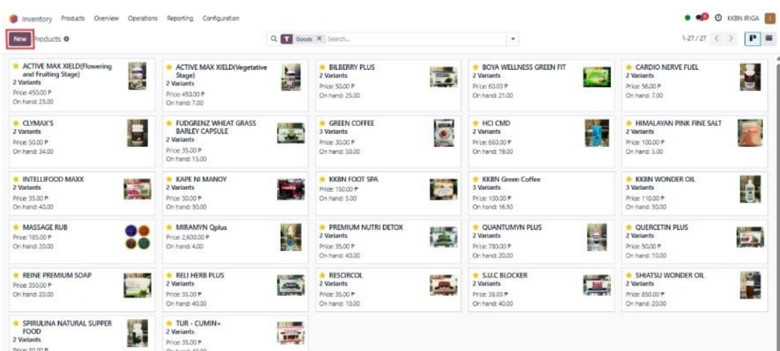
INTRODUCTION

Welcome to the odoo user manua l, focusing on the development and implementation of an Odoo ERP System for Kayamana Kalusugan Basta Natural, to enhance and automate the current business processes. The project shall help resolve inefficiencies developed from manual operations regarding sales, inventory, accounting, purchasing, and employee management. Such integration into one system enables the organization to realize better accuracy, faster transactions, and wiser decision-making. This study shall serve as a detailed guide to understand how the Odoo ERP system will help manage and facilitate the smooth conduct of business processes and, thus, will enable business growth. Ultimately, this project is expected to contribute to the enhancement of KKBN's overall productivity and operational efficiency through novel information system solutions.

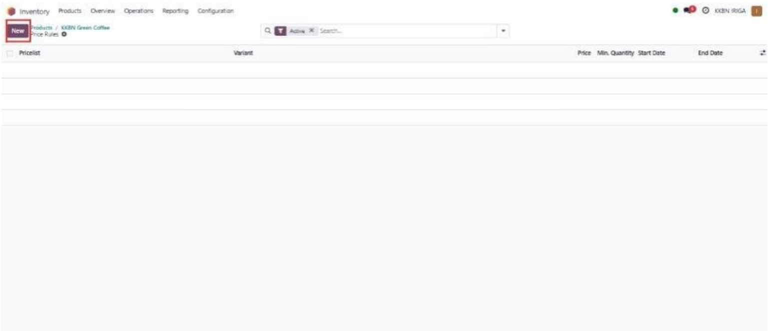
Step 1. Enter your email
Step 2. Enter your PASSWORD
Step 3. Click the LOG IN button to Access your account



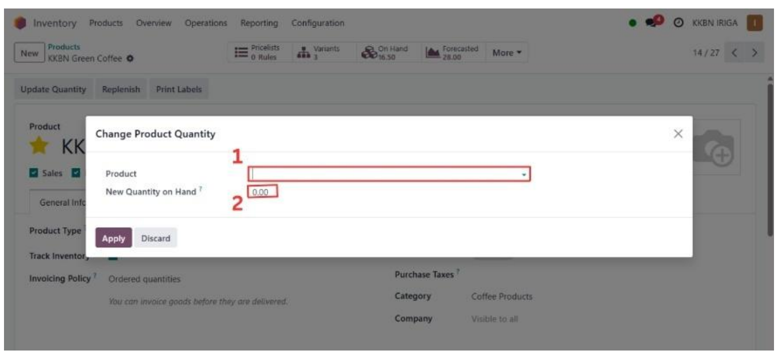
Step 1. look for the "red button" at the top right of the screen
Step 2. Click the button to "CHECK IN" to record the attendance



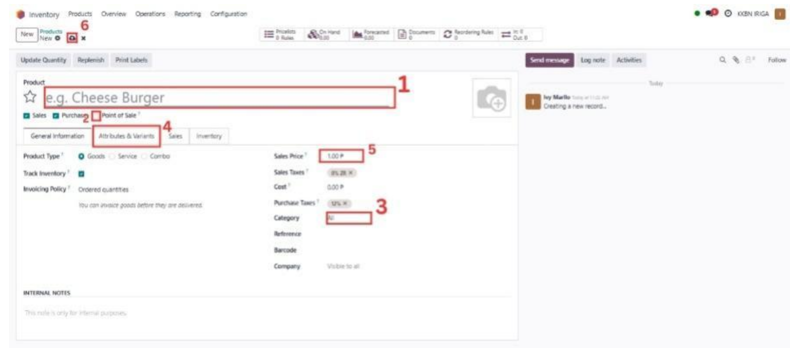
Inventory Module
Step 1. By clicking in "NEW" button, you can now create or add a new product in your inventory.



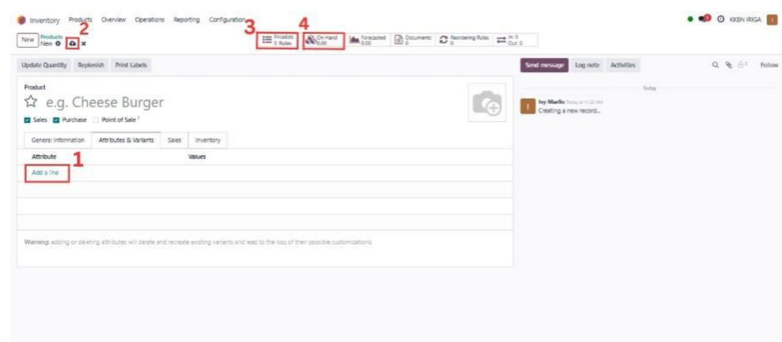
by clicking on "NEW" you will see if the product has a discount or not



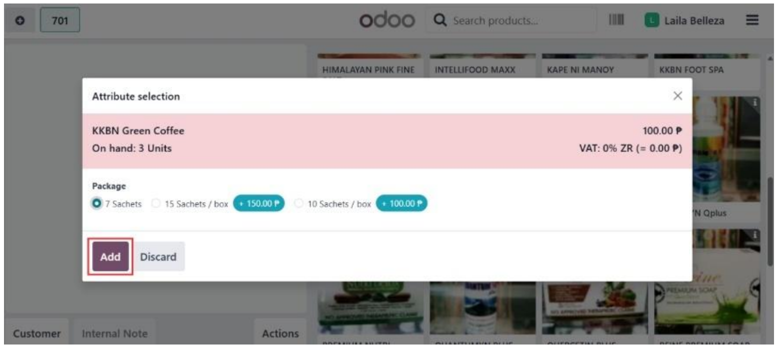
Step 1. Choose the specific product you want to update the quantity for from the list
Step 2. Enter the total updated quantity of the product



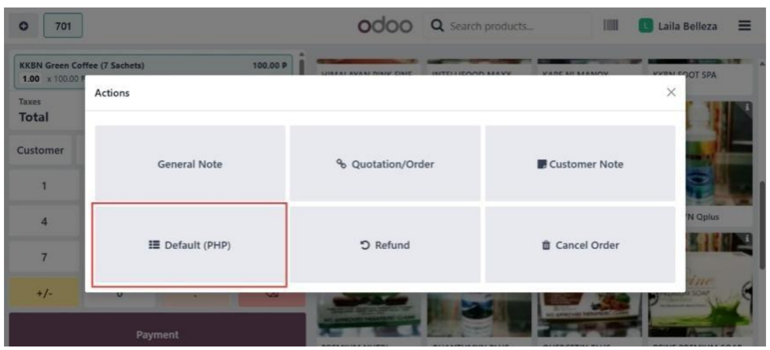
Step 1. Name of product to be created or added.
Step 2. select the Point of sales
Step 3. set the appropriate purchase taxes and category for the product
Step 4. Configure " Attributes & Variants" if the product has different options
Step 5. Enter the price which product will be sold
Step 6. Click Save



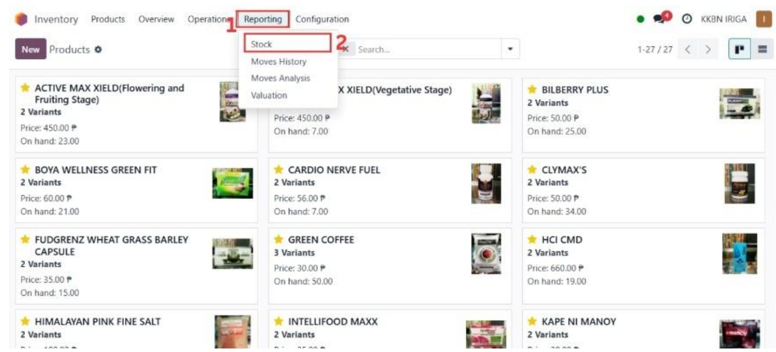
Step 1. Click a Add a line to add a new attribute for the product.
Step 2. Click Save.
Step 3. click the "pricelists & Rules" to manage different price rules for the product.
Step 4. Click the "ON HAND" to show the current quantity of the product in stock.



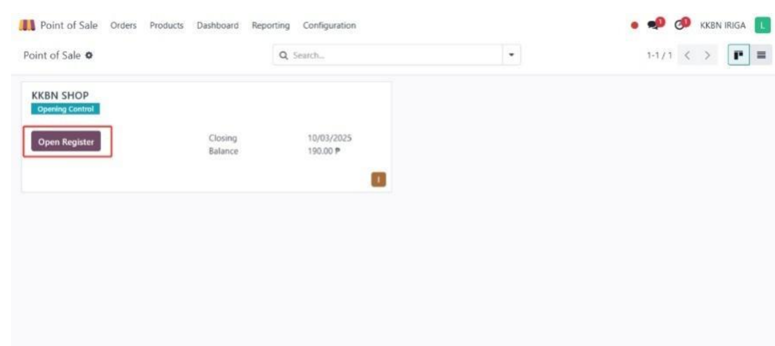
by Clicking on "ADD" button to include the selected item to the current transaction the one selected is the 7sachets



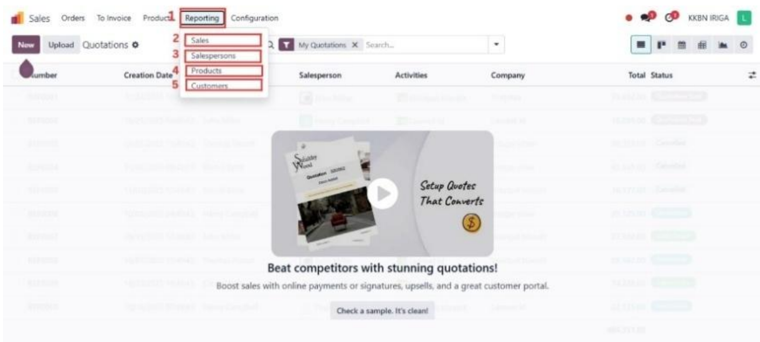
Selected the "DEFAULT PHP" you are setting the currency for the transaction to the product. Choose DEFAULT if product no discount, and DISCOUNTED PRICE if product have discount.



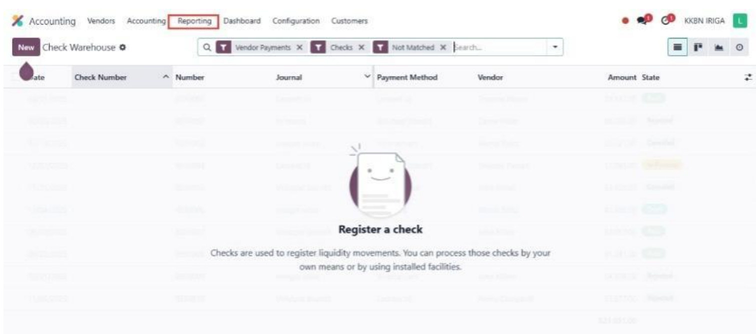
Step 1. Click on "REPORTING" to access various inventory report.
Step 2. Select "STOCK" to view report related to your current Stock level.



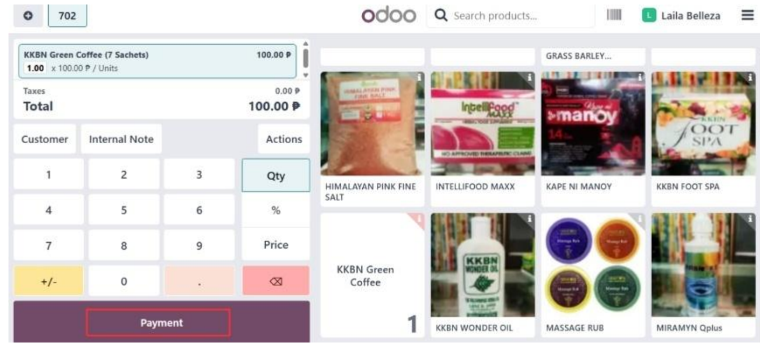
POS Module
By clicking on OPEN THE REGISTER, you will see the products that the customer will purchase. When the



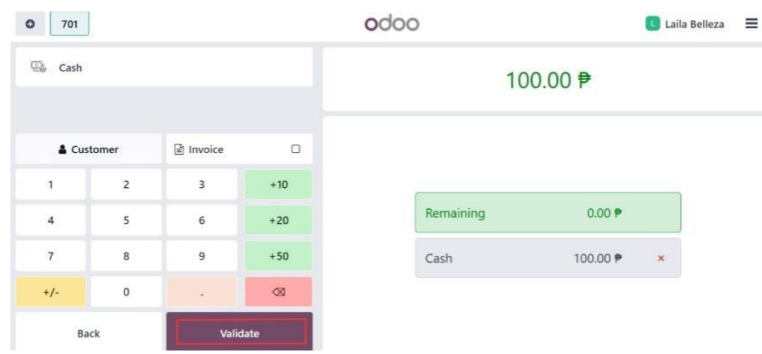
Sales Module
Step 1. Click "REPORTING" to access a different sales report.
Step 2. Choose a "SALES" to view overall sales performance.
Step 3. Choose a "SALESPERSONS" to see the reports base on individual salesperson
Step 4. Choose a "PRODUCT" to generate the reports for the specific product
Step 5. Choose a "CUSTOMER" to generate report base on customer data



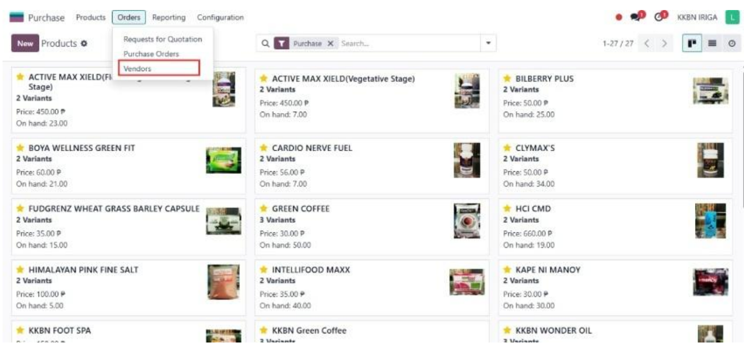
Accounting Module
By clicking reporting this is to access various financial reports related to the accounting data.



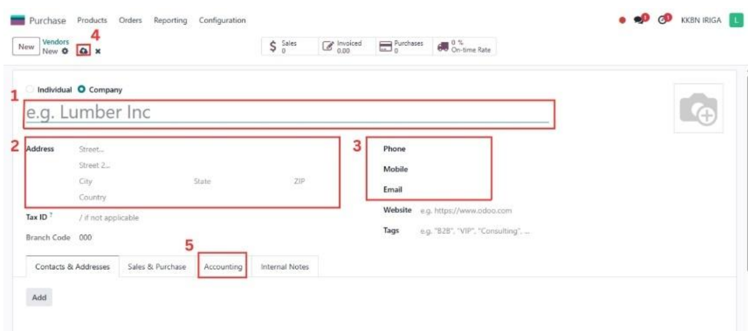
By Clicking on payment so you can pay for the product you selected.



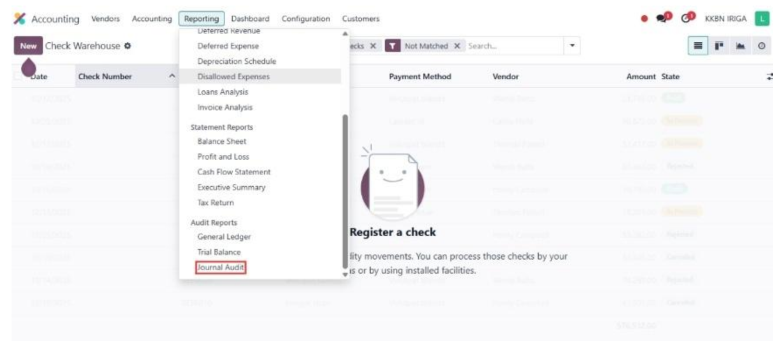
By clicking the "VALIDATE" button to finalize your transaction and to complete the sale



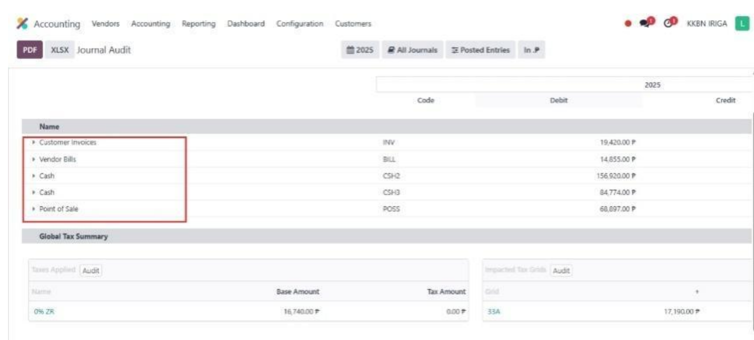
Purchase Module
Choose "Vendors" to manage and view a list of your suppliers. You can add new vendors, edit current ones, and view their contact details and purchase history.



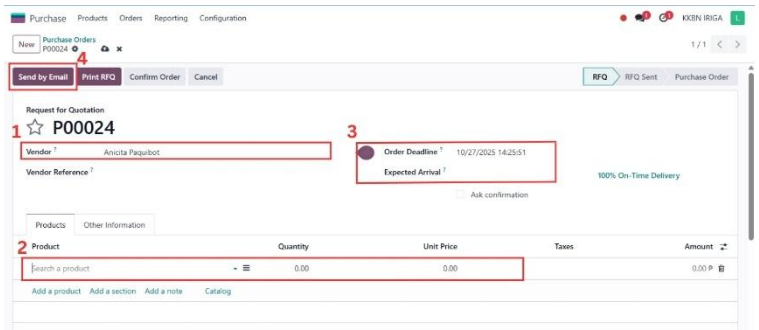
Step 1. Input the name of the vendor here.
Step 2. fill in the vendor's address details, including street, city, state, zip code, and country.
Step 3. Enter the vendor's contact information, such as phone number mobile number, and email address.
Step 4. Click Save.
Step 5. Click the Accounting to add accounting details



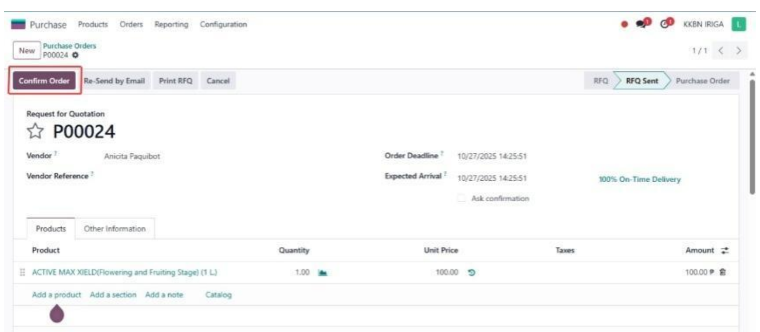
By clicking on "Journal Audit," you can view an in-depth report that allows you to review and authenticate all the accounting entries posted in your journals. This is needed to maintain accuracy and compliance in your books of account.



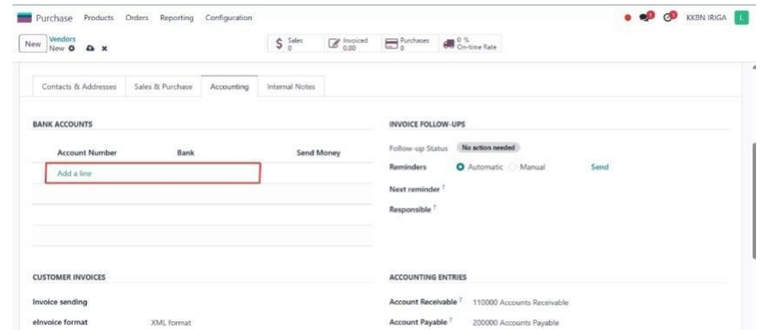
CUSTOMER INVOICES details of invoices issued to customer VENDOR BILLS records of bills received from the vendor, CASH transactions involving cash, POINT OF SALES records from point of sales transaction.



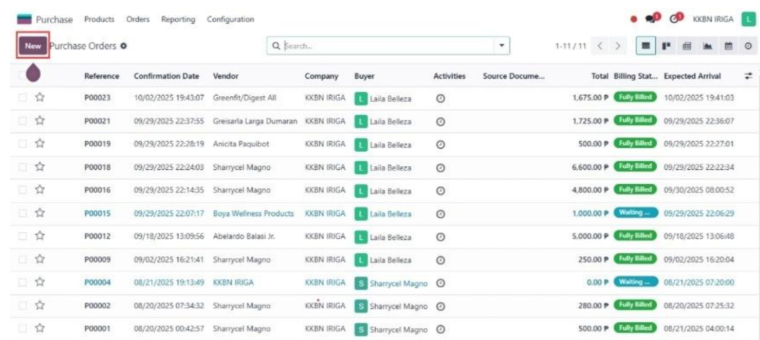
Step 1. Confirm vendor contact details and payment terms before sending the RFQ.
Step 2. clicking add a product where you add items you want to buy. For each line you select a product, enter quantity, confirm unit price, choose taxes, and the line total is calculated.
Step 3. scheduling information for the order: an Order Deadline or expected delivery date
Step 4. Clicking send by emails the RFQ (or PO) to the vendor using the vendor contact email.



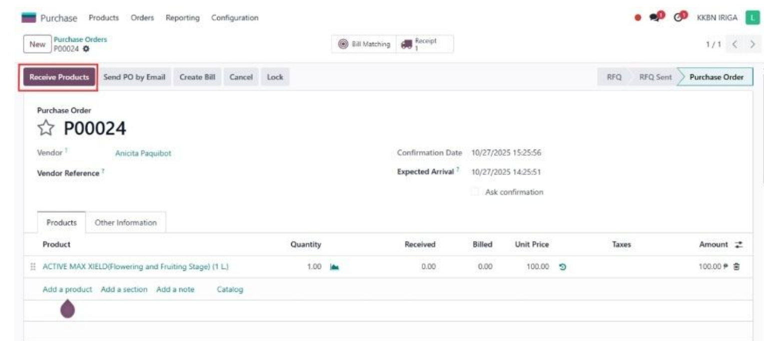
Converts the Request for Quotation (RFQ) into an official **Purchase Order**.



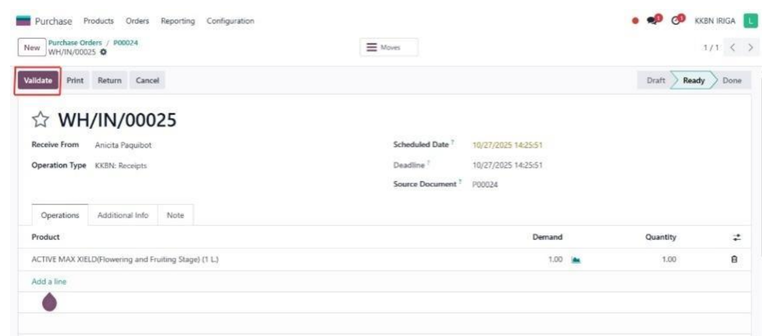
Clicking a "Add a line" to input the vendor bank account details, including the account number and bank name.



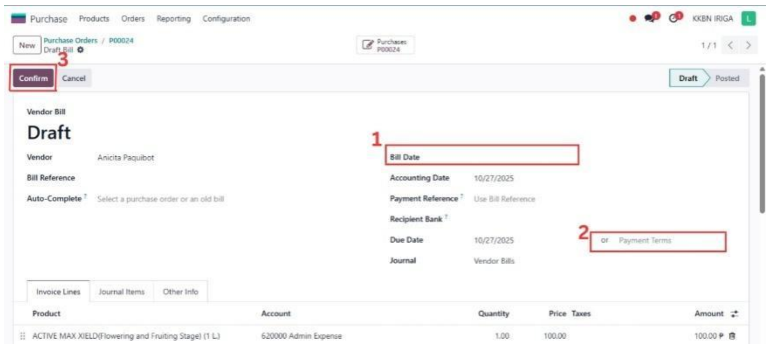
"NEW" Now the transaction belongs to the list of all Purchase order in sales module.



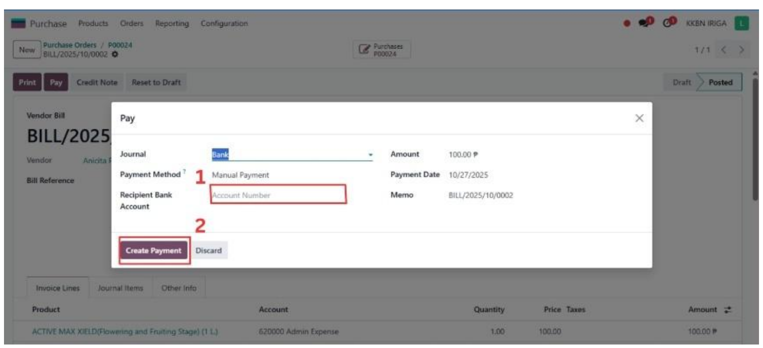
Clicking a "Receive Products" button to confirm the products have been received.



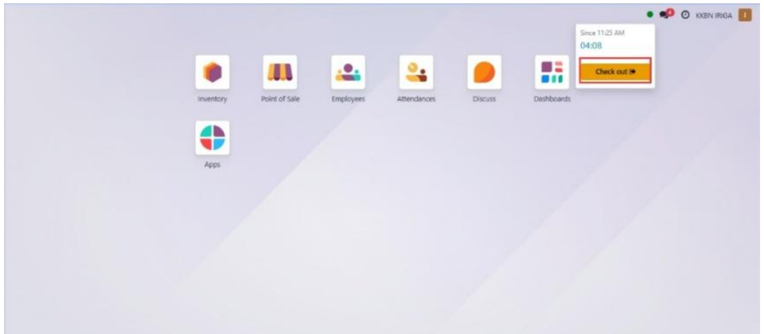
Clicking a "VALIDATE" to confirm the receipt of the product



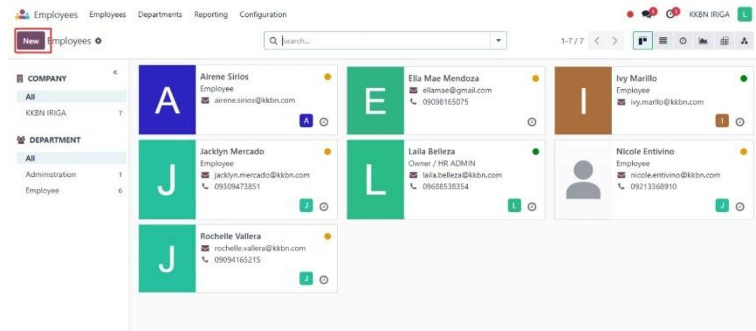
Step 1. Input the date of the bill
Step 2. set the payment terms of due date
Step 3. Click Confirm.



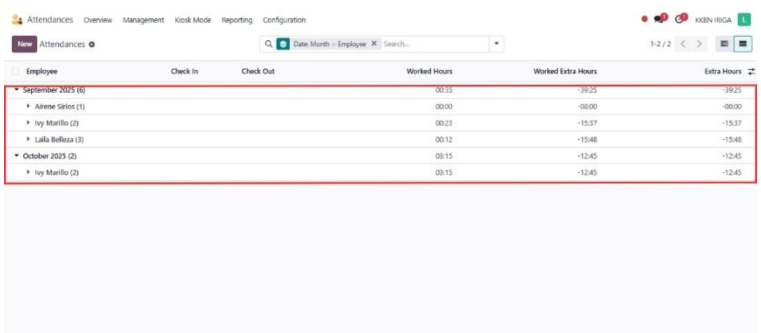
Step 1. Enter the recipient bank account number.
Step 2. Click the confirm payment to create the payment.



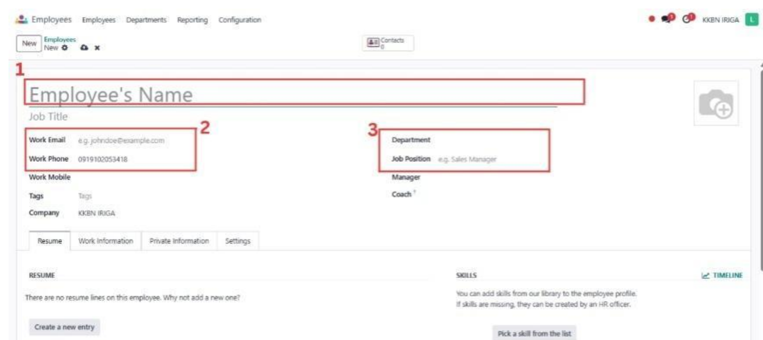
clicking a red button to "CHECK OUT" to record the end of your work period



clicking "NEW" to add a new employee



This column lists the names of employees who have attendance records.
Shows the exact time and date when an employee started their work or logged in.
Displays the time and date when the employee ended



Step 1. Enter the name of employee.
Step 2. Enter the employee work email address phone number.
Step 3. Select and set a department of employee belongs to their job

APPENDIX E

NON-DISCLOSURE AGREEMENT FORM

Non-Disclosure Agreement Form

EFFECTIVE DATE: Feb 26, 2025
(*TD submission date*)

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and **JOSEFINA H. LIAGAS** hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
   JOSEFINA H. LIAGAS
   Name and Expert

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
   MS. ROSEL O. ONESA, MIT
   Name and Expert

2. The confidential information disclosed under this Agreement is described as:
   Contents of the TCP by:
   Team Aport
   John Edward Hontalba, Gerald Hontalba, Sharrycel Magno

   which is entitled:

   Digital Transformation of KKBN Business Processes using ODOO: A Capstone Project

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the *effective date and one year following the date of the TCP submission.*
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
   a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
   b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:

a. which is or becomes publicly available without breach of this Agreement;
b. which is already known to the Recipient prior to its disclosure hereunder;
c. which is independently developed by the Expert.

8. The parties acknowledge that any technology, product or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:	CAMARINES SUR POLYTECHNIC COLLEGES
JOSEFINA H. LIAGAS **Expert** (Print name and Signature)	**MS. ROSEL O. ONESA, MIT** **CSPC Representative** (Print name and Signature)
DATE: Feb 26, 2025	DATE: FEB 28 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: Feb 26, 2025
(*TD submission date*)

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and **JEREMY JIREH S. NEO** hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
   JEREMY JIREH S. NEO
   Name and Expert

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
   MS. ROSEL O. ONESA, MIT
   Name and Expert

2. The confidential information disclosed under this Agreement is described as:
   Contents of the TCP by:

   Team Aport
   John Edward Hontalba, Gerald Hontalba, Sharrycel Magno

   which is entitled:
   Digital Transformation of KKBN Business Processes using ODOO: A Capstone Project

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
   a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
   b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:

a. which is or becomes publicly available without breach of this Agreement;
b. which is already known to the Recipient prior to its disclosure hereunder;
c. which is independently developed by the Expert.

8. The parties acknowledge that any technology, product or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:	CAMARINES SUR POLYTECHNIC COLLEGES
JEREMY JIREH S. NEO	**MS. ROSEL O. ONESA, MIT**
Expert	**CSPC Representative**
(Print name and Signature)	(Print name and Signature)
DATE: Feb 26, 2025	DATE FEB 28 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: Feb 26, 2025
(TD submission date)

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and **ANGELIE ELLA L. LAYNESA** hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
   MS. ANGELIE ELLA L. LAYNESA
   Name and Expert

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
   MS. ROSEL O. ONESA, MIT
   Name and Expert

2. The confidential information disclosed under this Agreement is described as:
   Contents of the TCP by:
   Team Aport
   John Edward Hontalba, Gerald Hontalba, Sharrycel Magno

   which is entitled:

   Digital Transformation of KKBN Business Processes using ODOO: A Capstone Project

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:

   a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
   b. *if disclosed in some other form or manner is identified as confidential*, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:

a. which is or becomes publicly available without breach of this Agreement;
b. which is already known to the Recipient prior to its disclosure hereunder;
c. which is independently developed by the Expert.

8. The parties acknowledge that any technology, product or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:

    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.

11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:	CAMARINES SUR POLYTECHNIC COLLEGES
MS. ANGELIE ELLA L. LAYNESA	**MS. ROSEL O. ONESA, MIT**
Expert	**CSPC Representative**
(Print name and Signature)	(Print name and Signature)
DATE: Feb 26, 2025	DATE: FEB 28 2025

APPENDIX F

JOINT AFFIDAVIT OF UNDERTAKING

Joint Affidavit of Undertaking

We, JOHN EDWARD HONTALBA, of legal age, single/ married and a resident of SAN NICOLAS,IRGA CITY CAMARINES SUR, SHARRYCEL MAGNO, of legal age, single/ married and a resident of SAN ISIDRO, IRIGA CITY CAMARINES SUR, GERALD BUERGO, of legal age, single/ married and a resident of ANTIPOLO OLD, NABUA CAMARINES SUR, after having been sworn to in accordance with law, do hereby take oath and state:

(i) That, we are officially enrolled for the thesis/capstone project on the topic titled DIGITAL TRANSFORMATION OF KKBN BUSINESS PROCESS USING ODOO: A CAPSTONE PROJECT in the COLLEGE OF COMPUTER STUDIES of CAMARINES SUR POLYTECHNIC COLLEGES.

(ii) That, the contents of our thesis/ capstone project submitted to the Camarines Sur Polytechnic Colleges, for award of BACHELOR OF SCIENCE IN INFORMATION SYSTEM Degree are original and our own work and is not plagiarized.

(iii) That, if, after completing the thesis/ capstone project, are found copied or come under plagiarism, we will be solely responsible for it and College shall have sole right to cancel my research work ab-initio.

(iv) That, this work has not been submitted for the award of any other Degree/Diploma in any other University/ Institute.

(v) That, we shall be responsible for any legal dispute/case(s) for violation of any provisions of the Copyright Act relating to our thesis/ capstone project.

IN WITNESS WHEREOF, I have hereunto set my name this________day of NOV 24 2025 202 in Nabua Camarines Sur, Philippines.

JOHN EDWARD HONTALBA
Affiant

SHARRYCEL MAGNO
Affiant

GERALD BUERGO
Affiant

ATTY. FEROZZA DELIA C. SIMBULAN NOTARY PUBLIC NABUA, CAMARINES SUR

SUBSCRIBED AND SWORN TO before me this______day of NOV 24 2025 NABUA, CAMARINES SUR, Philippines, affiants exhibiting to me their competent proofs of identity above stated.

Doc. No. 43
Page No. 10
Book No.: XIV
Series of 2025

ATTY. FEROZZA DELIA CASTRO SIMBULAN
NOTARY PUBLIC
Commission No. IR-174 valid until 31 December 2025
Roll of Attorneys No. 87451
PTR No. 3761925 / 01-06-2025 / Nabua, Camarines Sur
IBP No. 493309 / 01-[illegible] / Camarines Sur
2nd Flr. [illegible] Commercial
San Francisco, Nabua, Camarines Sur, 4434

APPENDIX G

PROJECT TEAM ASSIGNMENT FORM

PROJECT TEAM ASSIGNMENTS FORM

Team Alias	Aport
Course Code	IS 3212
CaPSA	ICHELLE F. BALUIS, MSIT

Name and Signature	Project Role	Email address/mobile#(s)
John Edward Hontalba	Project Head	Johontalba@my.cspc.edu.ph 095145888512
Gerald Buergo	Information Manager	gebuergo@my.cspc.edu.ph 09923887158
Sharrycel Magno	Manuscript	Shmagno@my.cspc.edu.ph 09098165075

APPENDIX H

ROLE ACCEPTANCE FORM

Role Acceptance Form

ROLE ACCEPTANCE FORM
College of Computer Studies

Date: 02-03-2025

To: JOSEFINA H. LIAGAS

We, the third year students of Camarines Sur Polytechnic Colleges pursuing a degree in **Bachelor Science Information System(BSIS)**, are currently enrolled in Capstone Project1.

We are writing to humbly request your service and expertise to serve as our **Adviser** for our capstone project. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our thesis/ capstone tentative title proposals for your kind reference.

Thank you and looking forward to your favorable response of our request.

Respectfully,

Team Aport
John Edward Hontalba
Gerald Buergo
Sharrycel Mgno

To: **ROSEL O. ONESA**
OIC Dean, CCS

This formally signifies that I ACCEPT/ REJECT the request to serve as **Adviser** of the **Team Aport**

As **Adviser**, I agree to perform my duties and responsibilities stipulated in Section 1.6 of the TCP Guidebook from Thesis 1 / Capstone Project 1 until Thesis 2 /Capstone Project 2.

Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the thesis/ capstone project.

Conformed:

JOSEFINA H. LIAGAS
Name and Signature

Role Acceptance Form

ROLE ACCEPTANCE FORM
College of Computer Studies

Date: 02-03-2025

To: JEREMY JIREH S. NEO

We, the third year students of Camarines Sur Polytechnic Colleges pursuing a degree in **Bachelor Science Information System(BSIS)**, are currently enrolled in Capstone Project1.

We are writing to humbly request your service and expertise to serve as our **Consultant** for our capstone project. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our thesis/ capstone tentative title proposals for your kind reference.

Thank you and looking forward to your favorable response of our request.

Respectfully,

Team Aport
John Edward Hontalba
Gerald Buergo
Sharrycel Mgno

To: **ROSEL O. ONESA**
OIC Dean, CCS

This formally signifies that I ACCEPT/ REJECT the request to serve as **Consultant** of the **Team Aport**

As **Consultant**, I agree to perform my duties and responsibilities stipulated in Section 1.6 of the TCP Guidebook from Thesis 1 / Capstone Project 1 until Thesis 2 /Capstone Project 2.

Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the thesis/ capstone project.

Conformed:

JEREMY JIREH S. NEO
Name and Signature

Role Acceptance Form

ROLE ACCEPTANCE FORM
College of Computer Studies

Date: 02-03-2025

To: ANGELIE ELLA L. LAYNESA

We, the third year students of Camarines Sur Polytechnic Colleges pursuing a degree in **Bachelor Science Information System(BSIS)**, are currently enrolled in Capstone Project1.

We are writing to humbly request your service and expertise to serve as our **Grammarian** for our capstone project. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our thesis/ capstone tentative title proposals for your kind reference.

Thank you and looking forward to your favorable response of our request.

Respectfully,

Team Aport
John Edward Hontalba
Gerald Buergo
Sharrycel Mgno

To: **ROSEL O. ONESA**
OIC Dean, CCS

This formally signifies that I ACCEPT/ REJECT the request to serve as **Grammarian** of the **Team Aport**

As **Grammarian**, I agree to perform my duties and responsibilities stipulated in Section 1.6 of the TCP Guidebook from Thesis 1 / Capstone Project 1 until Thesis 2 /Capstone Project 2.

Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the thesis/ capstone project.

Conformed:

ANGELIE ELLA L. LAYNESA
Name and Signature

APPENDIX I

FINAL PROJECT TITLE FORM

FINAL PROJECT TITLE FORM

Team Alias: Team Aport
Proponents/Researchers:

1.) John Edward Hontalba
2.) Gerald Buergo
3.) Sharrycel Magno

Proposed Thesis/ Capstone Project Title:

Digital Transformation of KKBN Business Processes using ODOO: A Capstone Project

Submitted by: JOHN EDWARD HONTALBA (Signature of Project Head over printed name) Date: 2/26/25	**Noted:** ICHELLE F. BALUIS (Signature of CAPSA over printed name) Date: 2/26/25
Recommending Approval: JOSEFINA H. LLAGAS (Signature of CAPA over printed name) Date: 2/26/25	**Approved:** ROSEL O. ONESA OIC Dean, CCS Date: 2/26/25

*** Accomplished in 3 copies

APPENDIX J

THESIS/CASPTONE PROJECT HEARING FORM

THESIS/ CAPSTONE PROJECT HEARING FORM

Date filed: 05-14-25 [✓] Title Proposal [ ] Pre-Oral [ ] Final Oral

Date of Hearing: May 15 2025 Time: 8am-10am Venue: Aircode

Department: COLLEGE OF COMPUTER STUDIES

Research Title:

Digital Transformation of KKBN Business Processes using ODOO: A Capstone Project

Proponent/s:

JOHN EDWARD HONTALBA **SHARRYCEL MAGNO**

GERALD BUERGO

Recommended by:

JOSEFINA H. LLAGAS, MIT
Thesis/ Capstone Project Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.

JAYVEE NIEL SIAS
PANEL MEMBER 1

DR. JOCELYN LIPATA
PANEL MEMBER 2

NIÑO JEFFREY LUZON, MIS
PANEL CHAIR

NOTED BY:

ICHELLE F. BALUIS
Subject Adviser

APPROVED:

ROSEL O. ONESA, MIT
OIC- DEAN, CCS

THESIS/ CAPSTONE PROJECT HEARING FORM

Date filed: October 7, 2025 [ ] Title Proposal [/] Pre-Oral [ ] Final Oral

Date of Hearing: October 15, 2025 Time: 8-10 AM Venue: Open Lab

Department: COLLEGE OF COMPUTER STUDIES

Research Title:

Digital Transformation of KKBN Business Processes using ODOO: A Capstone Project

Proponent/s:

John Edward Hontalba	Sharrycel Magno
Gerald Buergo	

Recommended by:

Josefina A. Plagas
Thesis/ Capstone Project Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research. [Please **PRINT NAME** and **SIGN**]

Jayvee Niel Sias
PANEL MEMBER 1

Dr. Jocelyn Lipata
PANEL MEMBER 2

Nino Jeffrey Luzon, MIS
PANEL CHAIR

NOTED BY:

ICHELLE F. BALUIS
Subject Adviser

APPROVED:

ROSEL O. ONESA, MIT
OIC- DEAN, CCS

THESIS/ CAPSTONE PROJECT HEARING FORM

Date filed: 11-07-25 [ ] Title Proposal [ ] Pre-Oral [/] Final Oral

Date of Hearing 11-07-25 Time: 9:00-12:00 Venue: Open Lab

Department: COLLEGE OF COMPUTER STUDIES

Research Title:

Digital Transformation of KKBN Business Processes using ODOO: A Capstone Project

Proponent/s:

John Edward Hontalba | **Sharrycel Magno**

Gerald Buergo

Recommended by:

Josefina H. Llagas, MIT
Thesis/ Capstone Project Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research. [Please **PRINT NAME** and **SIGN**]

Jayvee Niel Sias, MIT
PANEL MEMBER 1

Dr. Jocelyn Lipata
PANEL MEMBER 2

Niño Jeffrey Luzon, MIS
PANEL CHAIR

NOTED BY:

ICHELLE F. BALUIS
Subject Adviser

APPROVED:

ROSEL O. ONESA, MIT
OIC- DEAN, CCS

APPENDIX K

PANEL RCS (TD, POD, FOD)

Republic of the Philippines
CAMARINES SUR POLYTECHNIC COLLEGES
ISO 9001:2015 CERTIFIED

COLLEGE of COMPUTER STUDIE

Title : **Digital Transformation of KKBN Business Processes using ODOO: A Capstone Project**

Alias : Team Aport

Date : May 15, 2025, 8:00 AM | / | TD | | POD | | FOD

Secretary : Justine D. Labios

MANUSCRIPT			
CHAPTER	**PAGE NO.**	**RECOMMENDATIONS, SUGGESTIONS AND COMMENTS (RSC)**	**ACTION TAKEN**
1	2	**1.1 Project Context** • If possible, separate or provide another paragraph to show the necessity of the study. (felt.need)	We add another paragraph for the necessity of the study also we include the background of KKBN. [p.2]
1	3	**1.2 Purpose and Description of the Project** • 1. Analyze the business processes. • 2. ODOO modules required through identified processes.	We simplified the purpose and description of our study and analyze the business processes and list the modules that needs in KKBN for improving their Businesses. [p.3]
1	4	**1.3 Objectives of the Project** • 1. Identify the differences, examine the business processes and challenges. • 2. Hierarchy of processes • 2.1.2 ❖ Attendance • 2. Add the following: ❖ POS ❖ Accounting • 3.1.3 ❖ Check if included in ISO **1.4 Significance of the Study** • Business establishment/management • Other Business	We arrange the selected ODOO modules for KKBN based on the Hierarchy of processes and also we add the new module that will need for the improvement of processes of KKBN such as HR,POS, Accounting, and attendance for Employee Management. [p.4] For Significance of the Study As suggested we change the word from "Owner and Management" to "Business establishment/management" and we add Other Business. [p.4, 5]
1	5	**The following word should include an 'S':** • Customer • Researcher • Future Researcher	We add letter 'S' in the end of the word of the following word: • Customers • Researchers • Future Researchers

		1.5 Scope and Limitation • Improve • Specify the modules	And in limitation we specify the module of ODOO ERP the we will use in KKBN. [p.5, 6]
1	6	Promotion of ODOO focuses on the identified processes that can be accommodated by ODDO. **ISO 25010** • All parameters	We include all parameters of ISO 2510 Such as functional suitability, Performance Efficiency, Usability, Maintainability, Reliability, and Security. [p.4]
1	7	**ODDO** • open-source enterprise resource planning (ERP) **ADD** • Digital Transformation **Words that are found at the title and scope.** • Digital Transformation • KKBN • Employee Management/HR	We add another project dictionary such as Digital Transformation and KKBN. [p.7,8]
1	8	**Notes:** • Follow the proper format of the ACM Citation.	As suggested we follow the proper format of the ACM Citation.
2	17	**2.1.5 Customized Odoo Module Assessment/Evaluation** • That should be placed on the page.	As suggested we placed to other next page.
2	19	**2.2 Synthesis of the State-of-the-Art** • Same discussion	We removed a paragraph with the same discussion.
2	21	**Gap Bridge by the Study** • Complete sentence • Identify specific related studies and pinpoint a particular gap. **Additional studies:** • 1. Based on SOP 1, discuss the challenges of using manual processes. • 2. Processes/Operations subject to ERP. • 3. Customization of processes/operation using	We rewrote the section in full sentences, identified related studies with clear gaps, and added all suggested subtopics.

Republic of the Philippines
CAMARINES SUR
POLYTECHNIC COLLEGES
ISO 9001:2015 CERTIFIED

COLLEGE of
COMPUTER STUDIE

		ODOO, differentiate with other erp applications. • 4. ISO 25010, how helpful the evaluation to the customized operations as a whole.	
3	**29**	**SOP 1** • Process Identification • Process Discovery **SOP 2 hierarchy of operations** • Process Analysis **SOP 3** • Process Design **SOP 4** • Process Implementation • Process Monitoring	As suggested we added the structured SOPs and matched them with the corresponding process phases to ensure clarity and consistency with the methodology. [p.30]
3	**30**	**To researchers will…** • It should begin with 'The'	As suggested we change the word from "to researcher" to "the researcher"
3	**32**	**Sampling Technique** • SOP 1 - 5 participants • SOP 4 - 10 participants	We clearly indicated the number of participants per SOP as recommended. [p.33, 34]
3	**33**	**3.3.4 Data Analysis Procedure** • That should be placed on the page.	As suggested we placed to other next page.

NOTED BY:

NIÑO JEFFREY LUZON, MIS
PANEL, CHAIRMAN

JAYVEE NIEL SIAS
PANEL, MEMBER 1

DR. JOCELYN LIPATA
PANEL, MEMBER 2

Republic of the Philippines
CAMARINES SUR
POLYTECHNIC COLLEGES
ISO 9001:2015 CERTIFIED
COLLEGE of COMPUTER STUDIE

Title : **DIGITAL TRANSFORMATION OF KKBN BUSINESS PROCESS USING ODOO: A CAPSTONE PROJECT**

Alias : Team Aport

Date : October 15, 2025 | | TD | / | POD | | FOD

Secretary : Justine D. Labios

MANUSCRIPT			
CHAPTER	**PAGE NO.**	**RECOMMENDATIONS, SUGGESTIONS AND COMMENTS (RSC)**	**ACTION TAKEN**
1	1	**1.1 Project Context** • Basic paragraph, 3 sentences. → Introduction → Main → Conclucion	As Suggested we arrange our project context using the format Introduction,Main,and Conclusion.
1	3	**1.3 Objectives of the Project** • The word 'for' in the sentence 'Branch for business process for efficient work' should be replaced with 'and'.	As suggested we change for to and
1	6	**1.6 project Dictionary** • Include terms from SOP/Objectives. • ISO 25010 metrics ODOO components.	As suggested we include SOP 2 module such as Inventory, Sales, Accounting, POS, Purchase and Employee and Attendance. And we include all six components of ISO 25010.
1	7	Locale of the study specially Nabua and Iriga.	We add locale of the study in Nabua and Iriga.
1	9	**Notes:** • Title (Italize)	As suggestion we italize Notes
2	10	**2.1.1 Odoo for ERP Solution Implementation** • Discussion on ISO 25010 • Avoid using only one sentence in a paragraph.	As recommended we discuss the ISO25010 more than 2 senteces
2	17	**2.1.4 ERP software Impact on Business Operations**	As Suggested we remove the word accoring to Pandey et al and we change word content to contended

Republic of the Philippines
CAMARINES SUR
POLYTECHNIC COLLEGES
ISO 9001:2015 CERTIFIED

COLLEGE of COMPUTER STUDIE

		• Please remove According to Pandey et al. [2022] • In the second paragraph, change the word contended to contended in the first phrase.	
2	20	**2.2 Synthesis of the State-of-the-Art** • Used past tense in the discussion. • Change the words 'highlight' and 'explore' to 'highlighted' and 'explored'.	We change the word from present to past tense
2	22	No discussion of evaluation framework.	As Suggested we add a discussion of Framework
3	31	**3.2 Research Design** • Past tense	We used Past Tense in discussion
3	32	**3.3.2 Instrument** • Structured on non-structured • In section B, add the checklist, which will serve as the Observation Checklist. • Survey questionnaire?	As suggested we change the name in instruement from observation to checklist also in the survey questionaire
3	34	**Table 1** • For the percentage, change it to 20, 20, and 60, with a total of 100%.	As suggested we follow the percentage for Table 1.
3	38	Change Table 1 to Table 3.	we change the Table 1 to 3.
4		**BPMN** • Check format of tables of start and end events (noun, past tense) • Include an activity before the gateway, an activity that questions the label for the outgoing activity labels example. → Check if complete ➢ Complete ➢ Incompete • Including other actors in the swimlane. • Specify what process under HR. **ODOO** • Try the 3rd party accounting module.	As suggested we change start and end events noun, past tense We include an activity Include an activity before the gateway We include another actor from Point of sale We change the Employee Management/HR to Employee Management
4	42	Notes or supporting study	We add notes
4	44	Supporting Study?	We add notes
4	46	Accounting & purchasing [the some feature or different]	Purchasing is for buying goods and managing suppliers. Accounting, on the other hand, is for managing financial records, tracking income and vendor bills
4	48	Supporting study the same throughout the succeeding discussion.	We add notes

Republic of the Philippines
CAMARINES SUR
POLYTECHNIC COLLEGES
ISO 9001:2015 CERTIFIED
COLLEGE of COMPUTER STUD

4	57	**4.2.4** • Check • Add Customer	As suggested we add Actor customer
4	61	**4.2.6 Employee/HR** • One-way no interaction from the employee.	We add interaction form the employee
4	63	Change Table 1 to Table 4.	We change Table 1 to 4
4	65	Change Table 2 to Table 5.	We change Table 2 to 5
4	74	Change Table 3 to Table 6.	We change Table 3 to 6
4	83		
4	90	Change Table 4 to Table 7.	We change Table 4 to 7
4	91	Fix the indention.	We fix the idention
4	92	Change Table 5 to Table 8.	We Change Table 5 to 8
4	94	Change Table 6 to Table 9.	We Change Table 6 to 9
4	95	Fix the indention.	We fix the identtion
4	96	Change Table 7 to Table 10.	We Change Table 7 to 10
4	97	Change Table 8 to Table 11.	We change Table 8 to 11
4	99	Change Table 9 to Table 12.	We change Table 9 to 12
4	101	Change Table 10 to Table 13.	We change Table 10 to 13
5	106	The summary is quite long.	
5	109	**5.3 Conclusion** • Based on SOP Findings → 1-1 → 2-2 → 3-3	As Suggested we revise our conclusion ad we based it on our SOP Finding

CAPSTONE PROJECT OUTPUT	
RECOMMENDATIONS, SUGGESTIONS AND COMMENTS (RSC)	ACTION TAKEN

NOTED BY:

NIÑO JEFFREY LUZON, MIS
PANEL, CHAIRMAN

JAYVEE NIEL SIAS
PANEL, MEMBER 1

DR. JOCELYN LIPATA
PANEL, MEMBER 2

Title : **DIGITAL TRANSFORMATION OF KKBN BUSINESS PROCESS USING ODOO: A CAPSTONE PROJECT**

Alias : Team Aport

Date : November 07, 2025 | | TD | | POD | / | FOD

Secretary : Justine D. Labios

MANUSCRIPT			
CHAPTER	**PAGE NO.**	**RECOMMENDATIONS, SUGGESTIONS AND COMMENTS (RSC)**	**ACTION TAKEN**
		• Please include "MIT" after Ma'am Llagas's name. • Please add the letter "L" as the middle initial in Sir Luzon's name. **ACKNOWLEDGMENTS** • Please remove the extensions from the names. **ABSTRACT** • The section from determine down to the lower part should be formatted as features.	As suggestion we include MIT for ma'am Llagas name. We also include L. middle initial in sir luzon name. We also recome extensions from the name. We also include all features of module
3	33	Guide question. (Structured)	We change Structured of intruments used
4	53	**4.2.1 Inventory** Please refer to the manuscript; as Sir Jayvee has already demonstrated/explained how to do or what actions to take in BPM diagram, and he has also circled the sections that need to be modified.	We suggestion from sir jayvee we change the start event Inventory
4	55	**4.2.2 Sales** • Same in page 53 the BPM diagram.	We change a Start and End event
4	56	**4.2.3 Accounting** • Same in page 53 the BPM diagram.	We change a Start and End event
4	57	Two out.	We out two line from Point of sale BPM
4	60	**4.2.6 Employee Management and Attendance**	We change arrow a straight line

		• The arrow should be a straight line, not a broken line.	
4	72	**4.2.6 Employee Management and Attendance** • Before starting a sentence, there should be an indention.	We indention before starting a sentence
4	104	**Notes** • The information under (IMS) should contain complete details.	We include all information
5	112	**5.4 Recommendations** • Before starting a sentence, there should be an indention.	We indention before starting a sentence

CAPSTONE PROJECT OUTPUT	
RECOMMENDATIONS, SUGGESTIONS AND COMMENTS (RSC)	**ACTION TAKEN**

NOTED BY:

NIÑO JEFFREY L. LUZON, MIS
PANEL, CHAIRMAN

JAYVEE NIEL SIAS
PANEL, MEMBER 1

DR. JOCELYN LIPATA
PANEL, MEMBER 2

APPENDIX L

CONSULTATION LOG FORM

CONSULTATION LOGS FORM

Capstone Project Title:	Digital Transformation of KKBN Business Processes using ODOO: A Capstone Project
Alias:	TEAM APORT
PROPONENTS:	JOHN EDWARD HONTALBA GERALD BUERGO SHARRYCEL MAGNO

Draft of Chapter 1	Date of Consultation					Project Head's Signature	CAPA/Consultant/ Grammarian Signature
	3/7/2025					[signature]	[signature]

Remarks:

1) arrange the project context in the 1st paragraph it should be in general discussion then as the discussion progress it should be now in specified.

2. Purpose and Desc. of the project no hanging of page.
3. objectives of the proj. .
3.1 Examine the challenges of the current process in KKBN
3.2 to select the necessary Odoo modules . . .

4. to customize . . . .
5. Significance of the study
– arrange by who will benefit 1st the study
6. the sentences/ paragraph should be in future tense.
7. Check the template.
8. NCs should be in a separate page.

Revised Chapter 1	Date of Consultation					Project Head's Signature	CAPA/Consultant/ Grammarian Signature
							[signature]

Remarks:

– no hanging of pages
– use future tense. like " will aim to "

CONSULTATION LOGS FORM

Capstone Project Title:	Digital Transformation of KKBN Business Processes using ODOO: A Capstone Project
Alias:	TEAM APORT
PROPONENTS:	JOHN EDWARD HONTALBA SHARRYCEL MAGNO GERALD BUERGO

Draft of Chapter 2	**Date of Consultation**					**Project Head's Signature**	**CAPA/Consultant/ Grammarian Signature**
	04-14-25						
	Remarks: Good to go!						

Draft of Chapter 3	**Date of Consultation**					**Project Head's Signature**	**CAPA/Consultant/ Grammarian Signature**
	4-20-25						
	Remarks: 1.) Figure 1 must be bigger in size and title should not be in a separate page. 2) must be in future tense. 3.						

Chapter 1-3	**Date of Consultation**					**Project Head's Signature**	**CAPA/Consultant/ Grammarian Signature**
	5-9-25						
	Remarks: Good to go!						

Chapter 1-3	**Date of Consultation**					**Project Head's Signature**	**CAPA/Consultant/ Grammarian Signature**
	MAY 9						
	Remarks: GOOD TO GO						

CONSULTATION LOGS FORM

Capstone Project Title:	Digital Transformation of KKBN Business Processes using ODOO: A Capstone Project						
Alias:	TEAM APORT						
PROPONENTS:	JOHN EDWARD HONTALBA GERALD BUERGO SHARRYCEL MAGNO						
Chapter 1-3	**Date of Consultation**					**Project Head's Signature**	**CAPA/Consultant/ Grammarian Signature**
	05-14-25						
	Remarks: • Chapters 1 to 3 have been thoroughly reviewed for grammar, structure, and academic tone. Corrections have been made to ensure subject-verb agreement, proper verb tense (future tense for researcher actions), clarity of sentence construction, and consistency in academic language. The revised content adheres to formal writing conventions and avoids redundancy or ambiguity. • Recommended for TD.						
OUTPUT Chapter-4	**Date of Consultation**					**Project Head's Signature**	**CAPA/Consultant/ Grammarian Signature**
	09/10/15						
	Remarks: Good to go!						
OUTPUT	**Date of Consultation**					**Project Head's Signature**	**CAPA/Consultant/ Grammarian Signature**
	08-20-25						
	Remarks: Find scuma pos receipt OR As-IS: form, file, report Arrange module						

APPENDIX M

GRAMMARIAN'S CERTIFACATE

CERTIFACATE

This is to certify that the undersigned has reviewed and went through all the pages of the capstone project **entitled "Digital Transformation of KKBN Business Process Using Odoo: A Capstone Project"** against the set of structural rules that govern research writing in accord with the composition of sentences, phrases, and words in the English language.

Signed:

ANGELIE ELLA LL. LAYNESA
Grammarian
Date signed: December 10, 2025

APPENDIX N

SECRETARY'S CERTIFACATE

CERTIFICATION

This is to certify that the undersigned has provided true and correct recommendations, suggestions, and comments unanimously agreed and approved by the panel of examiners during the oral examination of the capstone project **entitled "Digital Transformation of KKBN Business Process Using Odoo: A Capstone Project"** prepared and submitted by John Edward Hontalba, Sharrycel Magno, Gerlad Buergo and that the same have not been amended, modified or obliterated.

Signed:

JUSTINE D. LABIOS
Secretary
Date signed: December 11, 2025

APPENDIX O

CERTIFACATE OF UTILIZATION

CAMARINES SUR POLYTECHNIC COLLEGES
NABUA, CAMARINES SUR

CERTIFICATE
OF UTILIZATION

This is to certify that the

"DIGITAL TRANSFORMATION OF KKBN BUSINESS PROCESS USING ODOO: A CAPSTONE PROJECT"

developed by John Edward Hontalba, Sharrycel Magno and Gerald Buergo, Colleges of Computer Studies (CCS) Students, as their undergraduate capstone project was adopted and utilized by the Kalusugan, Kayamanan, Basta Natural (KKBN).

This certification is issued this__Nov__day of __6__2025 at Kalusugan, Kayamanan, Basta Natural (KKBN) for whatever legitimate purpose in my serve.

Mrs. Laila Belleza
Business Owner

APPENDIX P

ACM FORMAT

DIGITAL TRANSFORMATION OF KKBN BUSINESS PROCESS USING ODOO: A CAPSTONE PROJECT

JOHN EDWARD HONTALBA

CAMARINES SUR POLYTECHNIC COLLEGES

johnedwardhontalba087@gmail.com

SHARRYCEL MAGNO

CAMARINES SUR POLYTECHNIC COLLEGES

sharrycel@gmail.com

GERALD BUERGO

CAMARINES SUR POLYTECHNIC COLLEGES

Gebuergo2my.cspc.edu.ph

In today's competitive environment, small and medium enterprises (SMEs) must embrace digital transformation to enhance operational efficiency and streamline processes. This study focused on implementing the Odoo Enterprise Resource Planning (ERP) system in Kayamanan, Kalusugan, Basta Natural (KKBN). Its main objectives were to analyze existing business operations, select and configure essential Odoo modules, and evaluate the system's acceptability based on ISO 25010 quality characteristics, including Functional Suitability, Performance Efficiency, Usability, Maintainability, Reliability, and Security. A mixed-method approach was utilized, combining surveys and interviews with stakeholders such as the business owner, secretary, sales personnel, ERP professionals, and IT specialists. The system's effectiveness was measured using a weighted mean to determine how well it met operational needs. Results revealed high acceptability across all quality criteria, with the strongest performance in security and functional suitability. The overall weighted mean of 4.75 indicated that the system surpassed expectations and complied fully with required standards. The study concludes that the customized Odoo ERP solution is highly effective in automating and optimizing SME operations. It further recommends exploring additional Odoo modules, enhancing customization to meet evolving business needs, and conducting comparative analyses with other ERP systems to identify best practices for SMEs.

CCS CONCEPTS • Information System _ Enterprise resource planning (ERP) •

Additional Keywords: Digital Transformation, Enterprise Resource Planning (ERP), Odoo, ISO 25010, Small and Medium Enterprises (SMEs), Business Operations, Odoo Modules, System Implementation

1 INTRODUCTION

Most small businesses today still face major challenges as they continue depending on outdated methods of operation. Traditional paper-based systems and manual processes dampen their efficiency, scalability, and accuracy, hence the urgent need for digital transformation to remain sustainable and competitive. This project focuses on addressing these issues in Kayamanan, Kalusugan, Basta Natural (KKBN), a business located in San Roque, Iriga, Camarines Sur, currently maintaining an inventory and sales reports through paper and manual systems, will find a solution to the inefficiencies brought about by such practices resulting in cases like stock shortages, inaccuracies, and missed sales opportunities. In resolving these issues, the project proposes the implementation of the Odoo ERP system, integrating disparate business processes into one digital platform. Odoo ERP is recognized as one of the most effective ERP solutions, particularly for small businesses [14]. Through this, KKBN will be able to improve efficiency, reduce errors, and generally enhance performance with Odoo. The main objective of this study is the implementation and configuration of Odoo ERP software by choosing modules suitable for KKBN's needs. The modular structure of Odoo allows access to real-time data, process automation, and a set of customizable tools that enhance productivity and support informed decisions. All ERP systems provide a common suite of finance and operations applications with a shared data model to ensure one version of the truth. Among them, Odoo is an efficient and flexible ERP that enables small businesses to be more productive by getting more in less time. Its benefits include faster decision-making through accurate data, higher productivity, integration across all departments, customization to business needs, reduced operational costs, and simplified management through a unified interface [6]. This study will establish the acceptability of the Odoo ERP system based on the ISO 25010 quality characteristics, namely: Functional Suitability, Performance Efficiency, Usability, Maintainability, Reliability, and Security. [8] modification will cover Odoo modules for inventory, purchasing, and sales, while its limitations are confined to the features within the system and a selected set of respondents comprising the business owner, key personnel, IT professionals, and ERP experts. The significance of the research is in relation to several parties: for the management of KKBN, since after implementation, improved efficiency, reduced errors, and increased decision-making capabilities are expected [2]. in the larger context of studies in information systems, the research goes a long way toward contributing to ERP implementation and digital transformation strategies in SMEs. Besides that, the results will further be useful in reference to future research and as a practical model for other small businesses in modernizing their operations by using ERP solutions like Odoo.

2 BACKGROUND

Enterprise Businesses today deal with a more complex and competitive business environment. Enterprise Resource Planning (ERP) systems are used by many firms to manage these difficulties and optimize operations. ERP systems give comprehensive data management and increase operational efficiency by combining several company activities into a single, integrated system [12]. Complexity in competing businesses and blurriness in product specifications necessitate efficiency as well as effective operational policies for corporations. To achieve the expected results, companies require an Integrated Information Technology System, commonly known as Enterprise Resource Planning (ERP). ERP consists of a collection of business software programs aimed at improving the management of a firm. By integrating all company processes into a single database, ERP systems enable organizations to gather and store data from various departments and locations through a standard user interface [3]. One way for SMEs to get beyond these obstacles and gain a competitive edge is through ERP. This study aims to investigate the ERP adoption process within the context of SMEs. The framework created for ERP implementation in an SME emphasizes addressing challenges such as defining the implementation scope, appropriate project planning, and minimizing customization. These elements are crucial for a smooth adoption process. The expectation is that the ERP system will become more integrated and effective for the firm after adoption. This study used the Business Process Model (BPM)Framework as a guide for implementing Enterprise Resource Planning. This framework aims to streamline the ERP implementation process, ensuring that the system meets the firms' specific needs and enhances overall operational efficiency [11]. The XYZ Weaving implemented one of ERP Software called Odoo in their company. It can support easier, more integrated, and more efficient management of business processes. To be able to sustain the business in contemporary markets, it therefore enhances customer pleasure and service. The produced ERP is freely available and requires no financial outlay for MSME businesses to implement. A company can gain tremendously from the ERP software by managing its resources, as it is easy to track, manage, and empower the company resources. The X Beauty business challenges of the manual process are an inventory process that causes frequent stockouts and inaccurate records, and sales process disrupting sales and reduces customer trust, and Purchase when a product is in stock [9]. Implementation of ERP Odoo at AH Mart integrates business processes across divisions, promotes data availability, and reduces response time [3]. The review of related literature showed that even today, many businesses face challenges due to their dependence on manual processes. Delays, errors, and inefficiencies compromise competitiveness and decision-making based on facts. Though most studies conducted in the local context have been on SAP ERP, this study places unique emphasis on the implementation of Odoo ERP, which is more adaptable and scalable. Odoo ERP straightens out the flow of business operations, decreases errors, and raises output by switching from manual methods to an entirely integrated system [4]. Odoo ERP has modules that can be customized for all aspects of business functions and thus can improve data accuracy, decision-making, and overall business performance [13]. This provides a modern solution with features that perfectly fit the needs of KKBN.

3 METHOLOGY

3.1 Business Process Management Framework

The study utilized the Business Process Management (BPM) life cycle to implement the Odoo ERP system at KKBN, aiming to enhance efficiency, productivity, and customer satisfaction. BPM uses a clear step-by-step plan, which reduces the chances of errors, delays, or going over budget during implementation. The framework consists of six phases of process identification, discovery, analysis, design, implementation, and monitoring which guided the researchers in analyzing KKBN's existing workflows, identifying areas for improvement, and ensuring continuous enhancement. Overall, the BPM frameworks helped the researcher systematically manage and the KKBN's business processes to improve their processes and run smoothly.

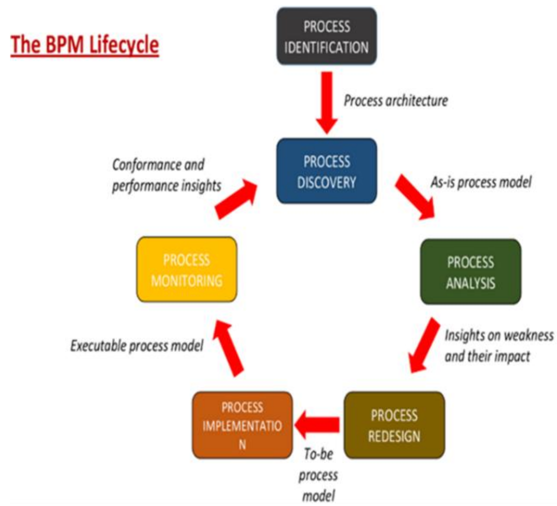


Figure 1: Business Process Management

Figure 1, show the Business Process Management (BPM) Framework used in the study, which consists of Six phases.

Process Identification. The researchers and the business owner defined the objectives and analyzed the current business operations through interviews.

Process Discovery. After conducting interviews, the researchers documented KKBN's existing (as-is) processes to gain a thorough understanding of the business.

Process Analysis. After obtaining a clear understanding of the existing KKBN processes, the researchers identified all the challenges and prioritized the issues that needed to be addressed.

Process Design. The researchers customized and configured the necessary Odoo modules to meet KKBN's operational needs, establishing the to-be processes.

Process Implementation. The researchers will implement the configured Odoo ERP system and assess its performance using ISO 25010 standards.

Process Monitoring. The researchers monitored the system monthly to evaluate its impact on KKBN's business.

3.2 Research Design

In line with the general and specific objectives of this study, the researchers will utilized mixed-methods research design, combining both qualitative and quantitative approaches.

3.3 Research Methods

The study applied the mixed-method approach, which combined both qualitative and quantitative research methodologies to achieve its objectives regarding the implementation of the Odoo ERP System at KKBN Iriga and Nabua branches.

3.3.1 Data Source

Primary data were collected directly from KKBN stakeholders. Interviews with the business owner, secretary/cashier, and sales lady provided insights into their daily workflows, responsibilities, and challenges before and after the ERP implementation. An observation checklist documented behaviors and processes. Surveys were distributed to assess the performance of the implemented ERP system.

3.3.2 Instruments

This subsection introduced the instrument used by the researcher, which helps ensure consistent and reliable data collection for the study.

Interview Guide. The interview guide will help the researchers ask focused questions to gather important information. Using it, they will interview the KKBN owner to understand current business processes and identify suitable Odoo modules. This will provide insights on workflows, system requirements, and the owner's goals for automation.

Checklist Checklist. The researchers will use an observation checklist to systematically record participants' actions and behaviors. This structured approach ensures consistent and reliable data collection. Even though it is manual, it provides a solid basis for implementing Odoo in the future.

Survey Questionnaire. The researchers will utilize this scale to assess the system's performance based on user feedback.

3.3.3 Data Analysis Procedures

This section will describe the procedures that the researchers will follow to gather the necessary data for the study.

Interview Guide. The interview serves as a primary method for collecting data across various areas. The responses from the interviewees were essential in addressing the first objective of the project. Through these interviews, the researchers aimed to gain an understanding of KKBN's business processes and identify key areas for improvement.

Observation Checklist. The researchers will use an observation checklist to collect data from participants. The checklist will include specific actions, behaviors, or reactions to observe, guiding the data-gathering process in a structured and consistent manner. Although the process will be manual, it will ensure reliable and organized data collection, forming a basis for a more systematic implementation of a tool like Odoo in the future and to identify challenges and areas for improvement in KKBN's current business processes.

Survey. The researchers will use surveys as a tool to collect data from the owner and other participants. The survey will aim to understand general perceptions and assess the level of acceptability of the new ERP system based on ISO 25010 standards. Specifically, the researchers will focus on evaluating the system Performance Efficiency, Functional Suitability, Usability, Maintainability, Reliability, and Security.

Sampling Technique

The respondents of the study consisted of ten individuals, including five KKBN including the owner, secretary/cashier, and sales personnel, three IT experts, and two ERP experts. Their involvement ensured both operational and technical evaluation of the Odoo ERP system, as shown in Tables 1 and 2.

Table 1: Distribution of Respondents for SOP 1

Respondents	No. Respondents	Percentage
Owner	1	20%
Secretary	1	20%
Staff	3	60%
Total	**5**	**100%**

Table 1 shows the distribution of respondents for SOP 1, which included the owner, secretary/cashier, and three staff members of KKBN. Their direct involvement in daily operations helped ensure that the data collected accurately reflected actual business processes, challenges, and user experiences through observation checklists and questionnaires.

Table 2: Distribution of Respondents for SOP 3

Respondents	No. Respondents	Percentage
Owner, secretary/cashier, and sales lady	5	50%
IT Experts	3	30%
ERP Experts	2	20%
Total	**10**	**100%**

Table 2 shows the distribution of respondents for SOP 3, which included five KKBN Owner, secretary/cashier, and sales lady, two ERP experts, and three IT experts. Their responses were analyzed using the weighted mean and Likert scale to evaluate the implementation of the Odoo ERP system and assess KKBN's readiness for system adoption.

Statistical Test

The researchers used statistical methods to analyze the data collected. The weighted mean was applied to determine the level of acceptability of the ERP system based on functional suitability, performance efficiency, usability, maintainability, reliability, and security.

$$WM = \frac{\Sigma(f_i \text{ x } w_i)}{N}$$

Where:

- f_i = frequency or the number of responses for each score
- w_i = weight or value of each score (from 1 to 5)
- N = total number of all responses

3.3.4 Data Analysis Procedure

This study will applied both qualitative and quantitative data analysis techniques where to address all specific objectives.

Thematic Analysis. Thematic analysis involves the search for and interpretation of patterns in qualitative information. The interview sessions conducted on the staff members of KKBN will be analyzed for opportunities and inefficiencies within the current processes.

BPMN Analysis. BPMN Analysis processes translates process steps, choices, and interactions into a visual representation. BPMN is based on standardized symbols, allowing processes to be understood, communicated, and optimized easily, which promotes efficient as well as clearer operations within a business.

Descriptive statistics. Descriptive statistics summarize and organize data in an easy-to-understand way. The survey responses, rated on a 5-point Likert scale, will be analyzed using mean, frequency, and percentage to assess how effective the Odoo ERP system is, based on ISO 25010 criteria.

4 RESULT AND DISSCUSION

4.1 Process Identification

In this phase, the researchers conduct an interview with the owner of KKBN (Kayamanan Kalusugan Basta Natural) to gather essential information about the business current operations. The Process Identification stage aimed to identify and understand the key business activities such as Inventory, Sales, Accounting, POS, Purchase, and Employee Management to gain a clear view of how tasks are performed and interconnected.

4.1.1 Inventory

Inventory management was crucial for businesses to ensure the availability of goods and raw materials to meet consumer demand.

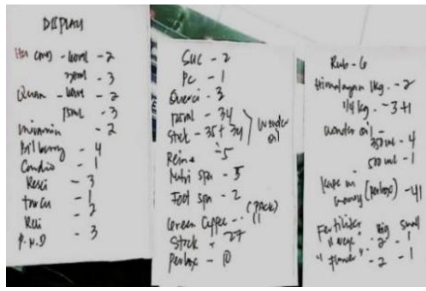


Figure 2: Manual Record of Product Stock

Figure 2 showed KKBN's handwritten inventory record used to track product quantities. This manual process was time-consuming and prone to errors, making it difficult to monitor fast-moving items and avoid stockouts.

4.1.2 Sales

Sales were considered one of the most vital processes in any business because they served as the main point where a company's products reached the customers.

Figure 3: Manual Record of Everyday Sales Report

Figure 3 showed KKBN's daily sales recorded by hand. Manual recording and calculations increased the risk of errors and made organizing and retrieving sales data difficult.

4.1.3 Purchase

In any organization, purchasing was an essential step that included careful planning, selecting trustworthy suppliers, negotiating fair prices, and ensuring that products were delivered on time.

Figure 6: Manual Record of Supplier Information

Figure 6 shows KKBN's handwritten supplier information records. Storing supplier details on paper increased the risk of outdated, lost, or incorrect information.

4.2 Process Discovery

The first step is in understanding a company's operations by identified, documented, and diagrammed day-to-day operations. It helps to understand who is responsible, connections between tasks provide a strong foundation for analyzing problems and finding improvement opportunities.

The As-Is Model represents a true picture of the way the business processes, along with their strengths, weaknesses, and pain points and can identify inefficiencies and redundancies. It helps identify areas for improvement by mapping actual workflows. Based on this, the To-Be Model includes redesigns and improvements processes for better performance and efficiency, ensuring that changes are strategic, organized, and aligned with business goals.

4.2.1 Inventory

KKBN currently manages inventory manually, tracking stock movements, updating records, and checking for discrepancies. This method can lead to inaccurate stock levels and delays in timely inventory management decisions.

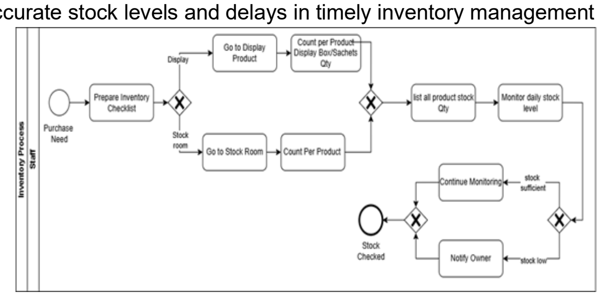


Figure 9: Business Process Modeling Notation for Inventory

Figure 9 illustrated the detailed manual inventory process utilized by of KKBN.

4.2.1 Sales

KKBN's sales process is handled manually, from customer purchases to daily and monthly sales reports. This approach causes inefficiencies, such as manual calculations and limited access to past sales data.

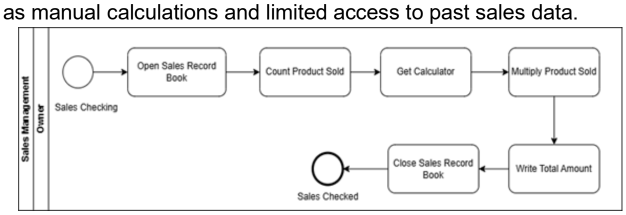


Figure 10: Business Process Modeling Notation for Sales

Figure 10 shows KKBN's current sales process. It starts with the owner opening the sales logbook to record transactions and review the list of products sold and their inventory.

4.2.4 Point of Sales

KKBN handles point-of-sale transactions manually, including processing purchases and generating receipts. This manual process is slow, does not update inventory automatically, and affects customer service.

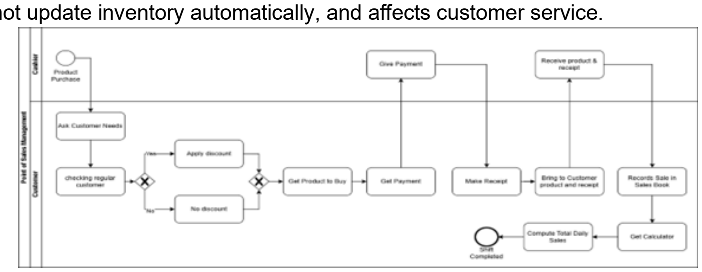


Figure 12: Business Process Modeling Notation for Point of Sales

Figure 12 illustrates the manual Point of Sale (POS) process being followed in the business.

4.3 Process Analysis

Process analysis was the step in which corporate processes were thoroughly investigated to determine their effectiveness, how they operated, and areas for improvement. Its goal is to identify inefficiencies, bottlenecks, and redundancies that hinder performance, enabling better decision-making, increased productivity, and improved overall operations.

Identified Problem

Process analysis was a critical step for finding and solving issues that hindered by manual process unclear staff responsibilities, errors in recording or tracking, and unnecessary delays customer service. KKBN relying heavily on manual workloads, the absence of integration such as sales, inventory, purchasing, and accounting, and inaccuracy of data. These challenges caused delays in decision-making and poor communication, emphasizing the need for an integrated and automated system to improve efficiency and accuracy.

Proposed Solution

The researchers developed recommendations to enhance the firm's operations, aiming to streamline routine tasks, reduce errors, and improve data tracking in areas such as inventory, sales, and purchases. Implementing these solutions saved time, increased productivity, and fostered a more organized system for staff development.

4.4 Process Redesign

The process redesign phase focused on creating the To-Be model to improve the current (As-Is) processes. It detailed how processes should work, using insights from the As-Is analysis, and aimed to make them more efficient, accurate, and aligned with the study's objectives, including the integration of new technologies.

The To-Be Model showed how the organization's processes, technology, and procedures could be improved to increase customer satisfaction, productivity, and efficiency. It provided a roadmap for creating more effective, automated, and user-friendly systems, serving as a strategic guide for achieving long-term goals.

4.4.1 Inventory

The redesigned inventory process automates KKBN's stock monitoring using the Odoo Inventory module, recording products and stock levels in real time. This ensures accurate, up-to-date information, improves efficiency, and helps prevent stock shortages or overstocking.

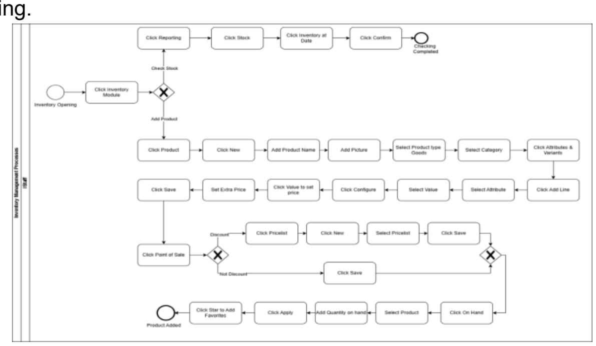


Figure 7: Business Process Modeling Notation for Inventory

Figure 7 showed the new process flow of Inventory in KKBN using Odoo. where staff can easily manage stock, add new products with complete details, set pricing, and ensure items are ready for sales transactions efficiently.

4.4.2 Sales

The redesigned sales process uses the Odoo Sales Module to automate orders, billing, and updates to inventory and accounting. This reduces manual work, minimizes errors, and provides accurate daily sales tracking for better management decisions.

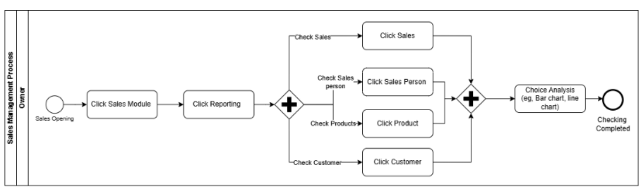


Figure 8: Business Process Modeling Notation for Sales

Figure 8 showed the new Sales process in KKBN using Odoo, where the Owner opens into the Sales module to view and analysis sales data by total sales, salesperson, product, or customer. The system provides visual tools like charts for easy analysis, and the process ends with closing the report ensuring that all information is structured and available for decision-making.

4.4.4 Point of Sales

The new POS process speeds up transactions and automatically updates inventory and accounting through the Odoo POS module. This ensures accurate records and provides a faster, smoother shopping experience for customers.

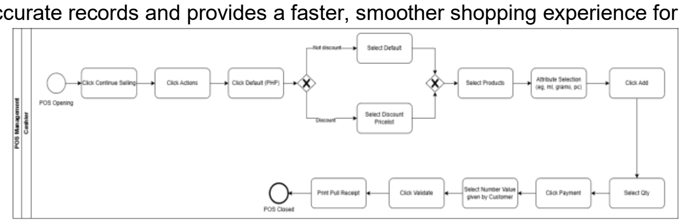


Figure 10: Business Process Modeling Notation for Point of Sales

Figure 10 showed the new process Point of Sale in KKBN using Odoo, where the cashier opens the POS module, selects items, applies discounts if applicable, and processes payment. The system printed receipt and streamlined transaction process that accommodated both discounted and regular sales.

4.5 Process Implementation

The implementation phase focused on applying the improved To-Be Model in real operations, ensuring the new processes were effective, well-integrated, and aligned with the objectives of the study.

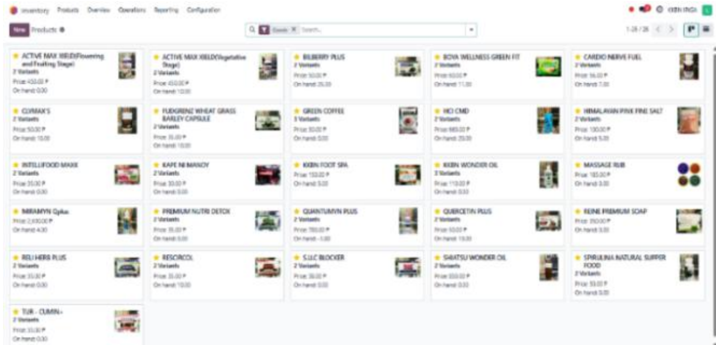
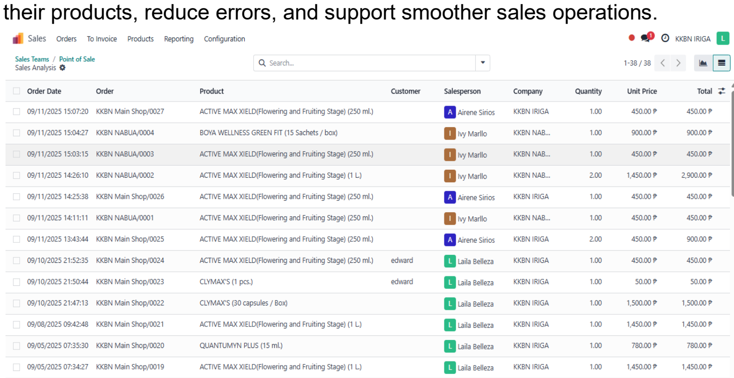


Figure 13: Products

Figure 13 showed the Product module in Odoo, which served as a digital system of all items stored in the company inventory each product including name, image, price, available quantity on hand, and number of variants was displayed. The system is designed to control their products, reduce errors, and support smoother sales operations.



Order Date	Order	Product	Customer	Salesperson	Company	Quantity	Unit Price	Total
09/11/2025 15:07:20	KKBN Main Shop/0027	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (250 ml.)		Airene Sirios	KKBN IRIGA	1.00	450.00 ₱	450.00 ₱
09/11/2025 15:04:27	KKBN NABUA/0004	BOYA WELLNESS GREEN FIT (15 Sachets / box)		Ivy Marllo	KKBN NAB...	1.00	900.00 ₱	900.00 ₱
09/11/2025 15:03:15	KKBN NABUA/0003	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (250 ml.)		Ivy Marllo	KKBN NAB...	1.00	450.00 ₱	450.00 ₱
09/11/2025 14:26:10	KKBN NABUA/0002	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (1 L.)		Ivy Marllo	KKBN NAB...	2.00	1,450.00 ₱	2,900.00 ₱
09/11/2025 14:25:38	KKBN Main Shop/0026	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (250 ml.)		Airene Sirios	KKBN IRIGA	1.00	450.00 ₱	450.00 ₱
09/11/2025 14:11:11	KKBN NABUA/0001	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (250 ml.)		Ivy Marllo	KKBN NAB...	1.00	450.00 ₱	450.00 ₱
09/11/2025 13:43:44	KKBN Main Shop/0025	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (250 ml.)		Airene Sirios	KKBN IRIGA	2.00	450.00 ₱	900.00 ₱
09/10/2025 21:52:35	KKBN Main Shop/0024	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (250 ml.)	edward	Laila Belleza	KKBN IRIGA	1.00	450.00 ₱	450.00 ₱
09/10/2025 21:50:44	KKBN Main Shop/0023	CLYMAX'S (1 pcs.)	edward	Laila Belleza	KKBN IRIGA	1.00	50.00 ₱	50.00 ₱
09/10/2025 21:47:13	KKBN Main Shop/0022	CLYMAX'S (30 capsules / Box)		Laila Belleza	KKBN IRIGA	1.00	1,500.00 ₱	1,500.00 ₱
09/08/2025 09:42:48	KKBN Main Shop/0021	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (1 L.)		Laila Belleza	KKBN IRIGA	1.00	1,450.00 ₱	1,450.00 ₱
09/05/2025 07:35:30	KKBN Main Shop/0020	QUANTUMYN PLUS (15 ml.)		Laila Belleza	KKBN IRIGA	1.00	780.00 ₱	780.00 ₱
09/05/2025 07:34:27	KKBN Main Shop/0019	ACTIVE MAX XIELD(Flowering and Fruiting Stage) (1 L.)		Laila Belleza	KKBN IRIGA	1.00	1,450.00 ₱	1,450.00 ₱

Figure 14: Sales Teams of Point of Sale

Figure 14 showed the Sales Teams feature of the Point-of-Sale module in Odoo ERP, which captured and tracked sales transactions such as Order Date, Order Number, Product, Customer, Salesperson, Company, Quantity, Unit Price, and Total Amount Sold. The system also manual calculation of sales totals, reducing errors and ensuring accurate records.

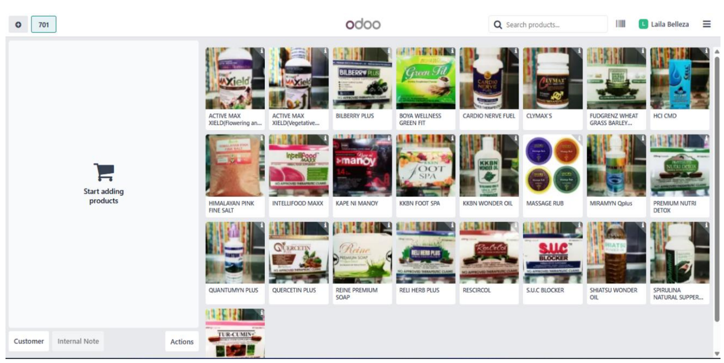


Figure 16: Point of Sale in Products

Figure 16 showed the Odoo Point of Sale (POS) system, which displayed a products available for selection, where each product presented image and name making it visually easy for cashier to identify and add items to customers. The system streamlined for hassle-free retailing, ensuring ease of use by personnel as well as efficiency in handling multiple sales transactions.

4.6 Process Monitoring

In this phase, the researchers implemented the new system at KKBN to resolve existing operational challenges. They evaluated the customized ERP software using the ISO 25010 standard, focusing on functional suitability, performance, usability, maintainability, reliability, and security.

Descriptions	Weighted Mean	Remarks
Functional Suitability	4.78	Highly Acceptable
Performance Efficiency	4.74	Highly Acceptable
Reliability	4.78	Highly Acceptable
Usability	4.68	Highly Acceptable
Maintainability	4.78	Highly Acceptable
Security	4.78	Highly Acceptable
Average	**4.75**	**Highly Acceptable**

The evaluation results show that the Customized Odoo ERP system achieved **highly acceptable ratings** across all six key aspects of software quality functional adequacy, performance efficiency, usability, maintainability, dependability, and security with an overall weighted mean of **4.75**, signifying strong user satisfaction and system effectiveness. All quality dimensions scored between **4.68 and 4.78**, aligning closely with the ISO/IEC 25010 software quality model, which emphasizes **functional suitability, performance efficiency, usability, reliability, maintainability, and security**. This suggests that the ERP system performs well in supporting operational requirements, task efficiency, stability, and data protection.

These findings reinforce the view of Peters & Aggrey [2020], who highlight the importance of these quality criteria in the successful implementation of ERP systems. High satisfaction levels imply that the system enhances organizational productivity, decision-making, and overall operational effectiveness.

However, the evaluation is limited as it relies solely on **user perceptions** within a specific timeframe, thereby not addressing potential long-term performance, scalability, or technical complexities that may arise during extended operations. Despite limitations, the high acceptability and satisfaction ratings signify that the system is effective, reliable, and supportive of organizational growth, providing a strong foundation for continued development and future system enhancements.

5 CONCLUSION

The findings emphasize that to increase operational efficiency, KKBN must think about process automation. inventory, sales, accounting, Point of Sales, purchasing, Employee management/HR and attendance. operations could be monitored in real time, workflows could be streamlined, and errors could be decreased by putting in place an integrated system like Odoo ERP. This study aims to design, and implement, and evaluate an integrated Odoo ERP system to address the problems that KKBN faced in their daily operations that caused by manual processes. The researchers analyzed the existing workflows, and configured the necessary Odoo ERP modules, and assessed their performance using ISO 25010 standards. Based on the results of the implementation and evaluation of the system, the following conclusions: Inventory Module, Sales Module, Accounting Module, POS module, Purchase module, Employee management and attendance module. The findings emphasize how crucial it is to assess ERP systems for user engagement and long-term flexibility in addition to functionality, as these factors have a big impact on adoption and sustainability. In conclusion, the deployment of an integrated Odoo ERP system has proved that automation of key business activities of KKBN has contributed largely to increased efficiencies in terms of data accuracy and real-time tracking. The high acceptability levels, measured using ISO 25010 criteria, also affirm that the system has removed weaknesses in manual processing procedures and, at the same time, remains friendly and stable for continuous functionality and usage. In addition, the effectiveness of integrating other modules such as inventory, sales, accounting, POS, purchasing, and human resource management modules, in Odoo, also emphasizes the effectiveness of Odoo ERP in small and medium-scale businesses.

6 ACKNOWLEDGEMENTS

The researchers sincerely express their heartfelt gratitude to everyone who extended their support, guidance, and encouragement in making this capstone project possible and enhanced the success of this study. Above all, the researchers offer their deepest thanks to the Almighty God for granting them wisdom, strength, and perseverance throughout the completion of this research.

7 REFERENCES

[1] Luay Anaya and Fadi Qutaishat. 2022. ERP systems drive businesses towards growth and sustainability. In Procedia Computer Science, 2022. Elsevier B.V., 854–861. https://doi.org/10.1016/j.procs.2022.08.103

[2] Ary Arvianto, Zainal Fanani Rosyada, Singgih Saptadi, Wiwik Budiawan, and Yasa Ewa Demilda. 2022. ERP ODOO IMPLEMENTATION IN SMALL RETAILERS. International Journal of Applied Science and Engineering Review 03, 06 (2022), 66–85. https://doi.org/10.52267/ijaser.2022.3605

[3] Hamzah Fatoni Efendi and Addin Aditya. 2022. Business Process Analysis and Implementation of Odoo Open Source ERP System in Inventory, Purchasing and Sales Activities (Case Study: Captain Gadget Store) Analisis Proses Bisnis dan Penerapan Sistem ERP Odoo Open Source Pada Aktivitas Persediaan, Pembelian dan Penjualan (Studi Kasus: Captain Gadget). Retrieved from https://pssh.umsida.ac.id.

[4] Teodora Ivanović and Mirjana Marić. 2021. Application of modern Enterprise Resource Planning (ERP) systems in the era of digital transformation. Strategic Management 26, 4 (2021), 28–36. https://doi.org/10.5937/straman2104028i

[5] H Panduwiyasa, M Saputra, Z F Azzahra, and A R Aniko. 2021. Accounting and Smart System: Functional Evaluation of ISO/IEC 25010:2011 Quality Model (a Case Study). IOP Conf Ser Mater Sci Eng 1092, 1 (March 2021), 012065. https://doi.org/10.1088/1757-899x/1092/1/012065

[6] Jejaring Penelitian, Pengabdian Masyarakat, and Arthit Kittisak. 2023. Challenges and Strategies for Inventory Management in Small and Medium-Sized Cosmetic Enterprises: A Review.

[7] Danang Setiawan, Muhammad Fahrezha, Nur Abdillah Bagus Prakoso, and Qurtubi Qurtubi. 2024. A Proposed Framework for ERP System Implementation in SMEs. International Journal of Artificial Intelligence Research 7, 2 (January 2024), 181. https://doi.org/10.29099/ijair.v7i2.1102

[8] Syamsuddin Syamsuddin, Bahrul Ulum Ilham, Kadek Agus Dwiwijaya, Dynes Rizky Navianti, and Reina A. Hadikusumo. 2023. Implementation of an Enterprise Resource Planning (ERP) System and its Impact on Manufacturing Company Operational Efficiency. Global International Journal of Innovative Research 1, 2 (November 2023), 194–199. https://doi.org/10.59613/global.v1i2.29

[9] Mega Putri Utami and Ivana Lucia Kharisma. Implementation of Enterprise Resource Planning (ERP) Based Information System Using Odoo Software. The Eastasouth Journal of Information System and Computer Science 1, 02 , 55–64. https://doi.org/10.58812/esiscs.v1i02

[10] Rona Sutra Dewangga Dyah Utami. 2024. Implementation of Odoo Based on Enterprise Resource Planning System with Sales and Purchasing Module Using Rapid Application Development. (2024).

APPENDIX Q

CERTIFICATE OF PLAGIARISM CHECK

REPUBLIC OF THE PHILIPPINES
CAMARINES SUR POLYTECHNIC COLLEGES
NABUA, CAMARINES SUR
RESEARCH AND DEVELOPMENT SERVICES OFFICE
RESEARCH@CSPC.EDU.PH CSPC.RESEARCHOFFICE

CERTIFICATION

Date of Release:	January 26, 2026
Submission ID:	23367:126896388
Output Title:	DIGITAL TRANSFORMATION OF KKBN BUSINESS PROCESS USING ODOO: A CAPSTONE PROJECT
Author(s):	John Edward M. Hontalba Sharrycel B. Magno Gerald C. Buergo
Program:	Bachelor of Science in Information System

Authenticity Report:

Similarity Index Report: **8%**

Internet Sources: *1%*

Publications: *1%*

Student Papers: *8%*

- *Interpretation:* The similarity index report means that **8%** of the output is similar to the sources in Turnitin's (authenticity software) online repository and databases.

AI-Generated Report: -

AI-generated only: -

AI-generated text that was AI-paraphrased: -

- *Interpretation:* The AI generated report means that the entire output is *less than 20%* and includes the *possibility* of false positives, although some text is *likely* AI-generated.

HAROLD JAN R. TERANO, Ph.D.
Director, Research & Development Services Office